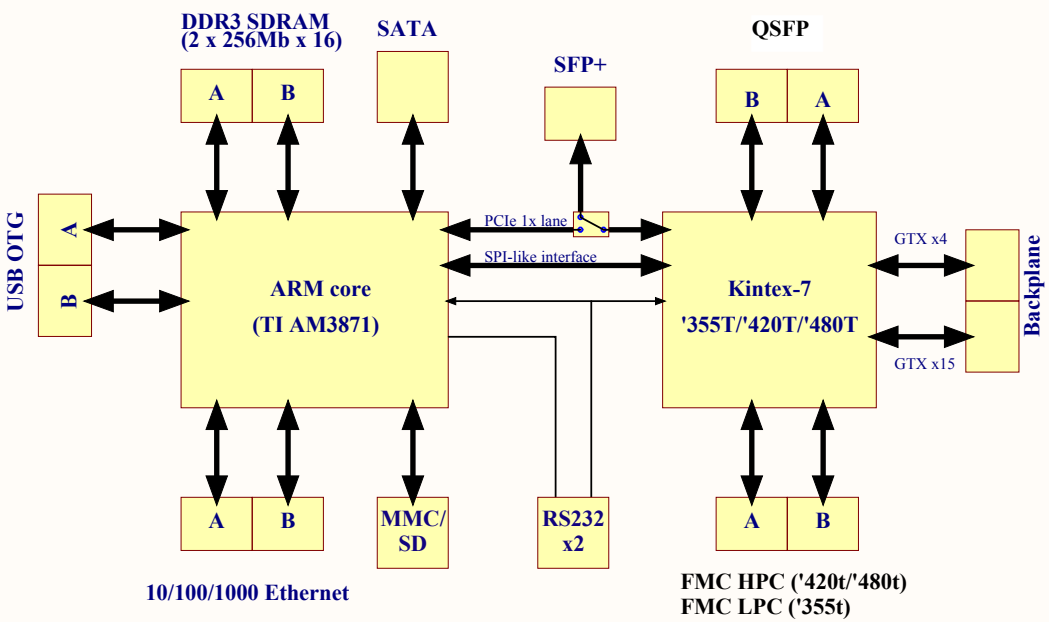
McGill Cosmology Instrumentation Library
ICE Board Schematics



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preamble_gtx.SchDoc	5	Cartoon: FPGA GTX Banking
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AM3871_gpio_3.SchDoc	25	ARM: GPIO bank 3
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SFP.SchDoc	41	SFP Port
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FMC_io.SchDoc	44	FMC: I/O
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power_master.SchDoc	46	Power: Master Sheet
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Power_BuckIndirect.SchDoc	48	Power: Buck Regulators (via intermediate 5v5)
Power_BuckSelect.SchDoc	49	Power: Buck Regulators (selectable)
BuckRegulator.SchDoc	50	Power: Buck Regulator (single)
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clock_generation.SchDoc	52	Clock: Generation
clock_pll.SchDoc	53	Clock: PLL
clock_replication.SchDoc	54	Clock: Replication
clock_passive_coupling.SchDoc	55	Clock: Passive Coupling
I2C_matrix.SchDoc	56	I2C: Interconnect Matrix
I2C_gpio.SchDoc	57	I2C: GPIOs
I2C_temp.SchDoc	58	I2C: Temperature Sensors
mechanical.SchDoc	59	Mechanical design elements (e.g. heat sinks)

http://mcgillcosmology.ca

http://kingspeak.physics.mcgill.ca

Unmarked capacitor packages are 0402s
Unmarked resistor packages are 0603s

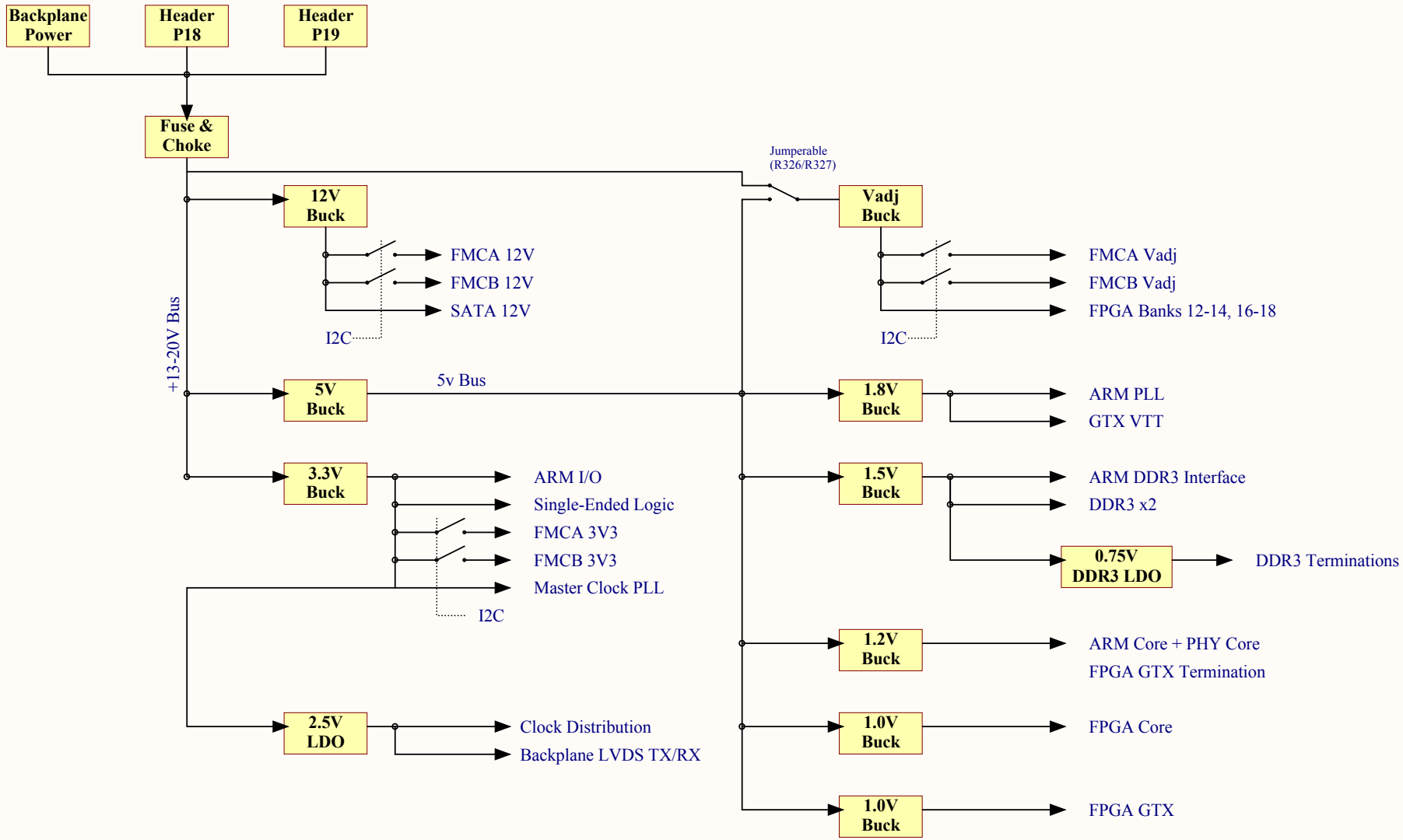
Title: Title Sheet		
Size: Letter	Number	Revision: 3-2014-01-07
Date: 1/9/2014		Sheet 1 of 59
File: C:\Users\...\title.SchDoc		Drawn By: Graeme Smecher

CONTAINER master.SchDoc;preamble_history.SchDoc;preamble_power.SchDoc;preamble_banking.SchDoc;preamble_gtx.SchDoc;preamble_clocking.SchDoc;preamble_addresses.SchDoc

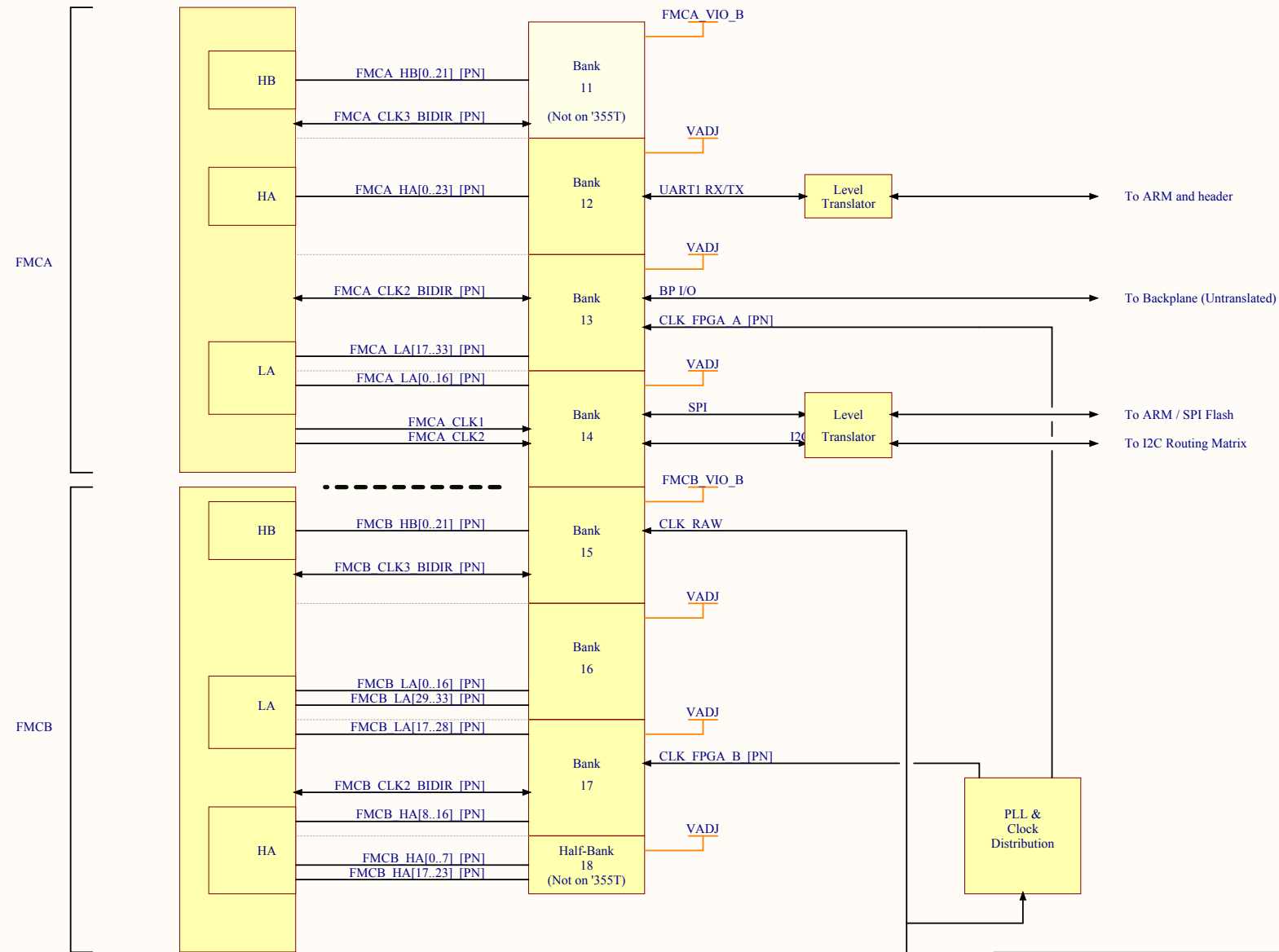
Date	Revision	Tag	Description
Feb. 22, 2013	0	rev0	Schematic capture complete. Preliminary placement. Fab run has significant limitations. FMCs rotated; IMPACT connector pads incorrectly manufactured. Several small, manual modifications required for successful power-up.
Nov. 15, 2013	2b	rev2b_1	Minor component modifications to Rev2 Found errors in Power stages - will recycle old PCBs and cut traces This commit has been used to generate the purchasing BOM

Date	Revision	Tag	Description

Title: Revision History		
Size: Letter	Number:	Revision: 3-2014-01-07
Date: 1/9/2014		Sheet 2 of 59
File: C:\Users\..\preamble_history.SchDoc		Drawn By: Graeme Smecher

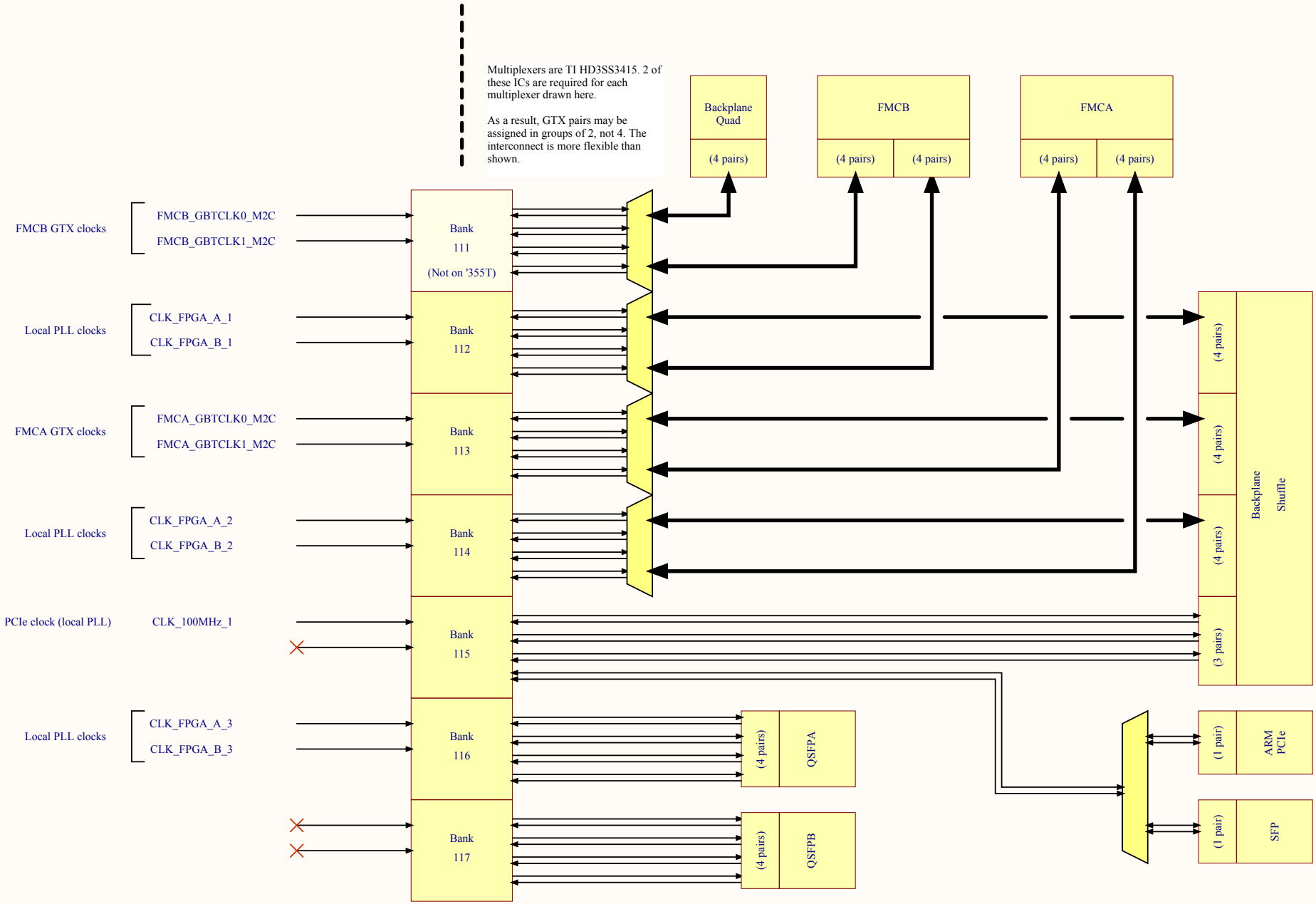


Title: Power Distribution		
Size: Letter	Number:	Revision: 3-2014-01-07
Date: 1/9/2014		Sheet 3 of 59
File: C:\Users\..\preamble_power.SchDoc		Drawn By: Graeme Smecher



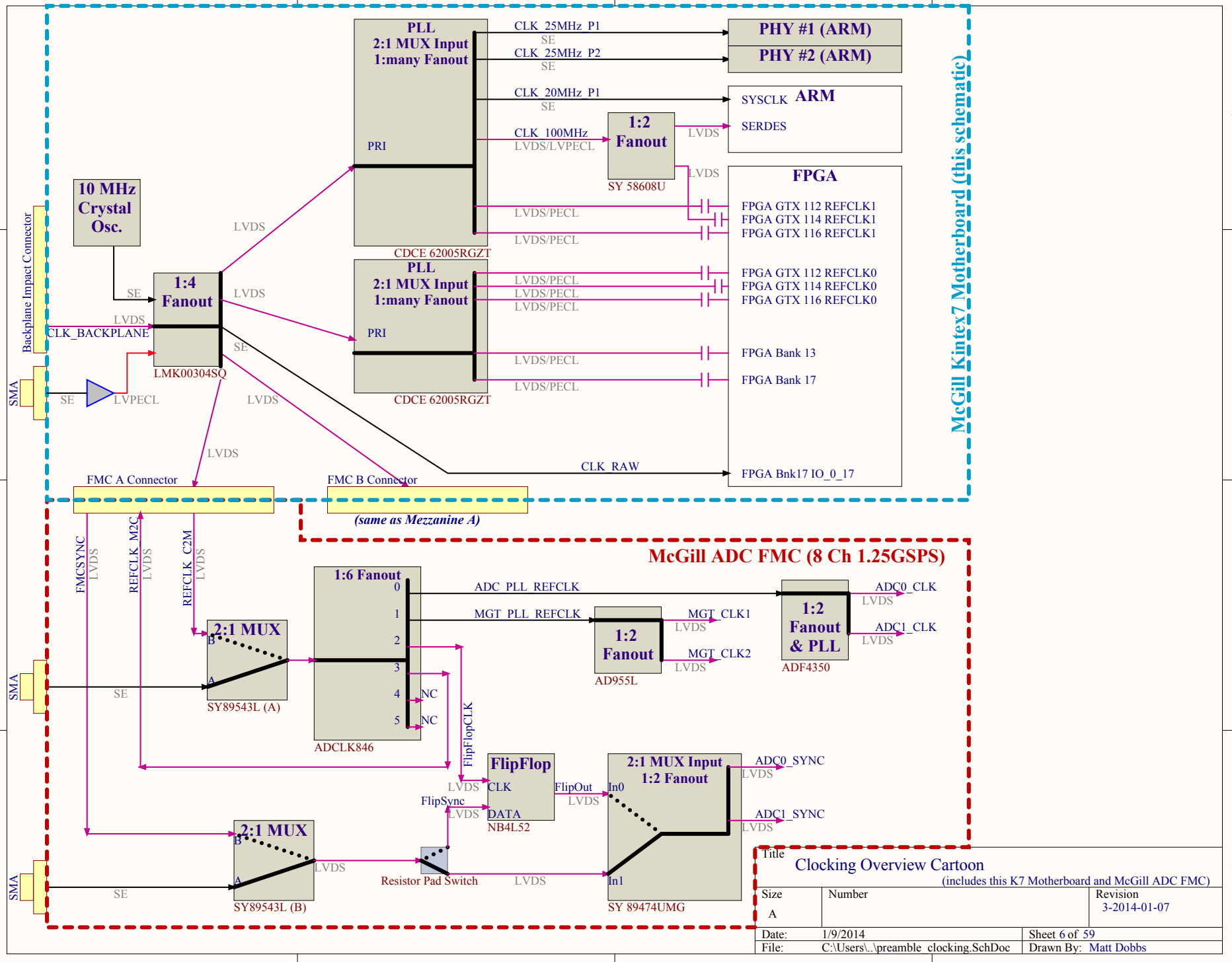
Low-speed FMC signals are controlled via the I2C matrix
FMCA has VREF signals attached to FPGA banks (e.g. for SSTL)

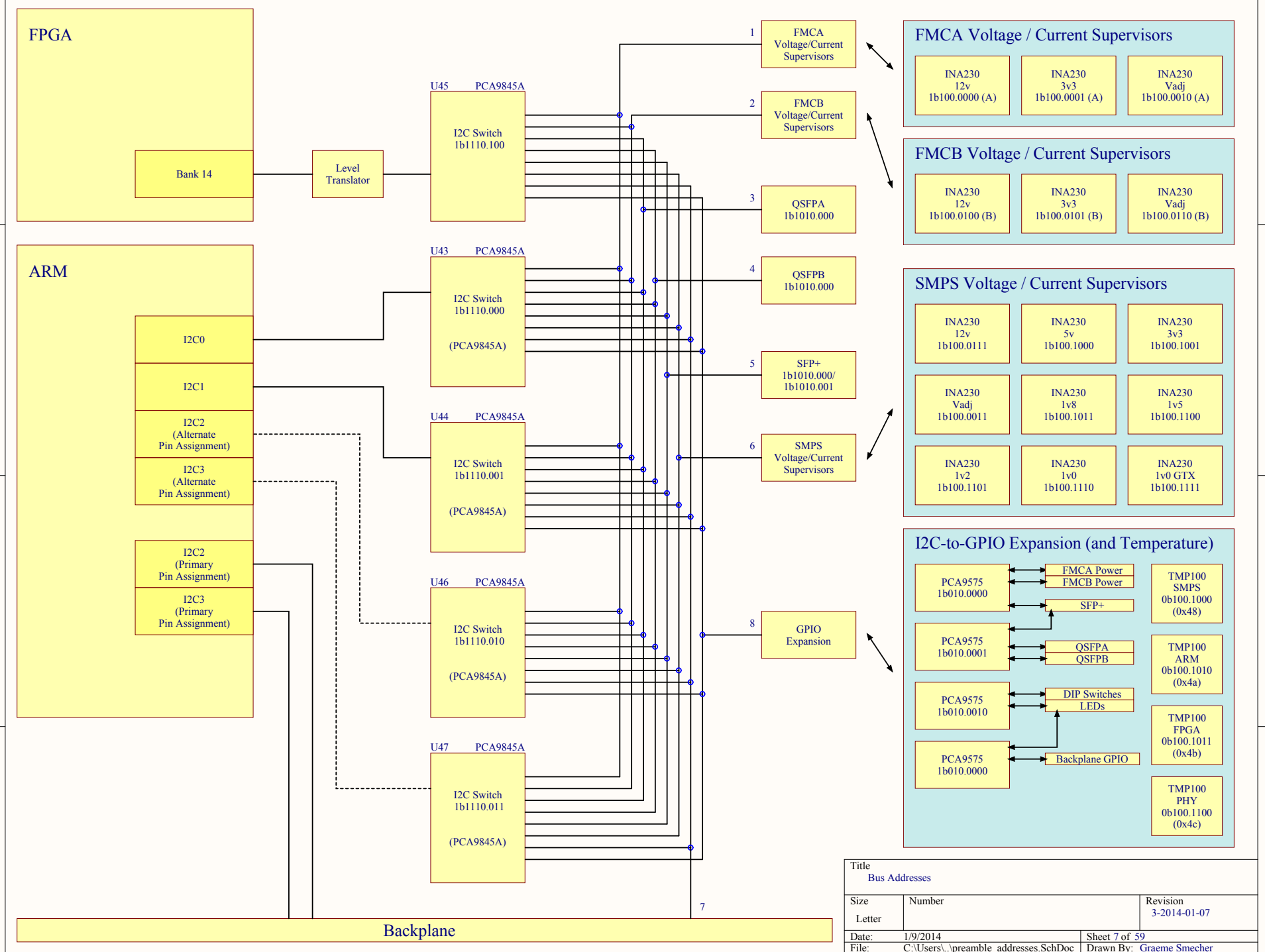
Title: FPGA Banking (Non-GTX)		
Size: Letter	Number:	Revision: 3-2014-01-07
Date: 1/9/2014		Sheet 4 of 59
File: C:\Users\..\preamble_banking.SchDoc		Drawn By: Graeme Smecher



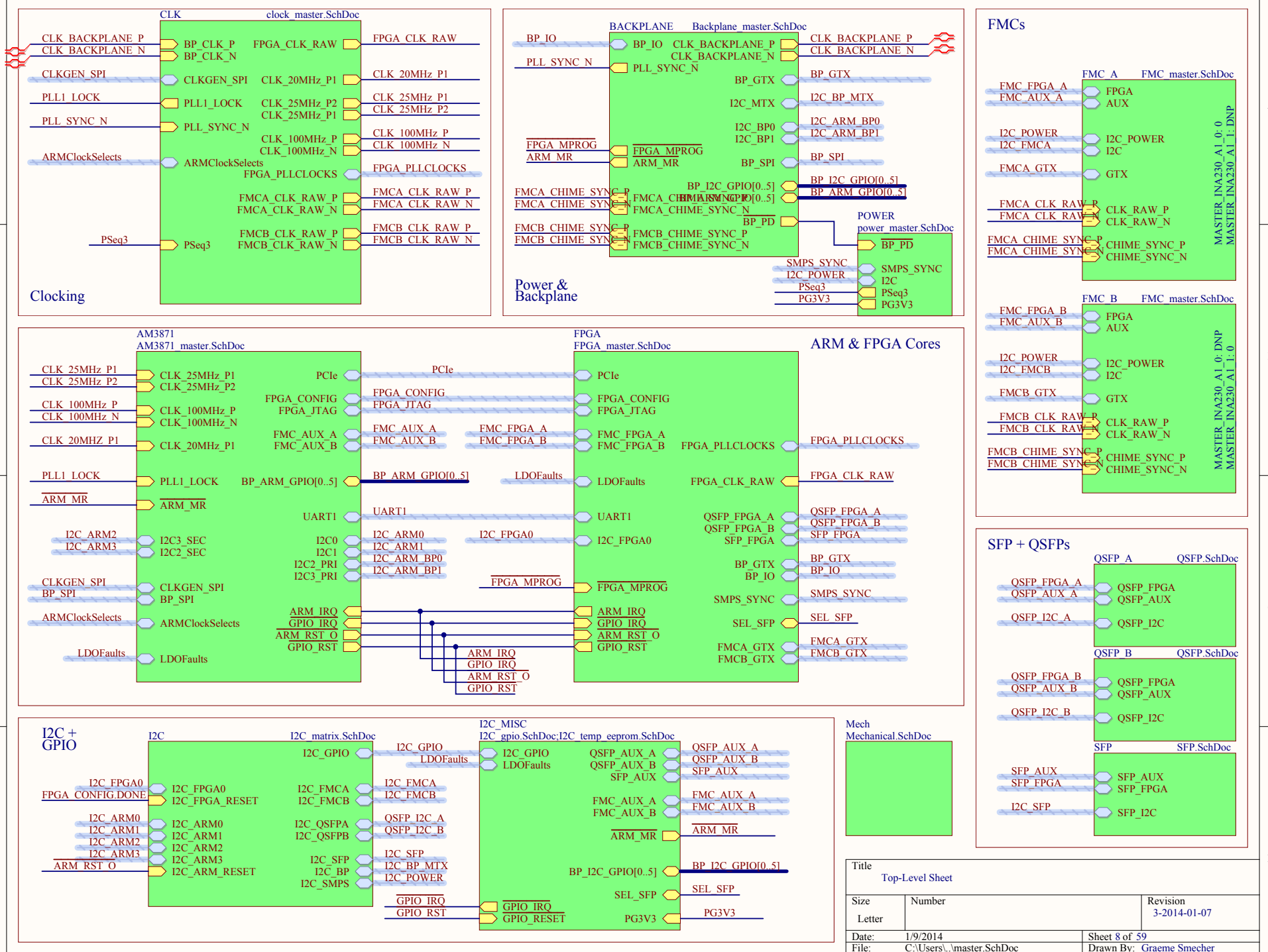
GTX clock interconnects. Each GTX clock can feed the 3 adjacent GTXes.

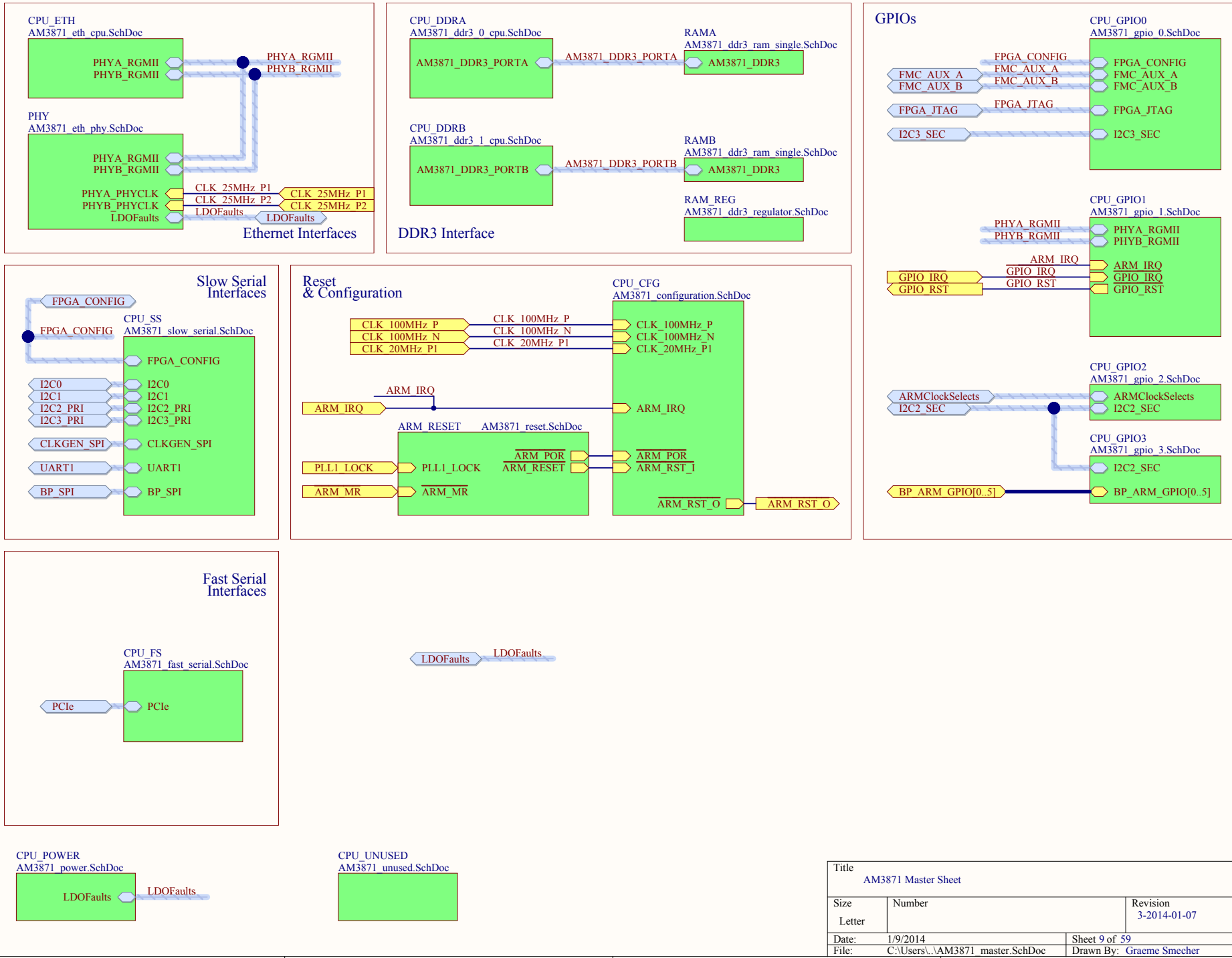
Title: Gigabit Transceivers: Clocking and Routing		
Size: Letter	Number:	Revision: 3-2014-01-07
Date: 1/9/2014		Sheet 5 of 59
File: C:\Users\..\preamble_gtx.SchDoc		Drawn By: Graeme Smecher

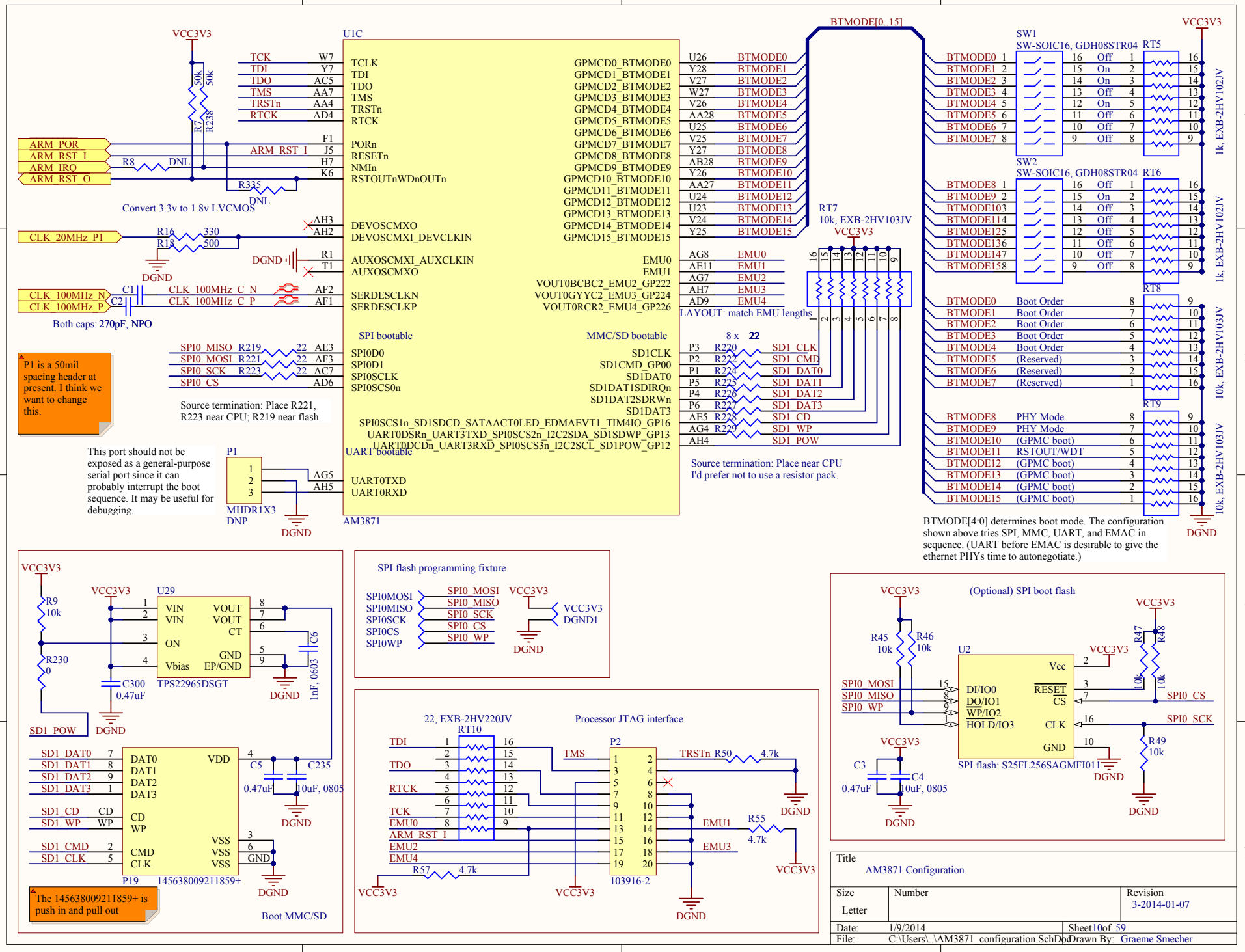

McGill Kintex7 Motherboard (this schematic)
PLL
2:1 MUX Input
1:many Fanout
PRI
CDCE 62005RGZT
CLK_25MHz_P1
CLK_25MHz_P2
SE
PHY #1 (ARM)
PHY #2 (ARM)
CLK_20MHz_P1
SYSCLK
ARM
SERDES
CLK_100MHz
LVDS/LVPECL
1:2
Fanout
SY 58608U
LVDS
FPGA
FPGA GTX 112 REFCLK1
FPGA GTX 114 REFCLK1
FPGA GTX 116 REFCLK1
LVDS/PECL
FPGA GTX 112 REFCLK0
FPGA GTX 114 REFCLK0
FPGA GTX 116 REFCLK0
FPGA Bank 13
FPGA Bank 17
10 MHz
Crystal
Osc.
1:4
Fanout
LMK00304SQ
Backplane Impact Connector
CLK_BACKPLANE
SMA
LVPECL
CLK_RAW
FPGA Bnk17 IO_0_17
FMC A Connector
FMC B Connector
(same as Mezzanine A)
McGill ADC FMC (8 Ch 1.25GSPS)
FMCSYNC
REFCLK_M2C
REFCLK_C2M
2:1 MUX
SY89543L (A)
1:6 Fanout
ADCLK846
NC
ADC_PLL_REFCLK
MGT_PLL_REFCLK
AD955L
MGT_CLK1
MGT_CLK2
1:2
Fanout
& PLL
ADF4350
ADC0_CLK
ADC1_CLK
FlipFlopCLK
FlipFlop
CLK
DATA
NB4L52
FlipSync
FlipOut
2:1 MUX Input
1:2 Fanout
In0
In1
SY 89474UMG
ADC0_SYNC
ADC1_SYNC
SY89543L (B)
Resistor Pad Switch
Title
Clocking Overview Cartoon
(includes this K7 Motherboard and McGill ADC FMC)
Size
A
Number
Revision
3-2014-01-07
Date:
1/9/2014
Sheet 6 of 59
File:
C:\Users\..\preamble_clocking.SchDoc
Drawn By:
Matt Dobbs

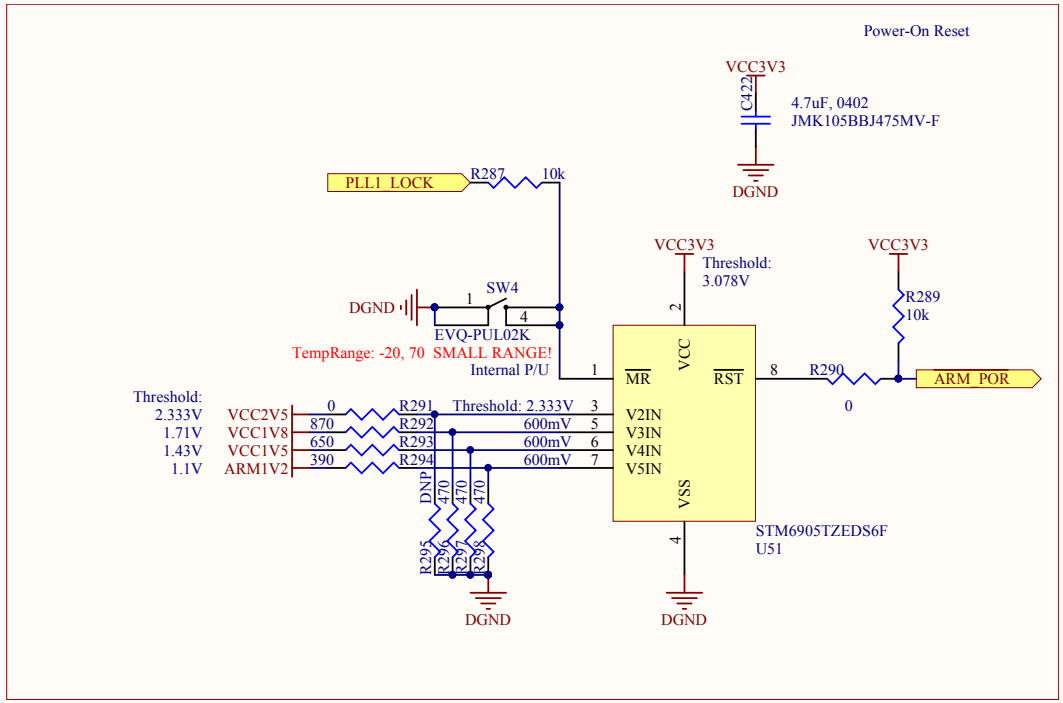


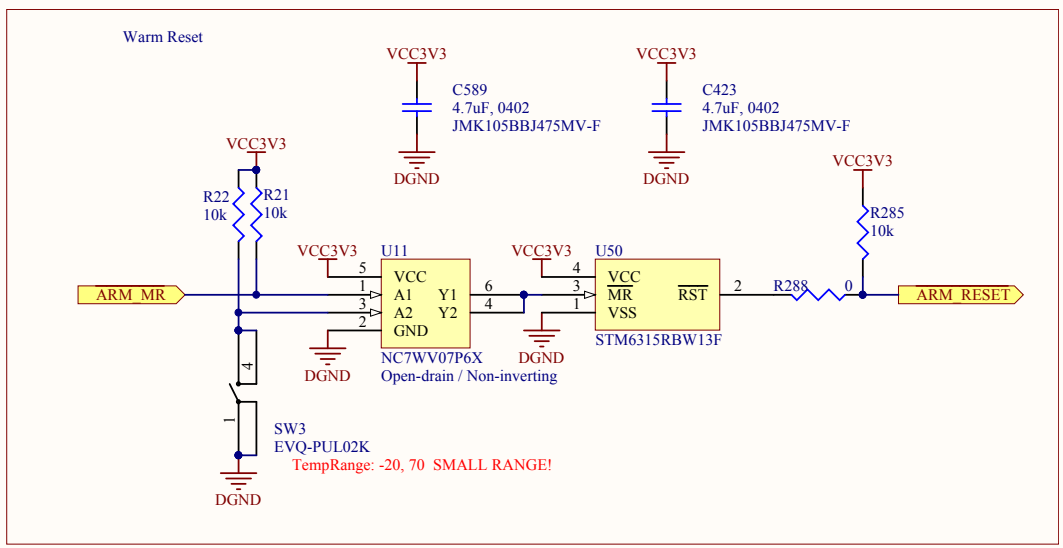
Title: Bus Addresses		
Size: Letter	Number:	Revision: 3-2014-01-07
Date: 1/9/2014		Sheet 7 of 59
File: C:\Users\..\preamble_addresses.SchDoc		Drawn By: Graeme Smecher


Clocking
CLK
clock_master.SchDoc
CLK_BACKPLANE_P
CLK_BACKPLANE_N
CLKGEN_SPI
PLL1_LOCK
PLL_SYNC_N
ARMClockSelects
PSeq3
BP_CLK_P
BP_CLK_N
CLKGEN_SPI
PLL1_LOCK
PLL_SYNC_N
ARMClockSelects
PSeq3
FPGA_CLK_RAW
CLK_20MHz_P1
CLK_25MHz_P2
CLK_25MHz_P1
CLK_100MHz_P
CLK_100MHz_N
FPGA_PLLCLOCKS
FMCA_CLK_RAW_P
FMCA_CLK_RAW_N
FMCB_CLK_RAW_P
FMCB_CLK_RAW_N
FPGA_CLK_RAW
CLK_20MHz_P1
CLK_25MHz_P1
CLK_25MHz_P2
CLK_100MHz_P
CLK_100MHz_N
FPGA_PLLCLOCKS
FMCA_CLK_RAW_P
FMCA_CLK_RAW_N
FMCB_CLK_RAW_P
FMCB_CLK_RAW_N
Power &
Backplane
BACKPLANE
Backplane_master.SchDoc
BP_IO
PLL_SYNC_N
FPGA_MPROG
ARM_MR
FMCA_CHIME_SYNC_P
FMCA_CHIME_SYNC_N
FMCB_CHIME_SYNC_P
FMCB_CHIME_SYNC_N
BP_IO
CLK_BACKPLANE_P
CLK_BACKPLANE_N
PLL_SYNC_N
BP_GTX
I2C_MTX
I2C_BP0
I2C_BP1
FPGA_MPROG
ARM_MR
BP_SPI
BP_I2C_GPIO[0..5]
BP_ARM_GPIO[0..5]
FMCA_CHIME_SYNC_P
FMCA_CHIME_SYNC_N
FMCB_CHIME_SYNC_P
FMCB_CHIME_SYNC_N
BP_PD
CLK_BACKPLANE_P
CLK_BACKPLANE_N
BP_GTX
I2C_BP_MTX
I2C_ARM_BP0
I2C_ARM_BP1
BP_SPI
BP_I2C_GPIO[0..5]
BP_ARM_GPIO[0..5]
POWER
power_master.SchDoc
BP_PD
SMPS_SYNC
I2C_POWER
PSeq3
PG3V3
SMPS_SYNC
I2C
PSeq3
PG3V3
FMCs
FMC_A
FMC_master.SchDoc
FMC_FPGA_A
FMC_AUX_A
I2C_POWER
I2C_FMCA
FMCA_GTX
FMCA_CLK_RAW_P
FMCA_CLK_RAW_N
FMCA_CHIME_SYNC_P
FMCA_CHIME_SYNC_N
FPGA
AUX
I2C_POWER
I2C
GTX
CLK_RAW_P
CLK_RAW_N
CHIME_SYNC_P
CHIME_SYNC_N
MASTER_INA230_A1_0: 0
MASTER_INA230_A1_1: DNP
FMC_B
FMC_master.SchDoc
FMC_FPGA_B
FMC_AUX_B
I2C_POWER
I2C_FMCB
FMCB_GTX
FMCB_CLK_RAW_P
FMCB_CLK_RAW_N
FMCB_CHIME_SYNC_P
FMCB_CHIME_SYNC_N
FPGA
AUX
I2C_POWER
I2C
GTX
CLK_RAW_P
CLK_RAW_N
CHIME_SYNC_P
CHIME_SYNC_N
MASTER_INA230_A1_0: DNP
MASTER_INA230_A1_1: 0
ARM & FPGA Cores
AM3871
AM3871_master.SchDoc
CLK_25MHz_P1
CLK_25MHz_P2
CLK_100MHz_P
CLK_100MHz_N
CLK_20MHZ_P1
PLL1_LOCK
ARM_MR
I2C_ARM2
I2C_ARM3
CLKGEN_SPI
BP_SPI
ARMClockSelects
LDOFaults
CLK_25MHz_P1
CLK_25MHz_P2
CLK_100MHz_P
CLK_100MHz_N
CLK_20MHz_P1
PLL1_LOCK
ARM_MR
I2C3_SEC
I2C2_SEC
CLKGEN_SPI
BP_SPI
ARMClockSelects
LDOFaults
PCIe
FPGA_CONFIG
FPGA_JTAG
FMC_AUX_A
FMC_AUX_B
BP_ARM_GPIO[0..5]
UART1
I2C0
I2C1
I2C2_PRI
I2C3_PRI
I2C_ARM0
I2C_ARM1
I2C_ARM_BP0
I2C_ARM_BP1
ARM_IRQ
GPIO_IRQ
ARM_RST_O
GPIO_RST
PCIe
FPGA_CONFIG
FPGA_JTAG
FMC_AUX_A
FMC_AUX_B
BP_ARM_GPIO[0..5]
UART1
I2C_ARM0
I2C_ARM1
I2C_ARM_BP0
I2C_ARM_BP1
ARM_IRQ
GPIO_IRQ
ARM_RST_O
GPIO_RST
FPGA
FPGA_master.SchDoc
PCIe
FPGA_CONFIG
FPGA_JTAG
FMC_FPGA_A
FMC_FPGA_B
LDOFaults
UART1
I2C_FPGA0
FPGA_MPROG
ARM_IRQ
GPIO_IRQ
ARM_RST_O
GPIO_RST
FPGA_PLLCLOCKS
FPGA_CLK_RAW
QSFP_FPGA_A
QSFP_FPGA_B
SFP_FPGA
BP_GTX
BP_IO
SMPS_SYNC
SEL_SFP
FMCA_GTX
FMCB_GTX
FMC_FPGA_A
FMC_FPGA_B
LDOFaults
I2C_FPGA0
FPGA_MPROG
FPGA_PLLCLOCKS
FPGA_CLK_RAW
QSFP_FPGA_A
QSFP_FPGA_B
SFP_FPGA
BP_GTX
BP_IO
SMPS_SYNC
SEL_SFP
FMCA_GTX
FMCB_GTX
ARM_IRQ
GPIO_IRQ
ARM_RST_O
GPIO_RST
SFP + QSFPs
QSFP_A
QSFP.SchDoc
QSFP_FPGA_A
QSFP_AUX_A
QSFP_I2C_A
QSFP_FPGA
QSFP_AUX
QSFP_I2C
QSFP_B
QSFP.SchDoc
QSFP_FPGA_B
QSFP_AUX_B
QSFP_I2C_B
QSFP_FPGA
QSFP_AUX
QSFP_I2C
SFP
SFP.SchDoc
SFP_AUX
SFP_FPGA
I2C_SFP
SFP_AUX
SFP_FPGA
SFP_I2C
I2C +
GPIO
I2C
I2C_matrix.SchDoc
I2C_FPGA0
FPGA_CONFIG.DONE
I2C_ARM0
I2C_ARM1
I2C_ARM2
I2C_ARM3
ARM_RST_O
I2C_GPIO
I2C_FMCA
I2C_FMCB
I2C_QSFPA
I2C_QSFPB
I2C_SFP
I2C_BP
I2C_SMPS
I2C_FPGA0
I2C_FPGA_RESET
I2C_ARM0
I2C_ARM1
I2C_ARM2
I2C_ARM3
I2C_ARM_RESET
I2C_GPIO
I2C_FMCA
I2C_FMCB
QSFP_I2C_A
QSFP_I2C_B
I2C_SFP
I2C_BP_MTX
I2C_POWER
GPIO_IRQ
GPIO_RST
I2C_MISC
I2C_gpio.SchDoc;I2C_temp_eeprom.SchDoc
I2C_GPIO
LDOFaults
GPIO_IRQ
GPIO_RESET
I2C_GPIO
LDOFaults
QSFP_AUX_A
QSFP_AUX_B
SFP_AUX
FMC_AUX_A
FMC_AUX_B
ARM_MR
BP_I2C_GPIO[0..5]
SEL_SFP
PG3V3
QSFP_AUX_A
QSFP_AUX_B
SFP_AUX
FMC_AUX_A
FMC_AUX_B
ARM_MR
BP_I2C_GPIO[0..5]
SEL_SFP
PG3V3
Mech
Mechanical.SchDoc
Title
Top-Level Sheet
Size
Letter
Number
Revision
3-2014-01-07
Date:
1/9/2014
Sheet 8 of 59
File:
C:\Users\..\master.SchDoc
Drawn By:
Graeme Smecher

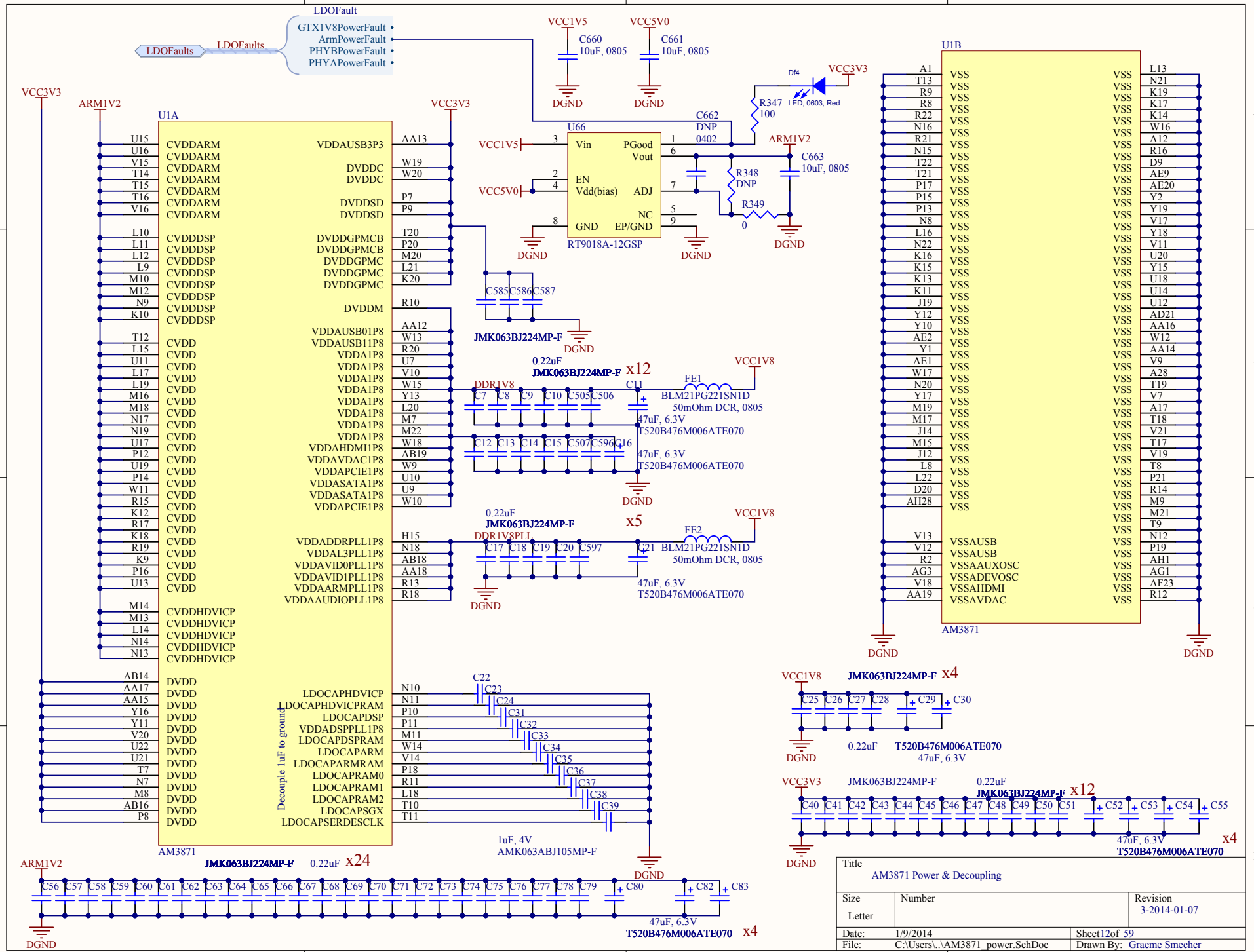

CPU_ETH
AM3871_eth_cpu.SchDoc
PHYA_RGMII
PHYB_RGMII
PHYA_RGMII
PHYB_RGMII
PHY
AM3871_eth_phy.SchDoc
PHYA_RGMII
PHYB_RGMII
PHYA_PHYCLK
PHYB_PHYCLK
LDOFaults
CLK_25MHz_P1
CLK_25MHz_P2
LDOFaults
CLK_25MHz_P1
CLK_25MHz_P2
LDOFaults
Ethernet Interfaces
CPU_DDRA
AM3871_ddr3_0_cpu.SchDoc
AM3871_DDR3_PORTA
AM3871_DDR3_PORTA
RAMA
AM3871_ddr3_ram_single.SchDoc
AM3871_DDR3
CPU_DDRB
AM3871_ddr3_1_cpu.SchDoc
AM3871_DDR3_PORTB
AM3871_DDR3_PORTB
RAMB
AM3871_ddr3_ram_single.SchDoc
AM3871_DDR3
RAM_REG
AM3871_ddr3_regulator.SchDoc
DDR3 Interface
GPIOs
CPU_GPIO0
AM3871_gpio_0.SchDoc
FPGA_CONFIG
FMC_AUX_A
FMC_AUX_B
FPGA_JTAG
I2C3_SEC
FPGA_CONFIG
FMC_AUX_A
FMC_AUX_B
FPGA_JTAG
I2C3_SEC
CPU_GPIO1
AM3871_gpio_1.SchDoc
PHYA_RGMII
PHYB_RGMII
ARM_IRQ
GPIO_IRQ
GPIO_RST
PHYA_RGMII
PHYB_RGMII
ARM_IRQ
GPIO_IRQ
GPIO_RST
CPU_GPIO2
AM3871_gpio_2.SchDoc
ARMClockSelects
I2C2_SEC
ARMClockSelects
I2C2_SEC
CPU_GPIO3
AM3871_gpio_3.SchDoc
I2C2_SEC
BP_ARM_GPIO[0..5]
BP_ARM_GPIO[0..5]
Slow Serial
Interfaces
FPGA_CONFIG
FPGA_CONFIG
CPU_SS
AM3871_slow_serial.SchDoc
FPGA_CONFIG
I2C0
I2C1
I2C2_PRI
I2C3_PRI
CLKGEN_SPI
UART1
BP_SPI
I2C0
I2C1
I2C2_PRI
I2C3_PRI
CLKGEN_SPI
UART1
BP_SPI
Reset
& Configuration
CPU_CFG
AM3871_configuration.SchDoc
CLK_100MHz_P
CLK_100MHz_N
CLK_20MHz_P1
CLK_100MHz_P
CLK_100MHz_N
CLK_20MHz_P1
ARM_IRQ
ARM_IRQ
ARM_IRQ
ARM_RESET
AM3871_reset.SchDoc
PLL1_LOCK
ARM_MR
PLL1_LOCK
ARM_MR
ARM_POR
ARM_RESET
ARM_POR
ARM_RST_I
ARM_RST_O
ARM_RST_O
Fast Serial
Interfaces
CPU_FS
AM3871_fast_serial.SchDoc
PCIe
PCIe
LDOFaults
LDOFaults
CPU_POWER
AM3871_power.SchDoc
LDOFaults
LDOFaults
CPU_UNUSED
AM3871_unused.SchDoc
Title
AM3871 Master Sheet
Size
Letter
Number
Revision
3-2014-01-07
Date:
1/9/2014
Sheet 9 of 59
File:
C:\Users\..\AM3871_master.SchDoc
Drawn By:
Graeme Smecher


U1C
AM3871
VCC3V3
TCK W7
TDI Y7
TDO AC5
TMS AA7
TRSTn AA4
RTCK AD4
ARM_POR
ARM_RST_I
ARM_IRQ
ARM_RST_O
R8 DNL
R7 50k
R238 50k
R335 DNL
Convert 3.3v to 1.8v LVCMOS
CLK_20MHz_P1
R16 330
R18 500
DGND
C1
C2
CLK_100MHz_C_N
CLK_100MHz_C_P
Both caps: 270pF, NPO
P1 is a 50mil spacing header at present. I think we want to change this.
SPI0_MISO R219 22
SPI0_MOSI R221 22
SPI0_SCK R223 22
SPI0_CS
Source termination: Place R221, R223 near CPU; R219 near flash.
This port should not be exposed as a general-purpose serial port since it can probably interrupt the boot sequence. It may be useful for debugging.
P1
MHDR1X3
DNP
TCLK
TDI
TDO
TMS
TRSTn
RTCK
PORn
RESETn
NMIn
RSTOUTnWDnOUTn
DEVOSCMXO
DEVOSCMXI_DEVCLKIN
AUXOSCMXI_AUXCLKIN
AUXOSCMXO
SERDESCLKN
SERDESCLKP
SPI bootable
SPI0D0
SPI0D1
SPI0SCLK
SPI0SCS0n
MMC/SD bootable
SD1CLK
SD1CMD_GP00
SD1DAT0
SD1DAT1SDIRQn
SD1DAT2SDRWn
SD1DAT3
SPI0SCS1n_SD1SDCD_SATAACT0LED_EDMAEVT1_TIM4IO_GP16
UART0DSRn_UART3TXD_SPI0SCS2n_I2C2SDA_SD1SDWP_GP13
UART0DCDn_UART3RXD_SPI0SCS3n_I2C2SCL_SD1POW_GP12
UART bootable
UART0TXD
UART0RXD
GPMCD0_BTMODE0
GPMCD1_BTMODE1
GPMCD2_BTMODE2
GPMCD3_BTMODE3
GPMCD4_BTMODE4
GPMCD5_BTMODE5
GPMCD6_BTMODE6
GPMCD7_BTMODE7
GPMCD8_BTMODE8
GPMCD9_BTMODE9
GPMCD10_BTMODE10
GPMCD11_BTMODE11
GPMCD12_BTMODE12
GPMCD13_BTMODE13
GPMCD14_BTMODE14
GPMCD15_BTMODE15
EMU0
EMU1
VOUT0BCBC2_EMU2_GP222
VOUT0GYYC2_EMU3_GP224
VOUT0RCR2_EMU4_GP226
BTMODE[0..15]
U26 BTMODE0
Y28 BTMODE1
V27 BTMODE2
W27 BTMODE3
V26 BTMODE4
AA28 BTMODE5
U25 BTMODE6
V25 BTMODE7
Y27 BTMODE8
AB28 BTMODE9
Y26 BTMODE10
AA27 BTMODE11
U24 BTMODE12
U23 BTMODE13
V24 BTMODE14
Y25 BTMODE15
AG8 EMU0
AE11 EMU1
AG7 EMU2
AH7 EMU3
AD9 EMU4
LAYOUT: match EMU lengths
RT7
10k, EXB-2HV103JV
VCC3V3
8 x 22
P3 R220 SD1_CLK
P2 R222 SD1_CMD
P1 R224 SD1_DAT0
P5 R225 SD1_DAT1
P4 R226 SD1_DAT2
P6 R227 SD1_DAT3
AE5 R228 SD1_CD
AG4 R229 SD1_WP
AH4 SD1_POW
Source termination: Place near CPU
I'd prefer not to use a resistor pack.
SW1
SW-SOIC16, GDH08STR04 RT5
SW2
SW-SOIC16, GDH08STR04 RT6
BTMODE0 Off
BTMODE1 On
BTMODE2 On
BTMODE3 Off
BTMODE4 On
BTMODE5 Off
BTMODE6 Off
BTMODE7 Off
BTMODE8 Off
BTMODE9 On
BTMODE103 Off
BTMODE114 Off
BTMODE125 Off
BTMODE136 Off
BTMODE147 Off
BTMODE158 Off
1k, EXB-2HV102JV
RT8
BTMODE0 Boot Order
BTMODE1 Boot Order
BTMODE2 Boot Order
BTMODE3 Boot Order
BTMODE4 Boot Order
BTMODE5 (Reserved)
BTMODE6 (Reserved)
BTMODE7 (Reserved)
10k, EXB-2HV103JV
RT9
BTMODE8 PHY Mode
BTMODE9 PHY Mode
BTMODE10 (GPMC boot)
BTMODE11 RSTOUT/WDT
BTMODE12 (GPMC boot)
BTMODE13 (GPMC boot)
BTMODE14 (GPMC boot)
BTMODE15 (GPMC boot)
10k, EXB-2HV103JV
BTMODE[4:0] determines boot mode. The configuration shown above tries SPI, MMC, UART, and EMAC in sequence. (UART before EMAC is desirable to give the ethernet PHYs time to autonegotiate.)
R9 10k
R230 0
C300 0.47uF
U29
TPS22965DSGT
VIN VOUT CT ON GND Vbias EP/GND
C6 1nF, 0603
SD1_POW
SD1_DAT0 SD1_DAT1 SD1_DAT2 SD1_DAT3
SD1_CD SD1_WP
SD1_CMD SD1_CLK
DAT0 DAT1 DAT2 DAT3 CD WP CMD CLK VDD VSS VSS VSS
C5 0.47uF
C235 10uF, 0805
P19 145638009211859+
The 145638009211859+ is push in and pull out
Boot MMC/SD
SPI flash programming fixture
SPI0MOSI SPI0MISO SPI0SCK SPI0CS SPI0WP
SPI0_MOSI SPI0_MISO SPI0_SCK SPI0_CS SPI0_WP
VCC3V3 DGND1
22, EXB-2HV220JV
RT10
Processor JTAG interface
P2
TDI TDO RTCK TCK EMU0 ARM_RST_I EMU2 EMU4
TMS
TRSTn R50 4.7k
EMU1 R55 4.7k
EMU3
R57 4.7k
103916-2
(Optional) SPI boot flash
R45 10k
R46 10k
R47 10k
R48 10k
R49 10k
U2
SPI flash: S25FL256SAGMFI011
Vcc RESET CS CLK GND DI/IO0 DO/IO1 WP/IO2 HOLD/IO3
SPI0_CS
SPI0_SCK
C3 0.47uF
C4 10uF, 0805
Title
AM3871 Configuration
Size Letter
Number
Revision 3-2014-01-07
Date: 1/9/2014
Sheet 10 of 59
File: C:\Users\..\AM3871_configuration.SchDoc
Drawn By: Graeme Smecher





Title: AM3871 Reset		
Size: Letter	Number:	Revision: 3-2014-01-07
Date: 1/9/2014		Sheet 11 of 59
File: C:\Users\..\AM3871_reset.SchDoc		Drawn By: Graeme Smecher

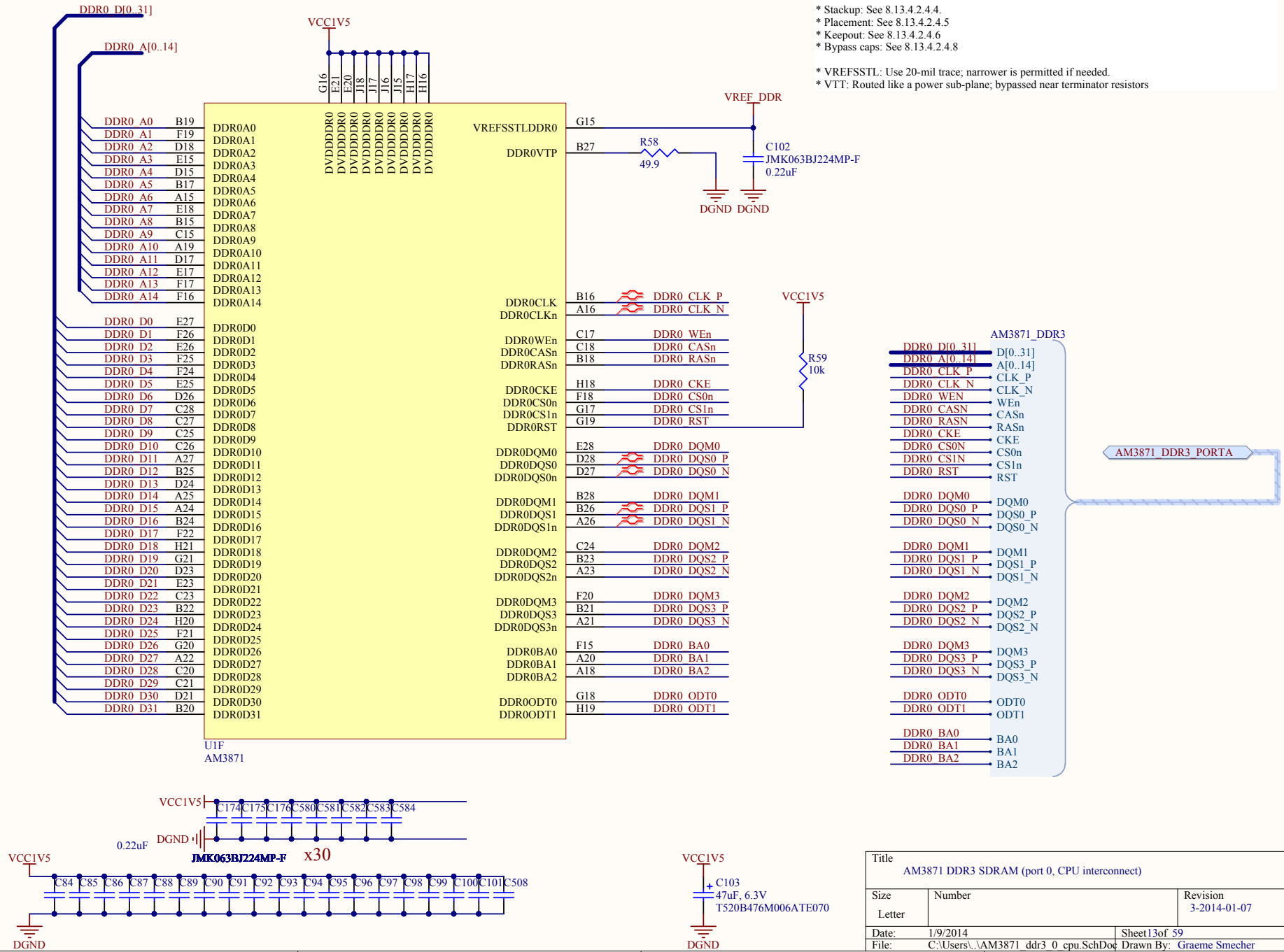


Title	AM3871 Power & Decoupling	
Size: Letter	Number	Revision: 3-2014-01-07
Date: 1/9/2014		Sheet 12 of 59
File: C:\Users\..\AM3871_power.SchDoc		Drawn By: Graeme Smecher

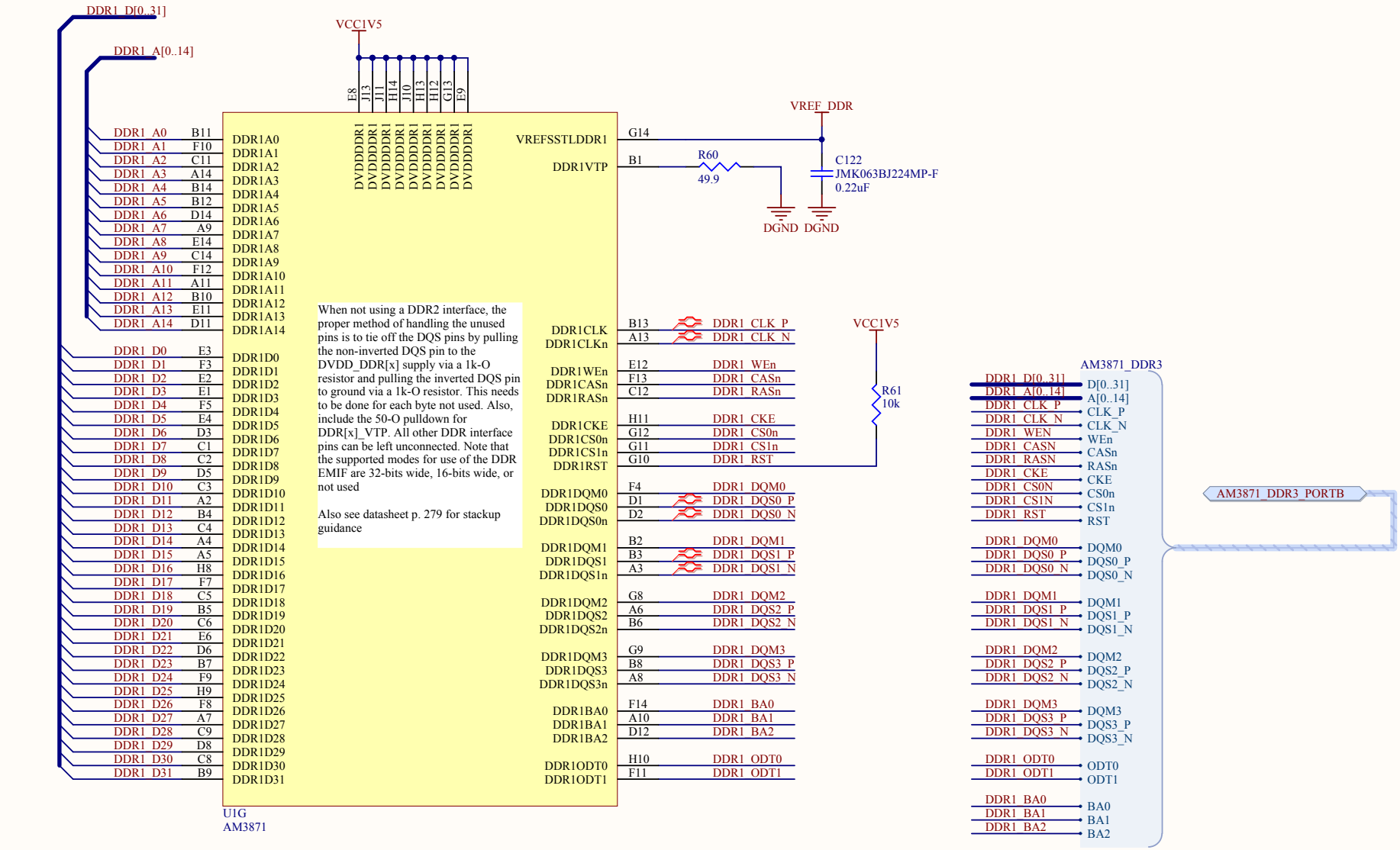
Datasheet (http://www.ti.com/lit/ds/symlink/am3871.pdf) contains extensive stackup/layout directions. Here are pointers:

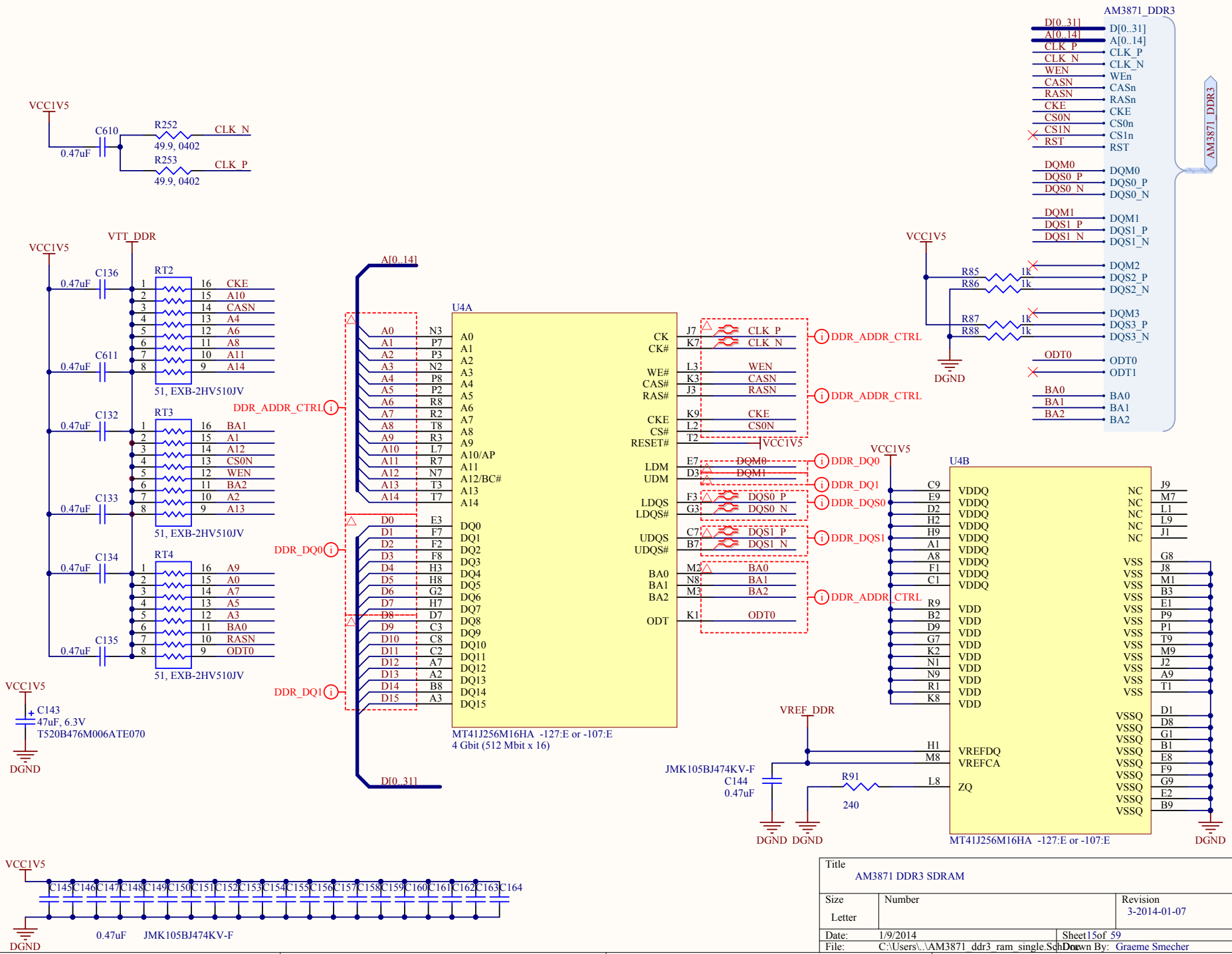
* Stackup: See 8.13.4.2.4.4.
* Placement: See 8.13.4.2.4.5
* Keepout: See 8.13.4.2.4.6
* Bypass caps: See 8.13.4.2.4.8

* VREFSSTL: Use 20-mil trace; narrower is permitted if needed.
* VTT: Routed like a power sub-plane; bypassed near terminator resistors

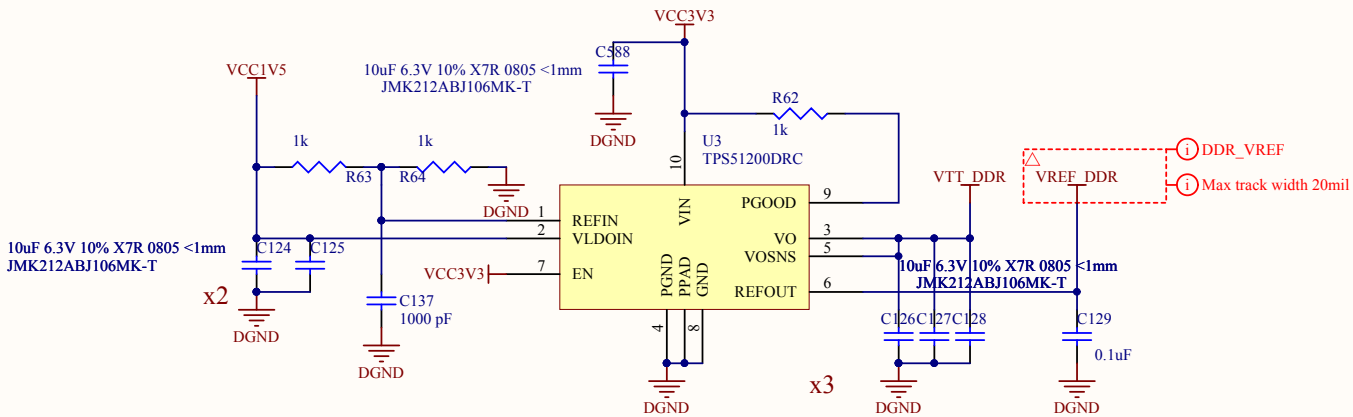


Title	AM3871 DDR3 SDRAM (port 0, CPU interconnect)	
Size: Letter	Number	Revision: 3-2014-01-07
Date: 1/9/2014	Sheet 13 of 59	
File: C:\Users\..\AM3871_ddr3_0_cpu.SchDoc	Drawn By: Graeme Smecher	


DDR1_D[0..31]
DDR1_A[0..14]
VCC1V5
E8 J13 J11 H14 J10 H13 H12 G13 E9
DVDDDDR1
VREF_DDR
VREFSSTLDDR1
DDR1VTP
R60
49.9
C122
JMK063BJ224MP-F
0.22uF
DGND
When not using a DDR2 interface, the proper method of handling the unused pins is to tie off the DQS pins by pulling the non-inverted DQS pin to the DVDD_DDR[x] supply via a 1k-O resistor and pulling the inverted DQS pin to ground via a 1k-O resistor. This needs to be done for each byte not used. Also, include the 50-O pulldown for DDR[x]_VTP. All other DDR interface pins can be left unconnected. Note that the supported modes for use of the DDR EMIF are 32-bits wide, 16-bits wide, or not used
Also see datasheet p. 279 for stackup guidance
DDR1_A0 B11
DDR1_A1 F10
DDR1_A2 C11
DDR1_A3 A14
DDR1_A4 B14
DDR1_A5 B12
DDR1_A6 D14
DDR1_A7 A9
DDR1_A8 E14
DDR1_A9 C14
DDR1_A10 F12
DDR1_A11 A11
DDR1_A12 B10
DDR1_A13 E11
DDR1_A14 D11
DDR1_D0 E3
DDR1_D1 F3
DDR1_D2 E2
DDR1_D3 E1
DDR1_D4 F5
DDR1_D5 E4
DDR1_D6 D3
DDR1_D7 C1
DDR1_D8 C2
DDR1_D9 D5
DDR1_D10 C3
DDR1_D11 A2
DDR1_D12 B4
DDR1_D13 C4
DDR1_D14 A4
DDR1_D15 A5
DDR1_D16 H8
DDR1_D17 F7
DDR1_D18 C5
DDR1_D19 B5
DDR1_D20 C6
DDR1_D21 E6
DDR1_D22 D6
DDR1_D23 B7
DDR1_D24 F9
DDR1_D25 H9
DDR1_D26 F8
DDR1_D27 A7
DDR1_D28 C9
DDR1_D29 D8
DDR1_D30 C8
DDR1_D31 B9
DDR1A0 DDR1A1 DDR1A2 DDR1A3 DDR1A4 DDR1A5 DDR1A6 DDR1A7 DDR1A8 DDR1A9 DDR1A10 DDR1A11 DDR1A12 DDR1A13 DDR1A14
DDR1D0 DDR1D1 DDR1D2 DDR1D3 DDR1D4 DDR1D5 DDR1D6 DDR1D7 DDR1D8 DDR1D9 DDR1D10 DDR1D11 DDR1D12 DDR1D13 DDR1D14 DDR1D15 DDR1D16 DDR1D17 DDR1D18 DDR1D19 DDR1D20 DDR1D21 DDR1D22 DDR1D23 DDR1D24 DDR1D25 DDR1D26 DDR1D27 DDR1D28 DDR1D29 DDR1D30 DDR1D31
G14
B1
DDR1CLK DDR1CLKn
DDR1WEn DDR1CASn DDR1RASn
DDR1CKE DDR1CS0n DDR1CS1n DDR1RST
DDR1DQM0 DDR1DQS0 DDR1DQS0n
DDR1DQM1 DDR1DQS1 DDR1DQS1n
DDR1DQM2 DDR1DQS2 DDR1DQS2n
DDR1DQM3 DDR1DQS3 DDR1DQS3n
DDR1BA0 DDR1BA1 DDR1BA2
DDR1ODT0 DDR1ODT1
B13 A13
E12 F13 C12
H11 G12 G11 G10
F4 D1 D2
B2 B3 A3
G8 A6 B6
G9 B8 A8
F14 A10 D12
H10 F11
DDR1_CLK_P DDR1_CLK_N
DDR1_WEn DDR1_CASn DDR1_RASn
DDR1_CKE DDR1_CS0n DDR1_CS1n DDR1_RST
DDR1_DQM0 DDR1_DQS0_P DDR1_DQS0_N
DDR1_DQM1 DDR1_DQS1_P DDR1_DQS1_N
DDR1_DQM2 DDR1_DQS2_P DDR1_DQS2_N
DDR1_DQM3 DDR1_DQS3_P DDR1_DQS3_N
DDR1_BA0 DDR1_BA1 DDR1_BA2
DDR1_ODT0 DDR1_ODT1
VCC1V5
R61
10k
U1G
AM3871
AM3871_DDR3
D[0..31]
A[0..14]
CLK_P CLK_N
WEn CASn RASn CKE CS0n CS1n RST
DQM0 DQS0_P DQS0_N
DQM1 DQS1_P DQS1_N
DQM2 DQS2_P DQS2_N
DQM3 DQS3_P DQS3_N
ODT0 ODT1
BA0 BA1 BA2
AM3871_DDR3_PORTB
VCC1V5
C141 C142 C165 C166 C167 C168 C169 C170 C171 C172 C173
DGND
0.22uF JMK063BJ224MP-F x30
C104 C105 C106 C107 C108 C109 C110 C111 C112 C113 C114 C115 C116 C117 C118 C119 C120 C121 C509
C123
47uF, 6.3V
T520B476M006ATE070
Title
AM3871 DDR3 SDRAM (port 1, CPU interconnect)
Size
Letter
Number
Revision
3-2014-01-07
Date: 1/9/2014
Sheet 14 of 59
File: C:\Users\..\AM3871_ddr3_1_cpu.SchDoc
Drawn By: Graeme Smecher

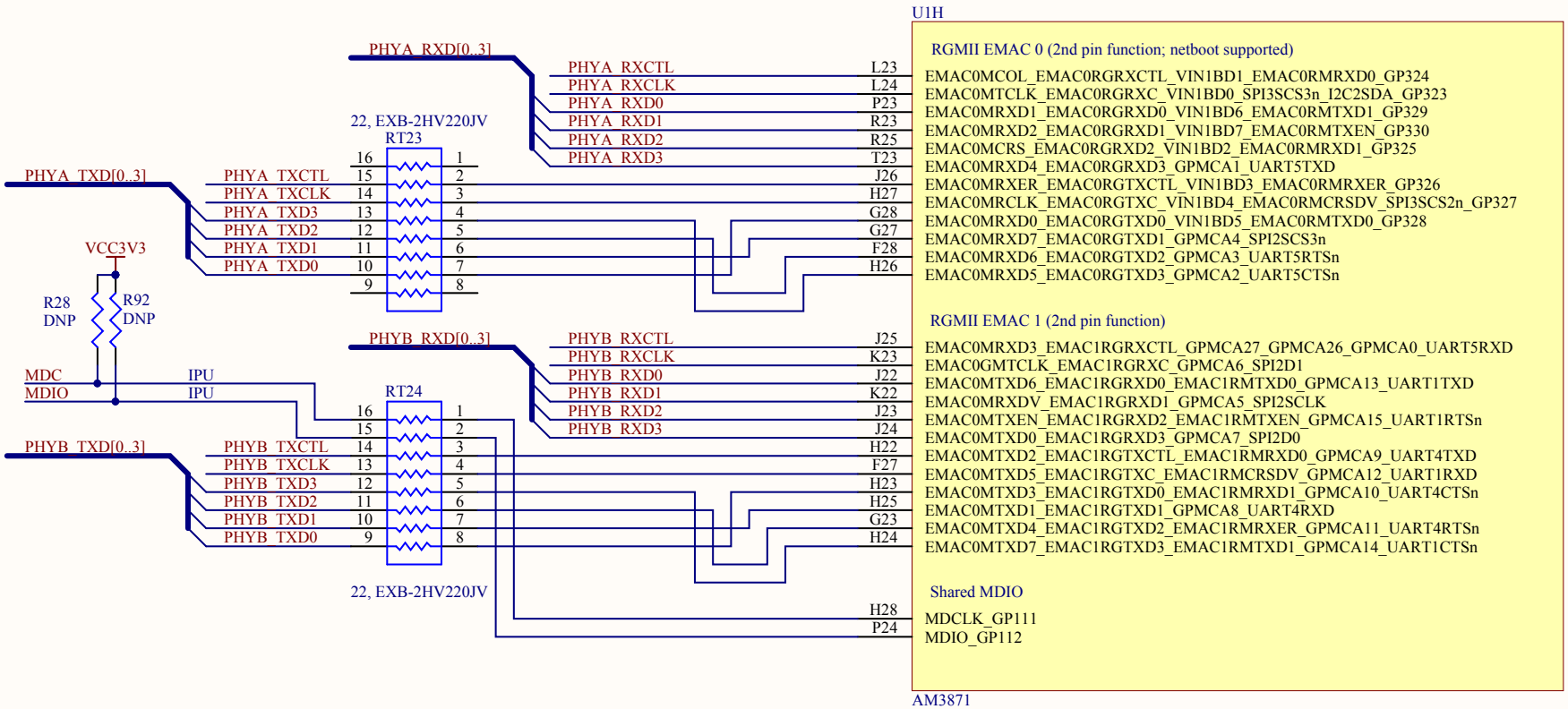

AM3871_DDR3
D[0..31]
A[0..14]
CLK_P
CLK_N
WEN
CASN
RASN
CKE
CS0N
CS1N
RST
DQM0
DQS0_P
DQS0_N
DQM1
DQS1_P
DQS1_N
DQM2
DQS2_P
DQS2_N
DQM3
DQS3_P
DQS3_N
ODT0
ODT1
BA0
BA1
BA2
VCC1V5
VTT_DDR
C610
0.47uF
R252
49.9, 0402
R253
49.9, 0402
C136
C611
C132
C133
C134
C135
RT2
RT3
RT4
51, EXB-2HV510JV
DDR_ADDR_CTRL
DDR_DQ0
DDR_DQ1
U4A
MT41J256M16HA -127:E or -107:E
4 Gbit (512 Mbit x 16)
R85
R86
R87
R88
1k
DGND
U4B
VREF_DDR
JMK105BJ474KV-F
C144
R91
240
C143
47uF, 6.3V
T520B476M006ATE070
C145 C146 C147 C148 C149 C150 C151 C152 C153 C154 C155 C156 C157 C158 C159 C160 C161 C162 C163 C164
Title
AM3871 DDR3 SDRAM
Size
Letter
Number
Revision
3-2014-01-07
Date: 1/9/2014
Sheet 15 of 59
File: C:\Users\..\AM3871_ddr3_ram_single.SchDoc
Drawn By: Graeme Smecher

VTT_DDR supply is 0.75v (1.5v/2)
Current could be 2 Amps

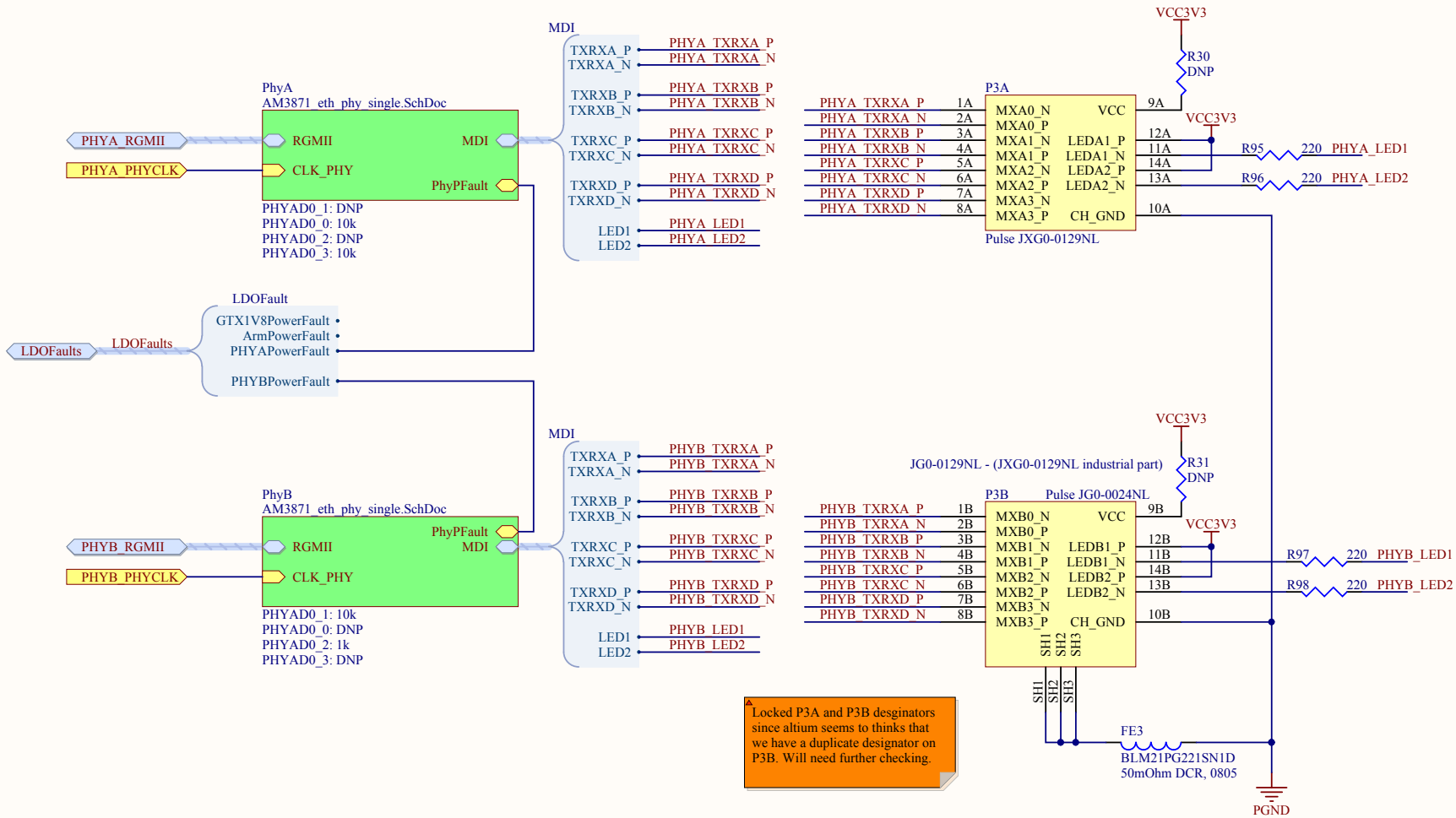


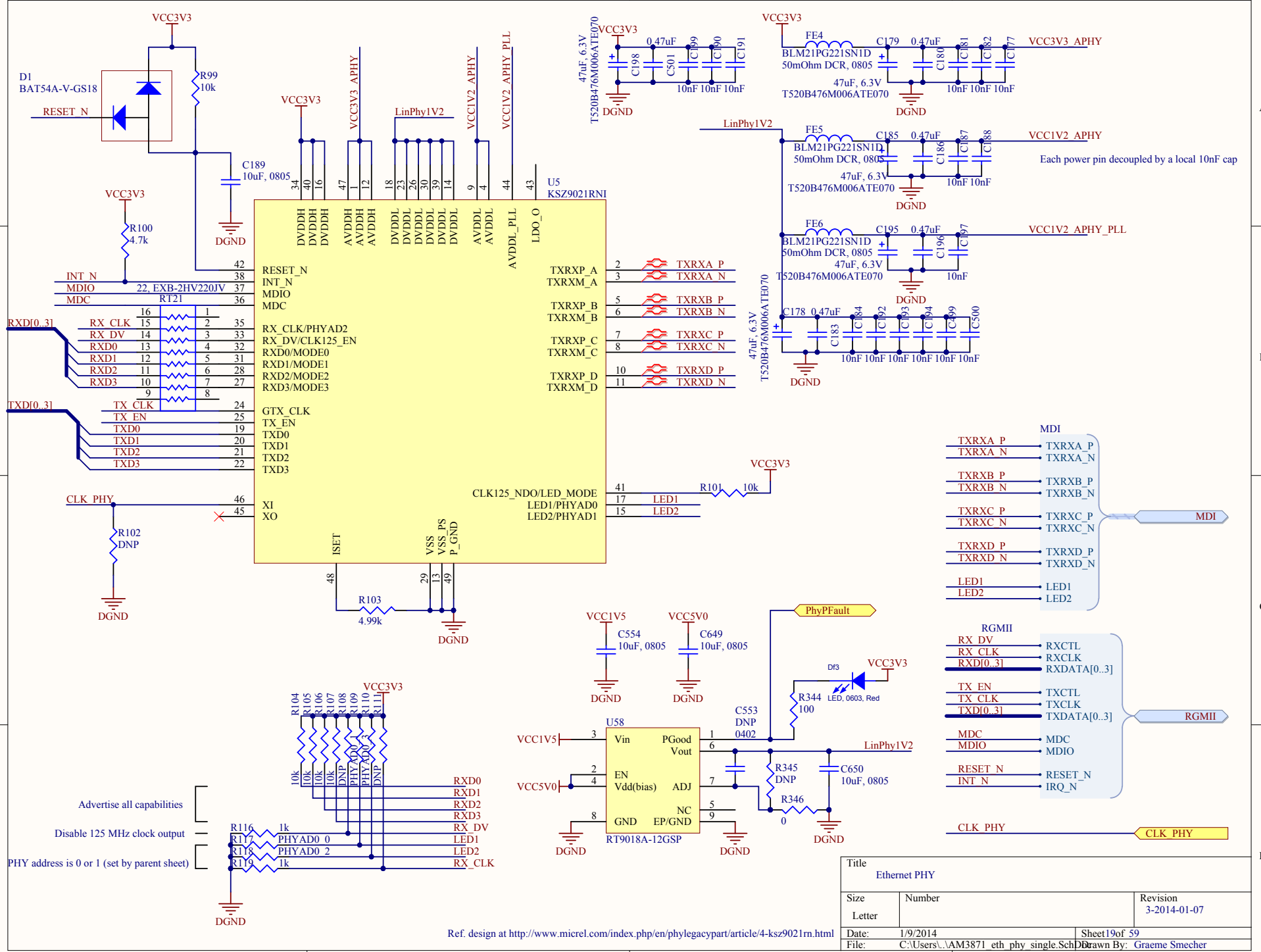
Layout considerations at http://www.ti.com/lit/ds/symlink/tps51200.pdf#page=17

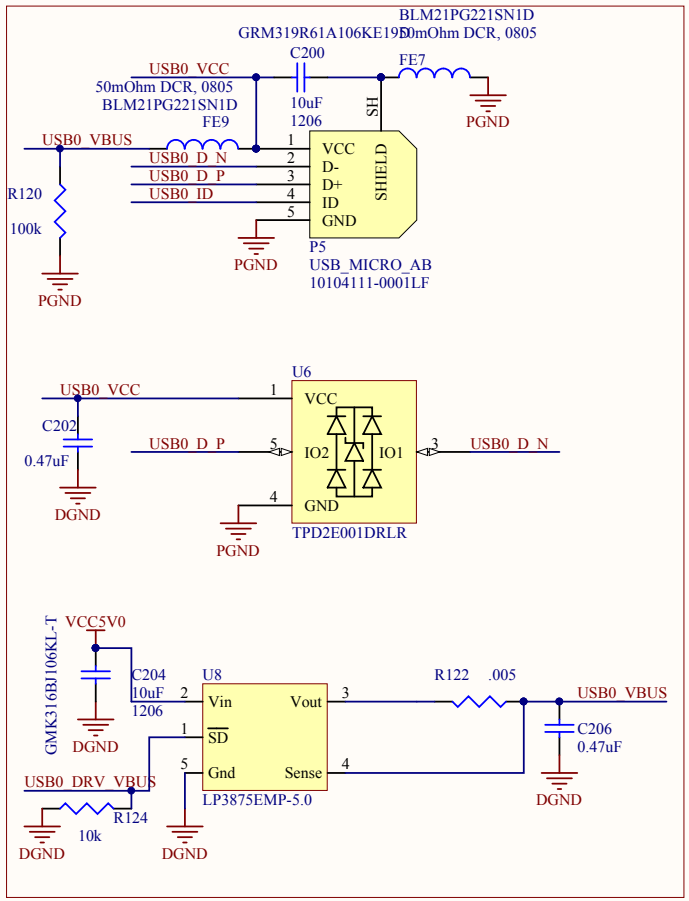
Title: AM3871 DDR3 Termination Regulator		
Size: Letter	Number:	Revision: 3-2014-01-07
Date: 1/9/2014	Sheet 16 of 59	
File: C:\Users\..\AM3871_ddr3_regulator.SchDoc	Drawn By: Graeme Smecher	

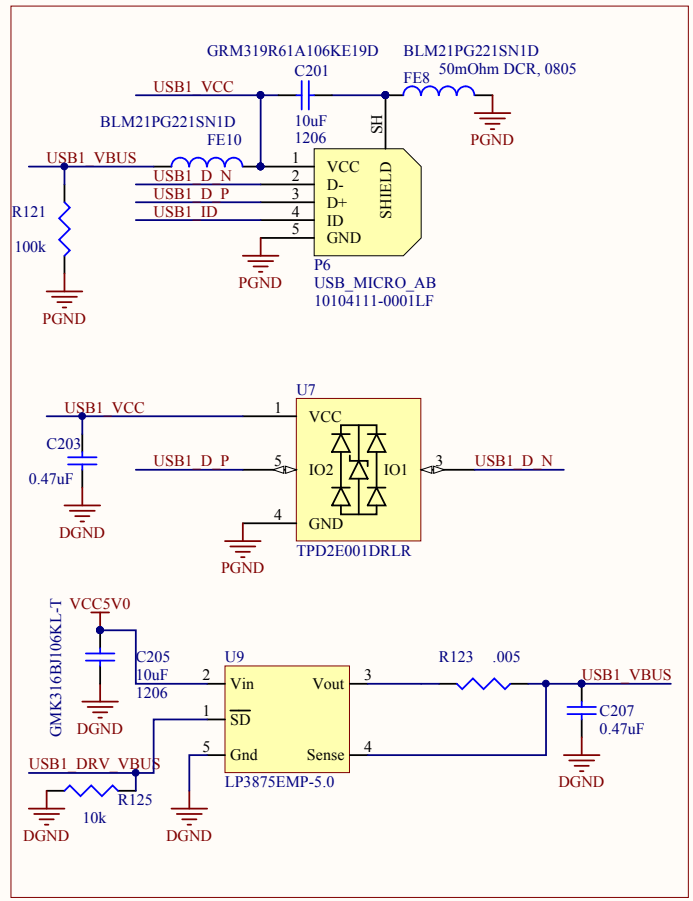


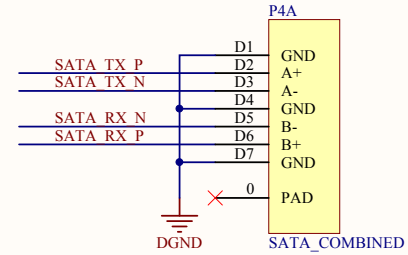
Title: Ethernet PHY		
Size: A	Number:	Revision: 3-2014-01-07
Date: 1/9/2014	Sheet 17 of 59	
File: C:\Users\..\AM3871_eth_cpu.SchDoc	Drawn By: Graeme Smecher	

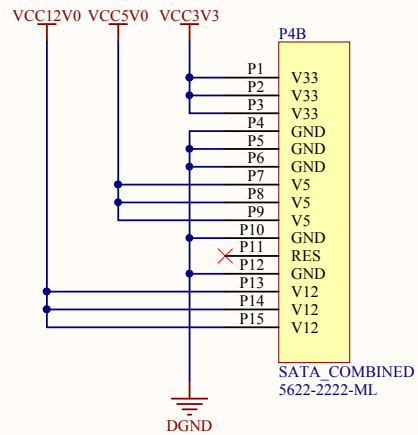

PhyA
AM3871_eth_phy_single.SchDoc
PHYA_RGMII
PHYA_PHYCLK
RGMII
CLK_PHY
MDI
PhyPFault
PHYAD0_1: DNP
PHYAD0_0: 10k
PHYAD0_2: DNP
PHYAD0_3: 10k
MDI
TXRXA_P
TXRXA_N
TXRXB_P
TXRXB_N
TXRXC_P
TXRXC_N
TXRXD_P
TXRXD_N
LED1
LED2
PHYA_TXRXA_P
PHYA_TXRXA_N
PHYA_TXRXB_P
PHYA_TXRXB_N
PHYA_TXRXC_P
PHYA_TXRXC_N
PHYA_TXRXD_P
PHYA_TXRXD_N
PHYA_LED1
PHYA_LED2
LDOFault
LDOFaults
GTX1V8PowerFault
ArmPowerFault
PHYAPowerFault
PHYBPowerFault
PhyB
AM3871_eth_phy_single.SchDoc
PHYB_RGMII
PHYB_PHYCLK
PHYAD0_1: 10k
PHYAD0_0: DNP
PHYAD0_2: 1k
PHYAD0_3: DNP
PHYB_TXRXA_P
PHYB_TXRXA_N
PHYB_TXRXB_P
PHYB_TXRXB_N
PHYB_TXRXC_P
PHYB_TXRXC_N
PHYB_TXRXD_P
PHYB_TXRXD_N
PHYB_LED1
PHYB_LED2
P3A
MXA0_N
MXA0_P
MXA1_N
MXA1_P
MXA2_N
MXA2_P
MXA3_N
MXA3_P
VCC
LEDA1_P
LEDA1_N
LEDA2_P
LEDA2_N
CH_GND
Pulse JXG0-129NL
VCC3V3
R30
DNP
R95
220_PHYA_LED1
R96
220_PHYA_LED2
JG0-0129NL - (JXG0-0129NL industrial part)
P3B
Pulse JG0-0024NL
R31
DNP
R97
220_PHYB_LED1
R98
220_PHYB_LED2
SH1
SH2
SH3
FE3
BLM21PG221SN1D
50mOhm DCR, 0805
PGND
Locked P3A and P3B desginators since altium seems to thinks that we have a duplicate designator on P3B. Will need further checking.
Title
Ethernet PHY
Size
Letter
Number
Revision
3-2014-01-07
Date:
1/9/2014
Sheet18of 59
File:
C:\Users\..\AM3871_eth_phy.SchDoc
Drawn By:
Graeme Smecher

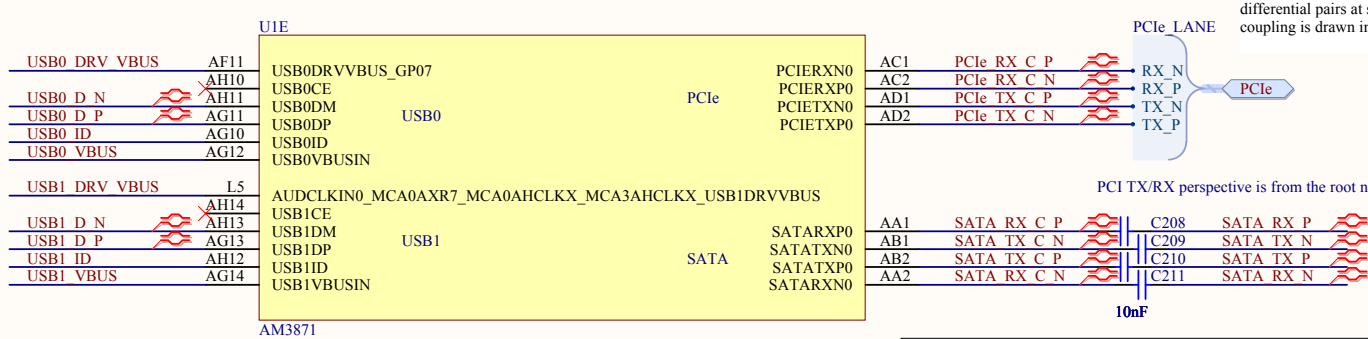

VCC3V3
D1
BAT54A-V-GS18
R99
10k
RESET_N
C189
10uF, 0805
DGND
R100
4.7k
INT_N
MDIO
MDC
22, EXB-2HV220JV
RT21
RXD[0..3]
RX_CLK
RX_DV
RXD0
RXD1
RXD2
RXD3
TXD[0..3]
TX_CLK
TX_EN
TXD0
TXD1
TXD2
TXD3
CLK_PHY
R102
DNP
R103
4.99k
VCC3V3_APHY
LinPhy1V2
VCC1V2_APHY
VCC1V2_APHY_PLL
U5
KSZ9021RNI
DVDDH
DVDDL
AVDDH
AVDDL
AVDDL_PLL
LDO_O
RESET_N
INT_N
MDIO
MDC
RX_CLK/PHYAD2
RX_DV/CLK125_EN
RXD0/MODE0
RXD1/MODE1
RXD2/MODE2
RXD3/MODE3
GTX_CLK
TX_EN
TXD0
TXD1
TXD2
TXD3
XI
XO
ISET
VSS
VSS_PS
P_GND
TXRXP_A
TXRXM_A
TXRXP_B
TXRXM_B
TXRXP_C
TXRXM_C
TXRXP_D
TXRXM_D
TXRXA_P
TXRXA_N
TXRXB_P
TXRXB_N
TXRXC_P
TXRXC_N
TXRXD_P
TXRXD_N
CLK125_NDO/LED_MODE
LED1/PHYAD0
LED2/PHYAD1
LED1
LED2
R101
10k
T520B476M006ATE070
C198
C501
C199
C190
C191
0.47uF
10nF
FE4
BLM21PG221SN1D
50mOhm DCR, 0805
C179
C180
C181
C182
C177
VCC3V3_APHY
47uF, 6.3V
FE5
C185
C186
C187
C188
VCC1V2_APHY
Each power pin decoupled by a local 10nF cap
FE6
C195
C196
C197
VCC1V2_APHY_PLL
C178
C183
C184
C192
C193
C194
C499
C500
MDI
RGMII
RX_DV
RX_CLK
RXD[0..3]
TX_EN
TX_CLK
TXD[0..3]
MDC
MDIO
RESET_N
INT_N
RXCTL
RXCLK
RXDATA[0..3]
TXCTL
TXCLK
TXDATA[0..3]
IRQ_N
PhyPFault
VCC1V5
VCC5V0
C554
10uF, 0805
C649
10uF, 0805
U58
RT9018A-12GSP
Vin
PGood
Vout
EN
Vdd(bias)
ADJ
NC
GND
EP/GND
C553
DNP
0402
C650
10uF, 0805
R344
100
Df3
LED, 0603, Red
R345
DNP
R346
0
R104
R105
R106
R107
R108
R109
R110
R111
PHYAD0_0
PHYAD0_1
PHYAD0_2
R116
R117
R118
R119
1k
Advertise all capabilities
Disable 125 MHz clock output
PHY address is 0 or 1 (set by parent sheet)
Ref. design at http://www.micrel.com/index.php/en/phylegacypart/article/4-ksz9021rn.html
Title
Ethernet PHY
Size
Letter
Number
Revision
3-2014-01-07
Date:
1/9/2014
Sheet19of 59
File:
C:\Users\..\AM3871_eth_phy_single.SchDoc
Drawn By:
Graeme Smecher











PCI convention is to capacitively couple differential pairs at sending side. This coupling is drawn in the GTX sheet.

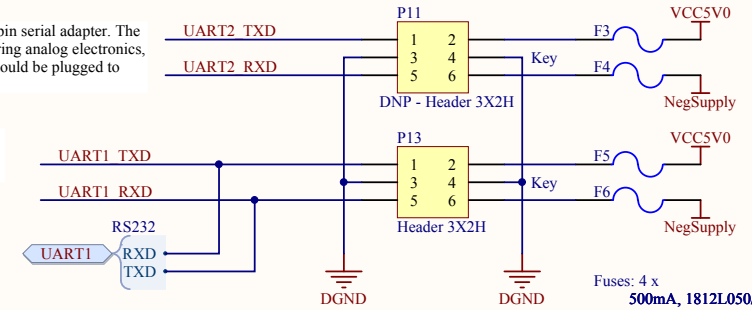
PCI TX/RX perspective is from the root node (the ARM core)

Title: AM3871 Fast Serial Interfaces		
Size: Letter	Number:	Revision: 3-2014-01-07
Date: 1/9/2014		Sheet 20 of 59
File: C:\Users\..\AM3871_fast_serial.SchDoc		Drawn By: Graeme Smecher

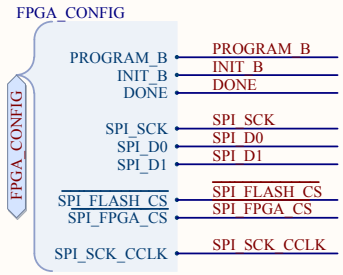
RS-232 Headers

These ports allow a standard, 3-pin serial adapter. The even pins are reserved for powering analog electronics, e.g. in an active dongle. Pin 4 should be plugged to prevent accidents.

UART1 also passes through to the FPGA. Note that TXD/RXD are reversed here.



FPGA_CONFIG

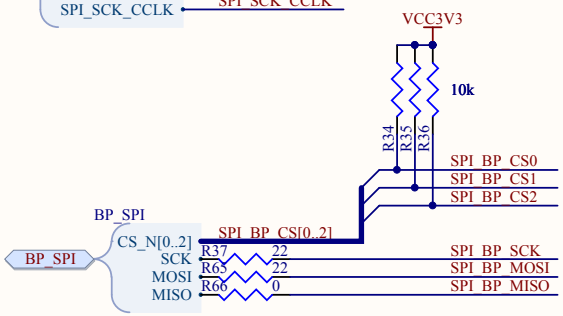


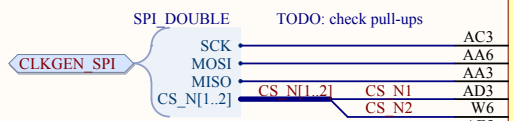
FPGA configuration status pins

Programming (SS + SPI flash) and data bus

Roles cannot be separated since the FPGA pins are combined

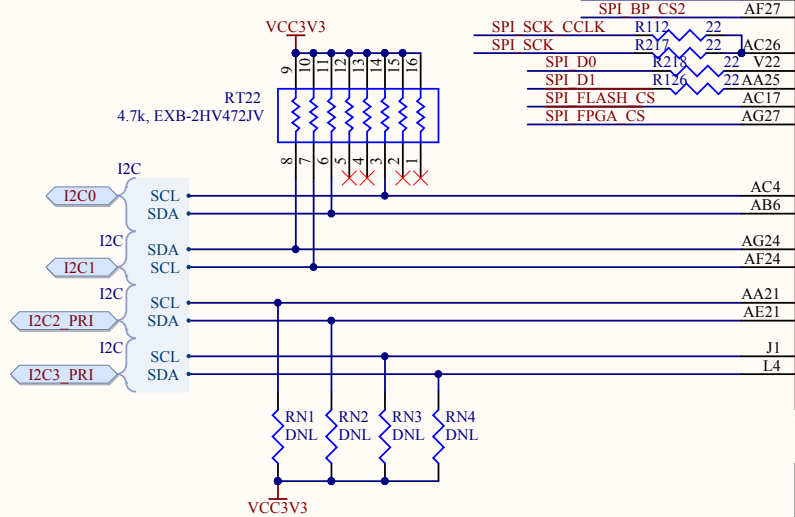
SPI chip-selects





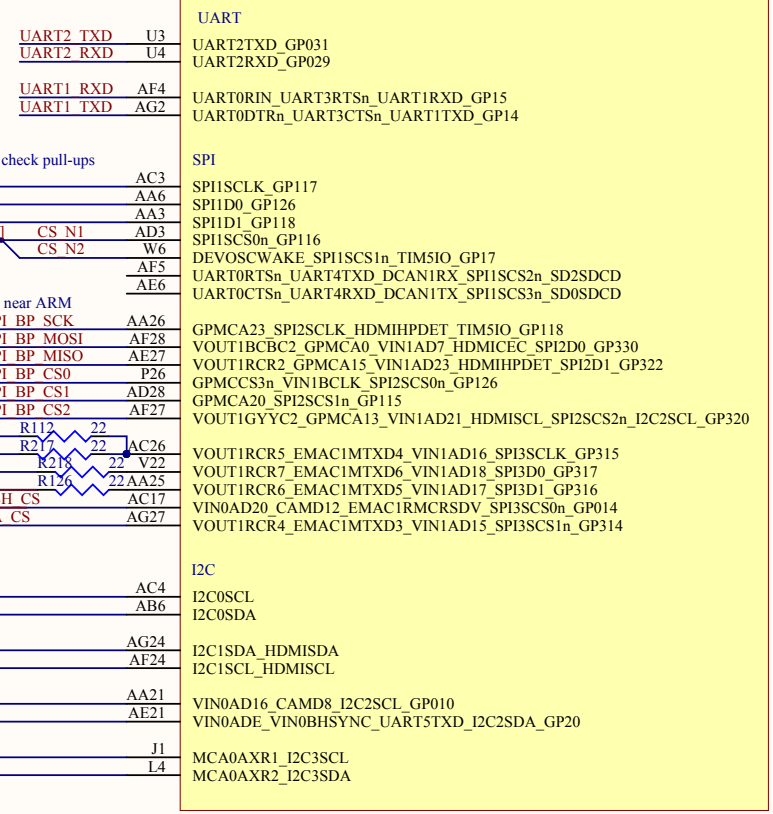
Source termination: Place R217, R218 near ARM

The FPGA needs two CLK references: SCK for user I/O, and CCLK for configuration. They are split here for signal integrity.



I2C2 and I2C3 have two pin-mux possibilities at the ARM: they can connect to neighbouring backplane slots (primary), or the I2C interconnect matrix (secondary).

U1D



AM3871

Title: AM3871 Slow Serial Interfaces		
Size: Letter	Number:	Revision: 3-2014-01-07
Date: 1/9/2014		Sheet21of 59
File: C:\Users\..\AM3871_slow_serial.SchDoc		Drawn By: Graeme Smecher

GPIOs are not fully populated, and redundant entries exist. Also note that the Ethernet sheet, for example, has some GPIOs populated; they can't be reused here.

LAYOUT DIRECTIONS: when redundant GPIO entries exist, you can swap one for another. For example: you can move a wire between two pins marked GP12.

FMC_AUX

FMC_AUX_A

EN_12V0
EN_3V3
EN_VADJ
TMS — FMCA_TMS
TCK — FMCA_TCK
TDO — FMCA_TDO
TDI — FMCA_TDI
TRST — FMCA_TRST
PG_M2C
PG_C2M
PRSNT_M2C_L
CLK_DIR

Unused pins: see I2C GPIO

VCC3V3

R27 4.7k
R29 4.7k

I2C
I2C3_SEC
SCL
SDA

I2C3's primary connection is to the backplane neighbour. Alternately, it connects to the I2C interconnect matrix.

FMC_AUX

FMC_AUX_B

EN_12V0
EN_3V3
EN_VADJ
TMS — FMCB_TMS
TCK — FMCB_TCK
TDO — FMCB_TDO
TDI — FMCB_TDI
TRST — FMCB_TRST
PG_M2C
PG_C2M
PRSNT_M2C_L
CLK_DIR

Unused pins: see I2C GPIO

JTAG

FPGA_JTAG

TMS — FPGA_TMS
TCK — FPGA_TCK
TDO — FPGA_TDO
TDI — FPGA_TDI
TRST

FPGA_CONFIG

FPGA_CONFIG

PROGRAM_B — PROGRAM_B
INIT_B — INIT_B
DONE — DONE
SPI_SCK
SPI_D0
SPI_D1
SPI_FLASH_CS
SPI_FPGA_CS
SPI_SCK_CCLK

(See ARM slow serial sheet)
(See ARM slow serial sheet)

FPGA Configuration
1 PROGRAM_B — Y6
2 INIT_B — N1
3 DONE — R7

FMCA
8 FMCA_TMS — R5
9 FMCA_TCK — H1
10 FMCA_TDO — U6
11 FMCA_TDI — AA5

GP014: SPI3 bit-banging
Y5
Y3
Y4
AB21
AF20
N2
AF21
V6
V5

FMCB
15 FMCA_TRST — H2
16 FMCB_TMS — AE18
17 FMCB_TCK — G6
18 FMCB_TDO — AC21
19 FMCB_TDI — AC16
20 FMCB_TRST — H4
AB17
G1
G2
AC15
AC22
F2
AD17
K7
AD18
H3
AC18
H6

FPGA JTAG
24 FPGA_TMS — R73 22 — J4
25 FPGA_TCK — R244 22 — AC19
26 FPGA_TDO — R245 22 — AA22
27 FPGA_TDI — R246 22 — J3
H5
AE23
AD23
L7
L6
AB23
T2

C615 DNL FPGA_TMS — DGND
C616 DNL FPGA_TCK — DGND
C617 DNL FPGA_TDO — DGND
C618 DNL FPGA_TDI — DGND
CC0402KRX7R9BB221

JTAG capacitors used to be 220pF

U1I

SD0CLK_GP01
SD0CMD_SD1CMD_GP02
SD0DAT0_SD1DAT4_GP03
SD0DAT1SDIRQn_SD1DAT5_GP04
SD0DAT2SDRWn_SD1DAT6_GP05
SD0DAT3_SD1DAT7_GP06

AUDCLKIN1_MCA0AXR8_MCA1AHCLKX_MCA4AHCLKX_EDMAEVT3_TIM2IO_GP08
AUDCLKIN2_MCA0AXR9_MCA2AHCLKX_MCA5AHCLKX_EDMAEVT2_TIM3IO_GP09
MCA2ACLKX_GP010
MCA2AFSX_GP011
VIN0AD17_CAMD9_EMAC1RMRXER_GP011
VIN0AD18_CAMD10_EMAC1RMRXD1_I2C3SCL_GP012
MCA2AXR0_SD0DAT6_UART5RXD_GP012
VIN0AD19_CAMD11_EMAC1RMRXD0_I2C3SDA_GP013
MCA2AXR1_SD0DAT7_UART5TXD_GP013
MCA2AXR2_MCA1AXR6_TIM2IO_GP014

MCA2AXR3_MCA1AXR7_TIM3IO_GP015
VIN0AD21_CAMD13_EMAC1RMTXD0_SPI3SCLK_GP015
MCA3ACLKX_GP016
VIN0AD22_CAMD14_EMAC1RMTXD1_SPI3D1_GP016
VIN0AD23_CAMD15_EMAC1RMTXEN_SPI3D0_GP017
MCA3AFSX_GP017
VIN0ADE_CAMD7_GP018
MCA3AXR0_TIM4IO_GP018
MCA3AXR1_TIM5IO_GP019
VIN0BDE_CAMD6_GP019
VIN0AFLD_CAMD5_GP020
MCA3AXR2_MCA1AXR8_GP020
VIN0BFLD_CAMD4_GP021
MCA4ACLKX_GP021
VOUT1GYYC1_CAMD3_GPMCA5_UART4RXD_GP022
MCA4AFSX_GP022
VOUT1GYYC0_CAMD2_GPMCA6_UART4TXD_GP023
MCA4AXR0_GP023
MCA4AXR1_TIM6IO_GP024
VOUT1RCR1_CAMD1_GPMCA7_UART4CTSn_GP024
VOUT1RCR0_CAMD0_GPMCA8_UART4RTSn_GP025
MCA5ACLKX_GP025
MCA5AFSX_GP026
VOUT1BCBC1_CAMHS_GPMCA9_UART2RXD_GP026
VOUT1BCBC0_CAMVS_GPMCA10_UART2TXD_GP027
MCA5AXR0_MCA4AXR2_GP027
MCA5AXR1_MCA4AXR3_TIM7IO_GP028
VOUT1FLD_CAMFLD_CAMWEn_GPMCA11_UART2CTSn_GP028
TCLKIN_GP030

GPIO bank 0

AM3871

Title: AM3871 GPIO		
Size: Letter	Number	Revision: 3-2014-01-07
Date: 1/9/2014		Sheet22of 59
File: C:\Users\..\AM3871_gpio_0.SchDoc		Drawn By: Graeme Smecher

RGMII

RXCTL
RXCLK
RXDATA[0..3]
(See ARM Ethernet sheet)

TXCTL
TXCLK
TXDATA[0..3]
(See ARM Ethernet sheet)

MDC
MDIO
(See ARM Ethernet sheet)

PHYA_RGMII

RESET_N — PHYA_RESET_N
IRQ_N — PHYA_IRQ_N

RGMII

RXCTL
RXCLK
RXDATA[0..3]
(See ARM Ethernet sheet)

TXCTL
TXCLK
TXDATA[0..3]
(See ARM Ethernet sheet)

MDC
MDIO
(See ARM Ethernet sheet)

PHYB_RGMII

RESET_N — PHYB_RESET_N
IRQ_N — PHYB_IRQ_N

Share PHY IRQs by default

PHYA_IRQ_N — R130 0 — PHYB_IRQ_N

GPIOs are not fully populated, and redundant entries exist. Also note that the Ethernet sheet, for example, has some GPIOs populated; they can't be reused here.

LAYOUT DIRECTIONS: when redundant GPIO entries exist, you can swap one for another. For example: you can move a wire between two pins marked GP12.

U1J

PHY IRQ/Reset

PHYA_IRQ_N — AH6
PHYA_RESET_N — AG6
PHYB_IRQ_N — W1
PHYB_RESET_N — W2

ARM_IRQ — P25
GPIO_IRQ — AE17
GPIO_RST — V1

V2
J27
AF9
AB11
AE28
M24
L26
AC27

GP115 is reserved for manual SPI_BP_CS1

M23
AC28
AB27

L25
N23
P22
R24
T28
K28
M25

R26
M26
U27
V28
W28

DCAN0TX_UART2TXD_I2C3SDA_GP10
DCAN0RX_UART2RXD_I2C3SCL_GP11

GP17
GP18
GPMCCS4n_SD2CMD_GP18
VIN0BCLK_CLKOUT0_GP19
GP19
GP110
EMACRMREFCLK_TIM2IO_GP110
VIN0AD0_GP111
VIN0AD1_GP112
GPMCA18_TIM2IO_GP113
SD2DAT1SDIRQn_GPMCA3_GP113
SD2DAT0_GPMCA4_GP114
GPMCA19_TIM3IO_GP114

SD2SCLK_GP115
GPMCA21_SPI2D0_GP116
GPMCA22_SPI2D1_HDMICEC_TIM4IO_GP117

GPIO bank 1

SD2DAT7_GPMCA24_GPMCA20_UART2RXD_GP119
SD2DAT6_GPMCA25_GPMCA21_UART2TXD_GP120
SD2DAT5_GPMCA26_GPMCA22_TIM6IO_GP121
SD2DAT4_GPMCA27_GPMCA23_GPMCCS7n_EDMAEVT0_TIM7IO_GP122
GPMCCS0n_GP123
GPMCCS1n_GPMCA25_GP124
GPMCCS2n_GPMCA24_GP125

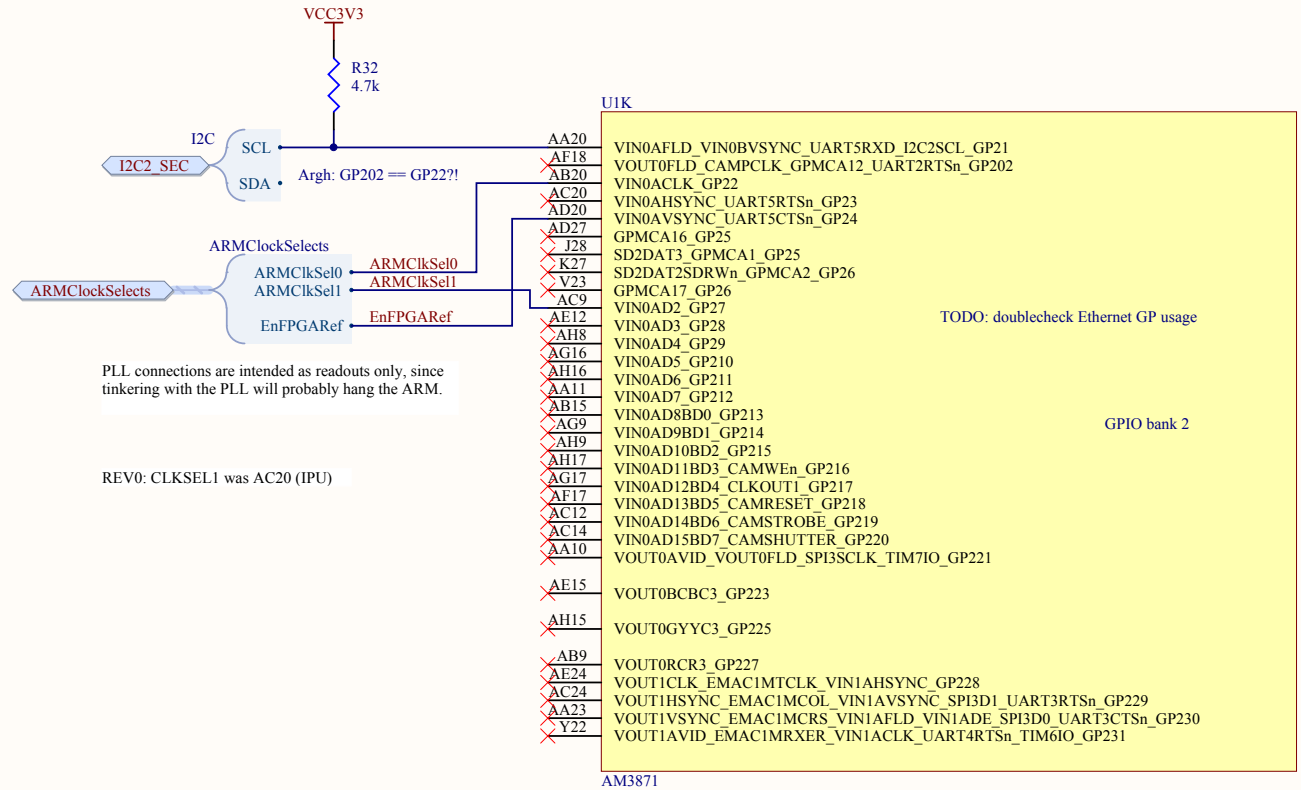
GPMCCLK_GPMCCS5n_GPMCWAIT1_CLKOUT1_EDMAEVT3_TIM4IO_GP127
GPMCADVnALE_GPMCCS6n_TIM5IO_GP128
GPMCBE0nCLE_GPMCA25_EDMAEVT2_TIM6IO_GP129
GPMCBE1n_GPMCA24_EDMAEVT1_TIM7IO_GP130
GPMCWAIT0_GPMCA26_EDMAEVT0_GP131

AM3871

Title: AM3871 GPIO		
Size: Letter	Number	Revision: 3-2014-01-07
Date: 1/9/2014		Sheet23of 59
File: C:\Users\..\AM3871_gpio_1.SchDoc		Drawn By: Graeme Smecher

GPIOs are not fully populated, and redundant entries exist. Also note that the Ethernet sheet, for example, has some GPIOs populated; they can't be reused here.

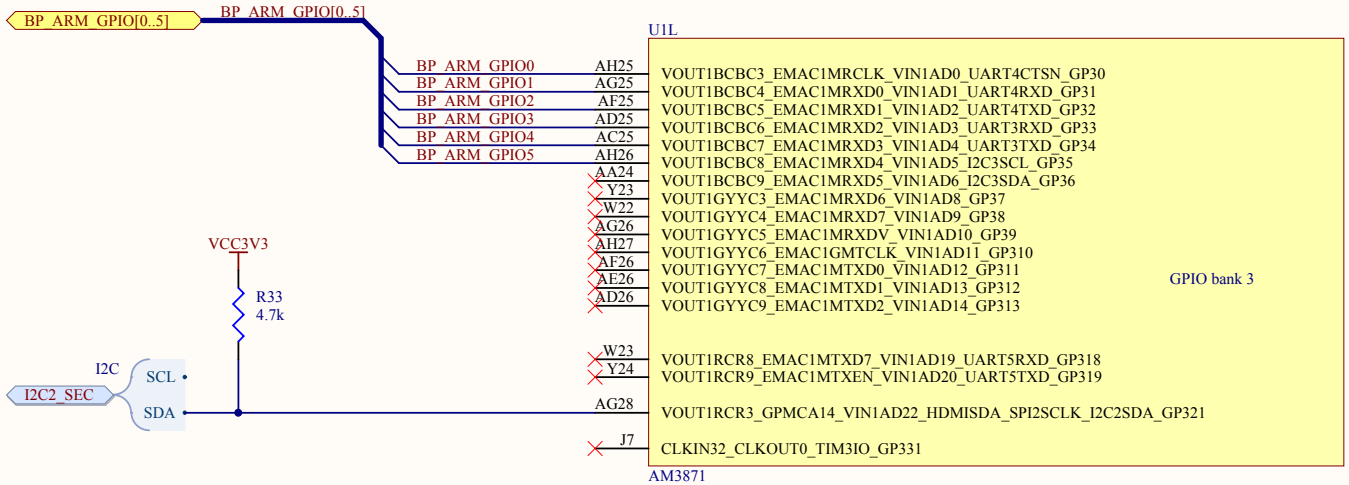
LAYOUT DIRECTIONS: when redundant GPIO entries exist, you can swap one for another. For example: you can move a wire between two pins marked GP12.



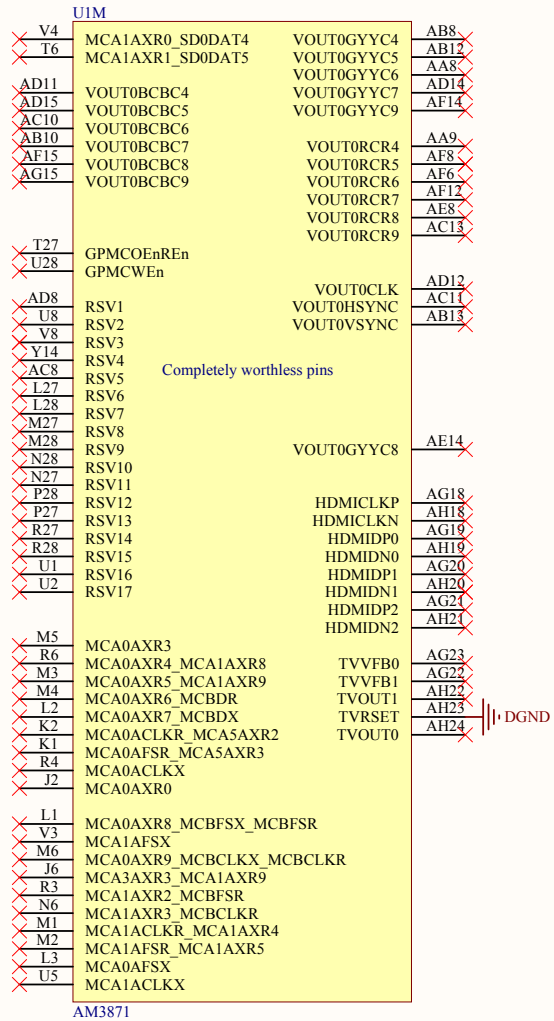
Title: AM3871 GPIO		
Size: Letter	Number:	Revision: 3-2014-01-07
Date: 1/9/2014		Sheet 24 of 59
File: C:\Users\..\AM3871_gpio_2.SchDoc		Drawn By: Graeme Smecher

GPIOs are not fully populated, and redundant entries exist. Also note that the Ethernet sheet, for example, has some GPIOs populated; they can't be reused here.

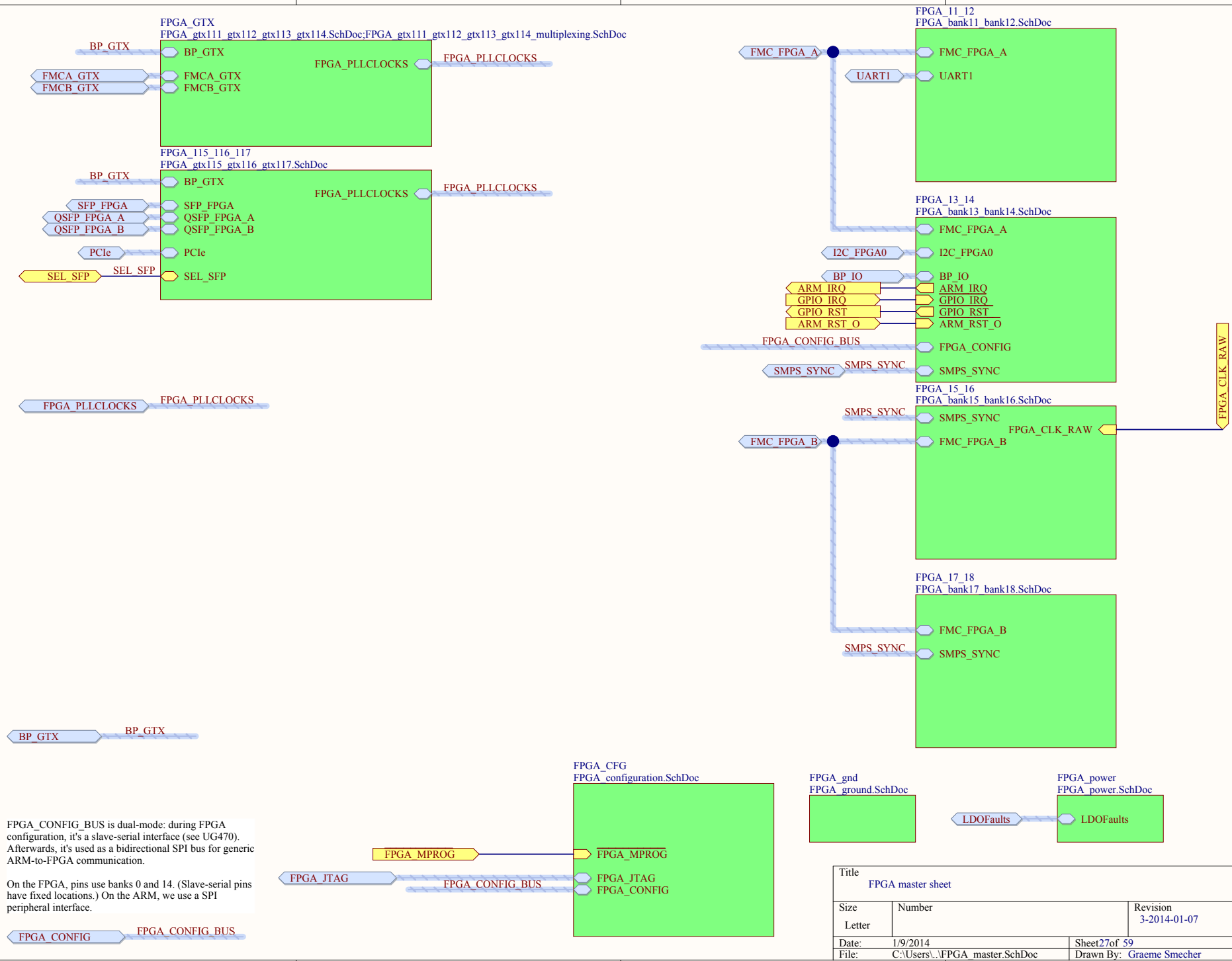
LAYOUT DIRECTIONS: when redundant GPIO entries exist, you can swap one for another. For example: you can move a wire between two pins marked GP12.



Title: AM3871 GPIO		
Size: Letter	Number:	Revision: 3-2014-01-07
Date: 1/9/2014		Sheet25of 59
File: C:\Users\..\AM3871_gpio_3.SchDoc		Drawn By: Graeme Smecher



Title: AM3871 Useless Pins		
Size: Letter	Number:	Revision: 3-2014-01-07
Date: 1/9/2014		Sheet 26 of 59
File: C:\Users\..\AM3871_unused.SchDoc		Drawn By: Graeme Smecher


FPGA_GTX
FPGA_gtx111_gtx112_gtx113_gtx114.SchDoc;FPGA_gtx111_gtx112_gtx113_gtx114_multiplexing.SchDoc
BP_GTX
FMCA_GTX
FMCB_GTX
FPGA_PLLCLOCKS
FPGA_115_116_117
FPGA_gtx115_gtx116_gtx117.SchDoc
SFP_FPGA
QSFP_FPGA_A
QSFP_FPGA_B
PCIe
SEL_SFP
FPGA_11_12
FPGA_bank11_bank12.SchDoc
FMC_FPGA_A
UART1
FPGA_13_14
FPGA_bank13_bank14.SchDoc
I2C_FPGA0
BP_IO
ARM_IRQ
GPIO_IRQ
GPIO_RST
ARM_RST_O
FPGA_CONFIG_BUS
FPGA_CONFIG
SMPS_SYNC
FPGA_15_16
FPGA_bank15_bank16.SchDoc
FPGA_CLK_RAW
FMC_FPGA_B
FPGA_17_18
FPGA_bank17_bank18.SchDoc
FPGA_CFG
FPGA_configuration.SchDoc
FPGA_MPROG
FPGA_JTAG
FPGA_gnd
FPGA_ground.SchDoc
FPGA_power
FPGA_power.SchDoc
LDOFaults
FPGA_CONFIG_BUS is dual-mode: during FPGA configuration, it's a slave-serial interface (see UG470). Afterwards, it's used as a bidirectional SPI bus for generic ARM-to-FPGA communication.
On the FPGA, pins use banks 0 and 14. (Slave-serial pins have fixed locations.) On the ARM, we use a SPI peripheral interface.
Title
FPGA master sheet
Size
Letter
Number
Revision
3-2014-01-07
Date: 1/9/2014
Sheet27of 59
File: C:\Users\..\FPGA_master.SchDoc
Drawn By: Graeme Smecher

The configuration bus is muddled amongst three regimes:
* FPGA is configured from the SPI flash
* FPGA is configured from the ARM, via slave serial (SS)
* ARM communicates with the FPGA, using standard SPI

The SPI flash is a fallback we don't intend to use. Note that it's effectively off-limits after configuration, unless we figure out multi-mastering (to keep the ARM and FPGA from competing for the bus).

U27B
xc7k420tffg901

JTAG

FPGA_CONFIG

FPGA Reset

Triggers an FPGA reset. In SS mode, the ARM needs to notice that PROGRAM_B has fallen and initiate its own programming procedure.

NC7WV07P6X
Open-drain / Non-inverting

STM6315RBW13F

SW5
EVQ-PUL02K
TempRange: -20, 70 SMALL RANGE!

LED Front Panel GYR
DS10A

Q1
FDY301NZ

C363
100uF
10V, 1206
C3216X5R1A107M160AC

Configuration Mode Select

SW9
SW-SOIC8, GDH04STR04

M[2:0]=000: Master serial
M[2:0]=001: Master SPI
M[2:0]=010: Master BPI
M[2:0]=011:
M[2:0]=100: Master SelectMAP
M[2:0]=101: JTAG
M[2:0]=110: Slave SelectMAP
M[2:0]=111: Slave Serial

JTAG

RT19
10k, EXB-2HV103JV

For TRST_N, to match FMC JTAG

Changed above resistors from 22Ohms to 0 Ohms

Header, 7-Pin

CC0402KRX7R9BB221

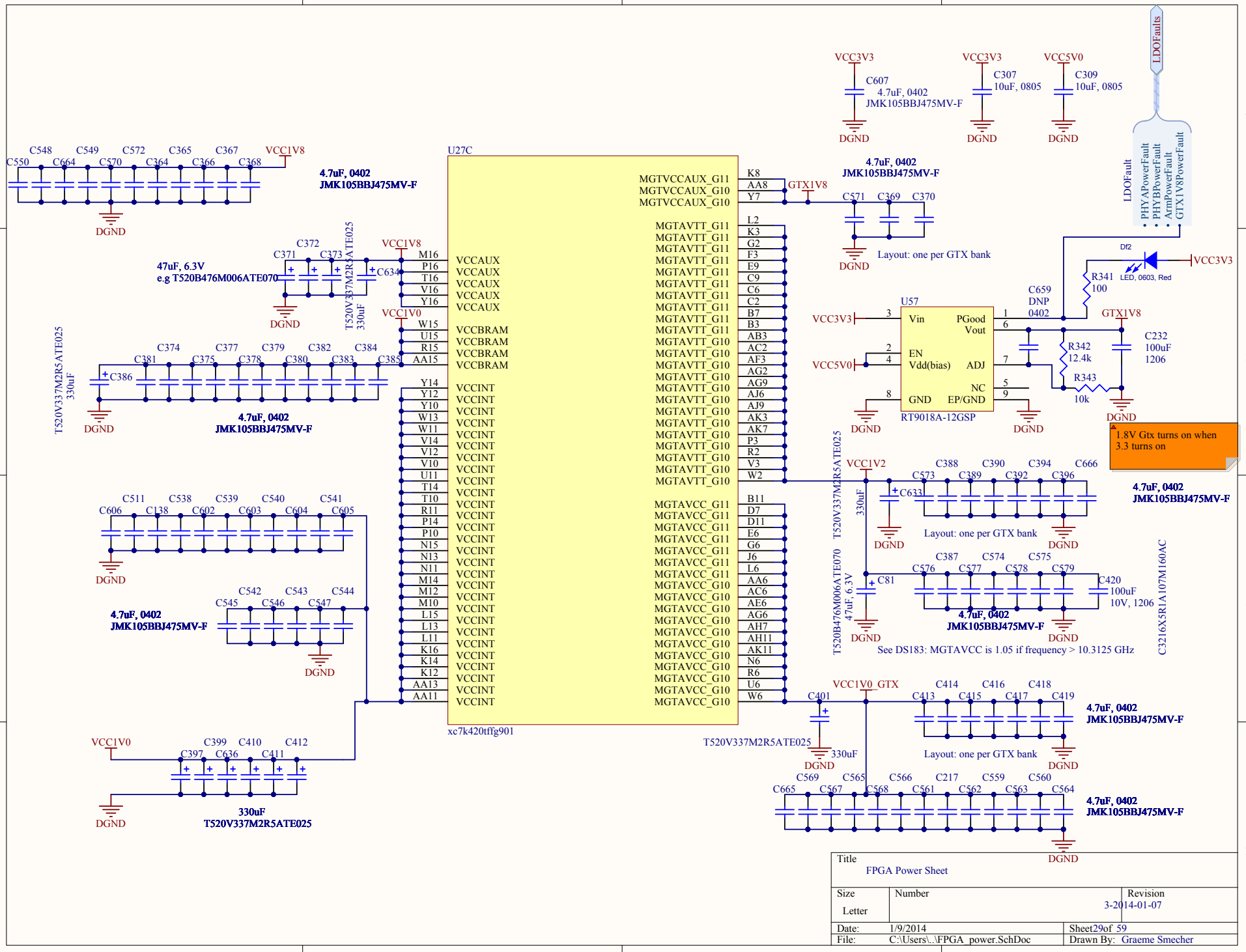
(Optional) SPI bitstream flash

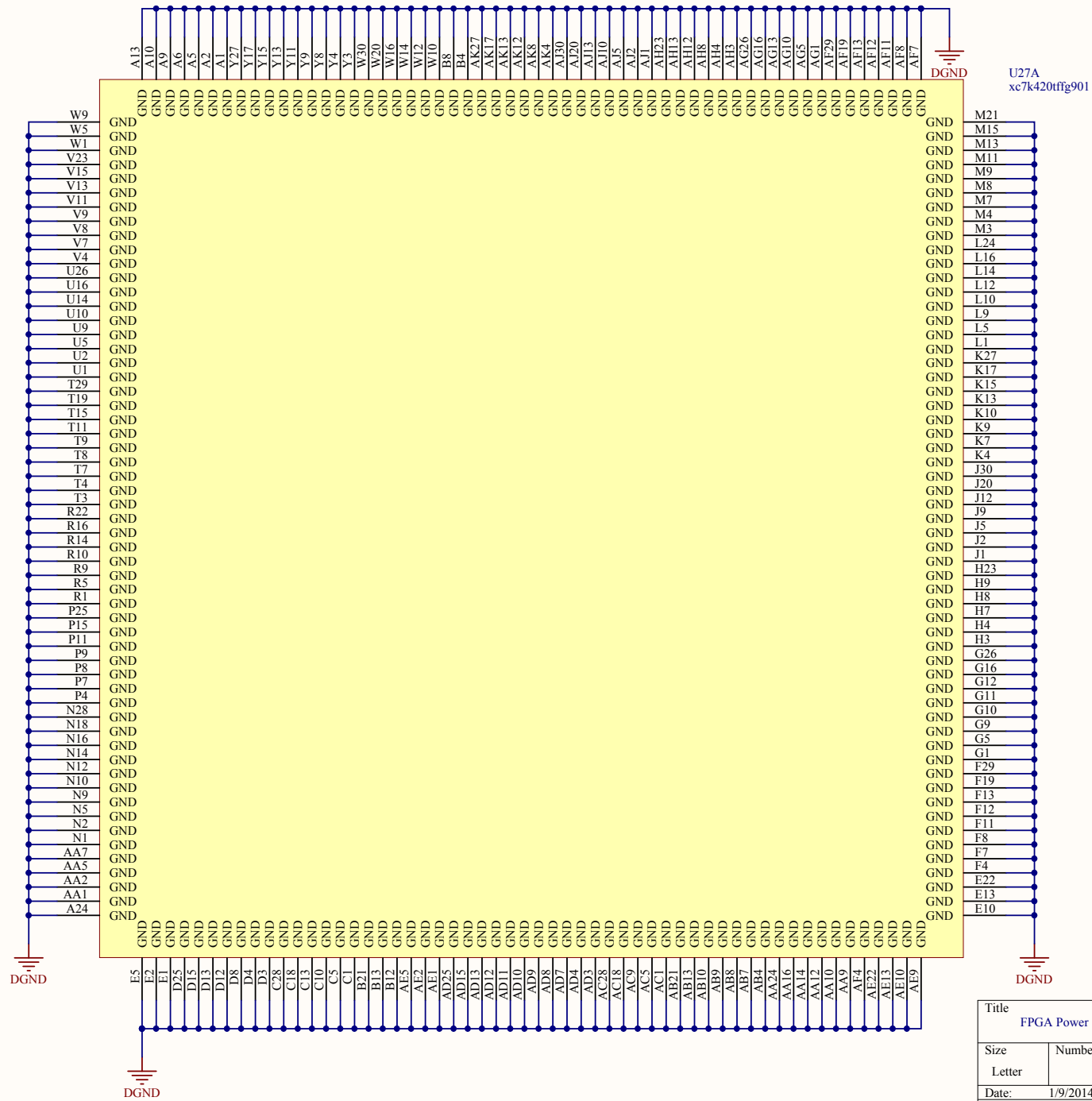
SPI flash: S25FL256SAGMFI011

10k pull-up baked into TXS0102

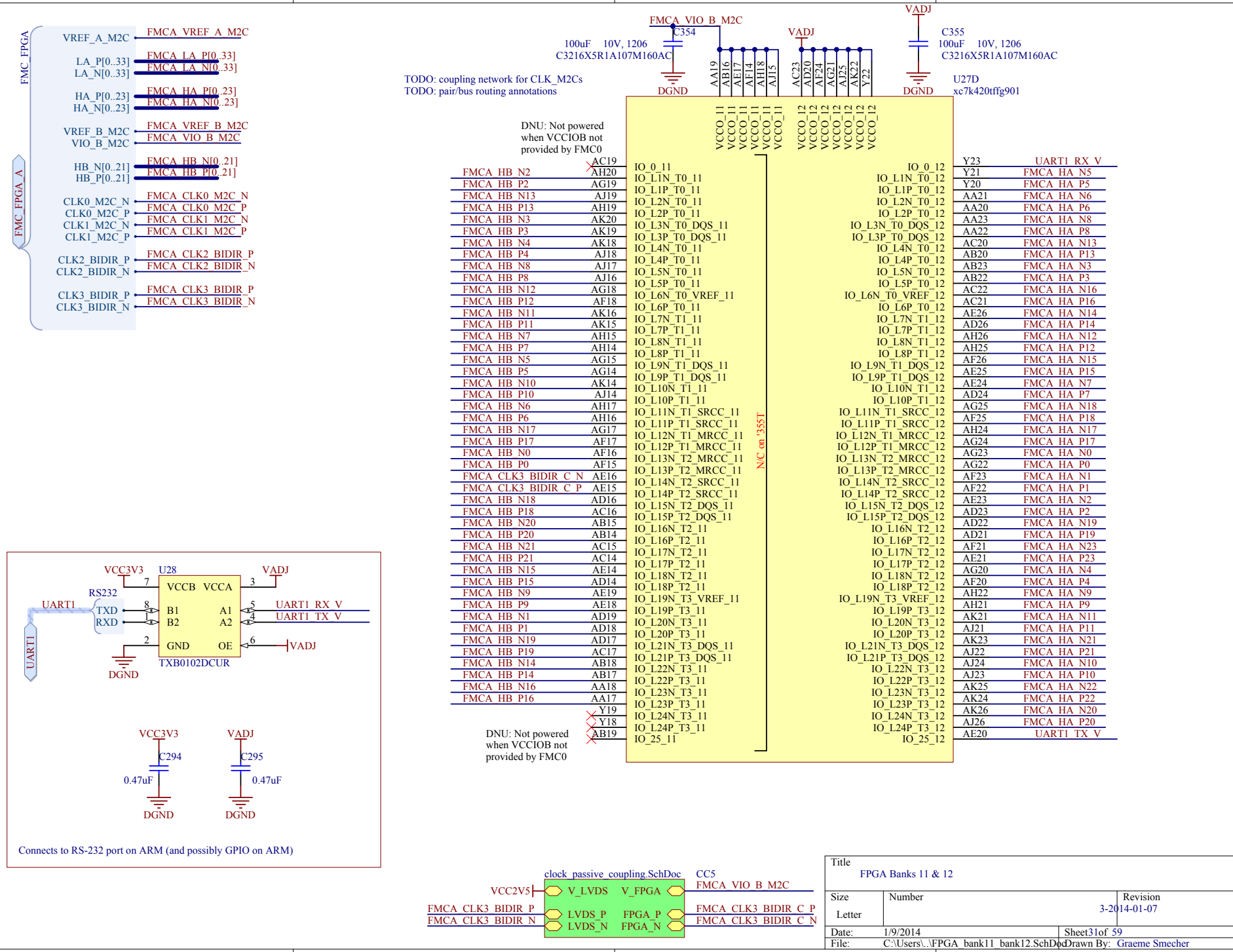
Test points and J1 allow flash programming, even while board isn't powered.

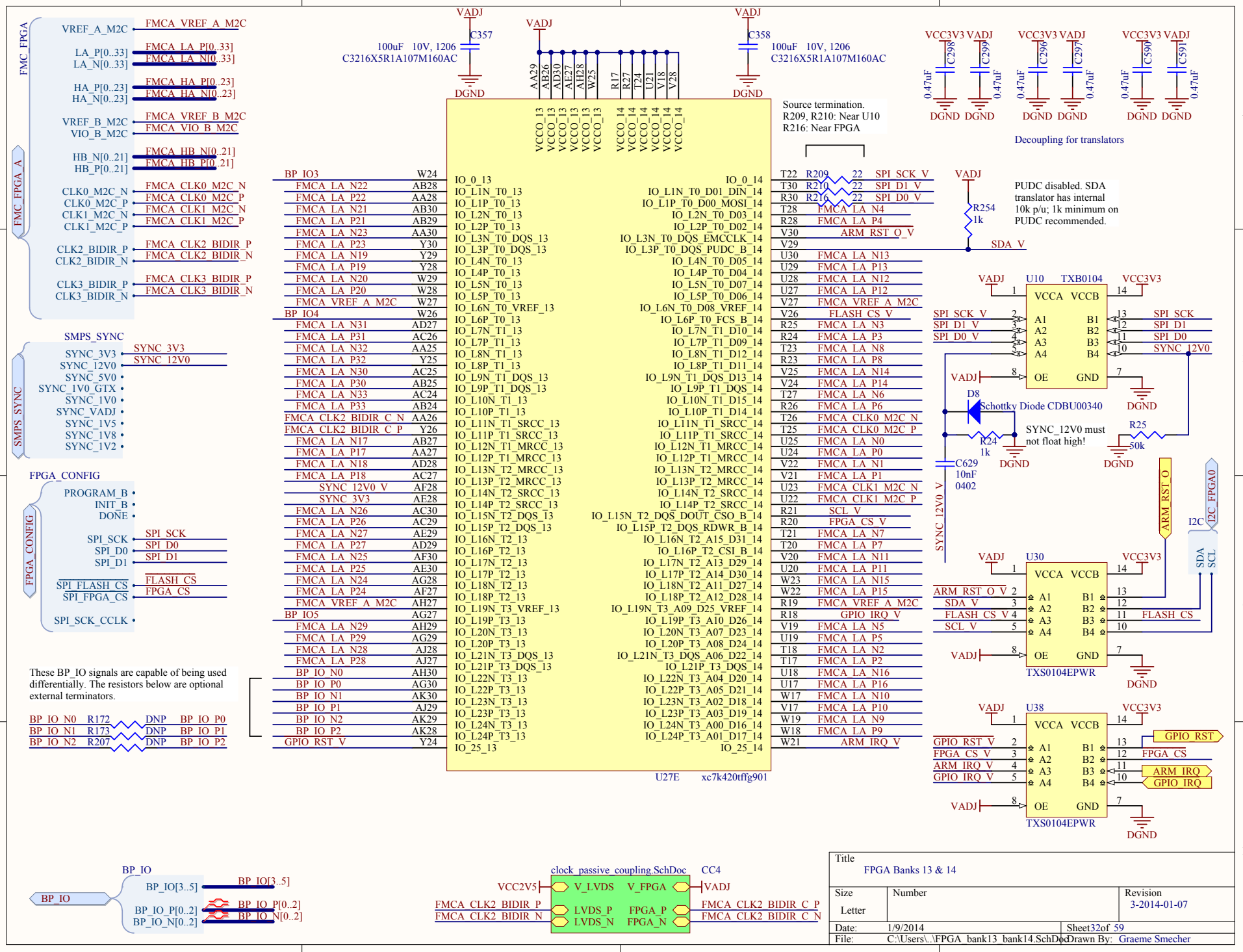
Title: FPGA configuration sheet		
Size: Letter	Number	Revision: 3-2014-01-07
Date: 1/9/2014		Sheet 28 of 59
File: C:\Users\..\FPGA_configuration.SchDoc		Drawn By: Graeme Smecher


VCC3V3
C607
4.7uF, 0402
JMK105BBJ475MV-F
DGND
VCC3V3
C307
10uF, 0805
DGND
VCC5V0
C309
10uF, 0805
DGND
LDOFaults
LDOFault
PHYAPowerFault
PHYBPowerFault
ArmPowerFault
GTX1V8PowerFault
C548 C549 C572 C365 C367
C550 C664 C570 C364 C366 C368
VCC1V8
4.7uF, 0402
JMK105BBJ475MV-F
DGND
U27C
MGTVCCAUX_G11
MGTVCCAUX_G10
MGTVCCAUX_G10
K8
AA8
Y7
GTX1V8
4.7uF, 0402
JMK105BBJ475MV-F
C571 C369 C370
DGND
Layout: one per GTX bank
C372
C371 C373
C634
47uF, 6.3V
e.g T520B476M006ATE070
T520V337M2R5ATE025
330uF
DGND
VCC1V8
VCC1V0
VCCAUX
VCCBRAM
VCCINT
M16 P16 T16 V16 Y16
W15 U15 R15 AA15
Y14 Y12 Y10 W13 W11 V14 V12 V10 U11 T14 T10 R11 P14 P10 N15 N13 N11 M14 M12 M10 L15 L13 L11 K16 K14 K12 AA13 AA11
MGTAVTT_G11
MGTAVTT_G10
L2 K3 G2 F3 E9 C9 C6 C2 B7 B3 AB3 AC2 AF3 AG2 AG9 AJ6 AJ9 AK3 AK7 P3 R2 V3 W2
MGTAVCC_G11
MGTAVCC_G10
B11 D7 D11 E6 G6 J6 L6 AA6 AC6 AE6 AG6 AH7 AH11 AK11 N6 R6 U6 W6
xc7k420tffg901
U57
Vin
EN
Vdd(bias)
GND
PGood
Vout
ADJ
NC
EP/GND
RT9018A-12GSP
VCC3V3
VCC5V0
C659
DNP
0402
R341
100
Df2
LED, 0603, Red
VCC3V3
GTX1V8
R342
12.4k
R343
10k
C232
100uF
1206
DGND
1.8V Gtx turns on when
3.3 turns on
C374 C377 C379 C382 C384
C381 C375 C378 C380 C383 C385
C386
T520V337M2R5ATE025
330uF
DGND
4.7uF, 0402
JMK105BBJ475MV-F
VCC1V2
C388 C390 C394 C666
C573 C389 C392 C396
C633
330uF
T520V337M2R5ATE025
DGND
Layout: one per GTX bank
4.7uF, 0402
JMK105BBJ475MV-F
C511 C538 C539 C540 C541
C606 C138 C602 C603 C604 C605
DGND
C542 C543 C544
C545 C546 C547
4.7uF, 0402
JMK105BBJ475MV-F
DGND
C387 C574 C575
C576 C577 C578 C579
C81
T520B476M006ATE070
47uF, 6.3V
C420
100uF
10V, 1206
C3216X5R1A107M160AC
4.7uF, 0402
JMK105BBJ475MV-F
DGND
DGND
See DS183: MGTAVCC is 1.05 if frequency > 10.3125 GHz
VCC1V0
C399 C410 C412
C397 C636 C411
DGND
330uF
T520V337M2R5ATE025
VCC1V0_GTX
C401
T520V337M2R5ATE025
330uF
DGND
C414 C416 C418
C413 C415 C417 C419
4.7uF, 0402
JMK105BBJ475MV-F
DGND
Layout: one per GTX bank
C569 C565 C566 C217 C559 C560
C665 C567 C568 C561 C562 C563 C564
4.7uF, 0402
JMK105BBJ475MV-F
DGND
Title
FPGA Power Sheet
Size
Letter
Number
Revision
3-2014-01-07
Date: 1/9/2014
Sheet29of 59
File: C:\Users\..\FPGA_power.SchDoc
Drawn By: Graeme Smecher

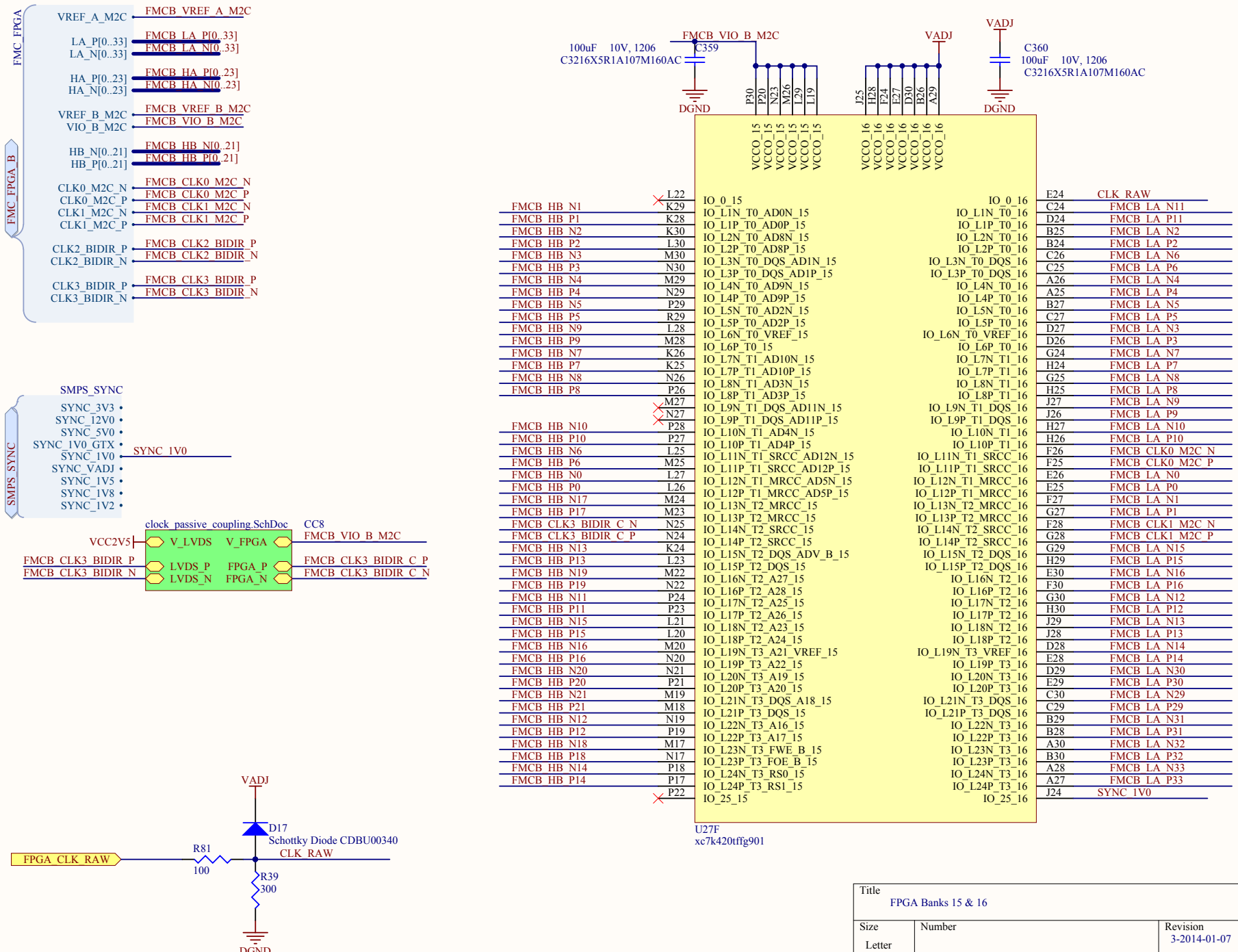


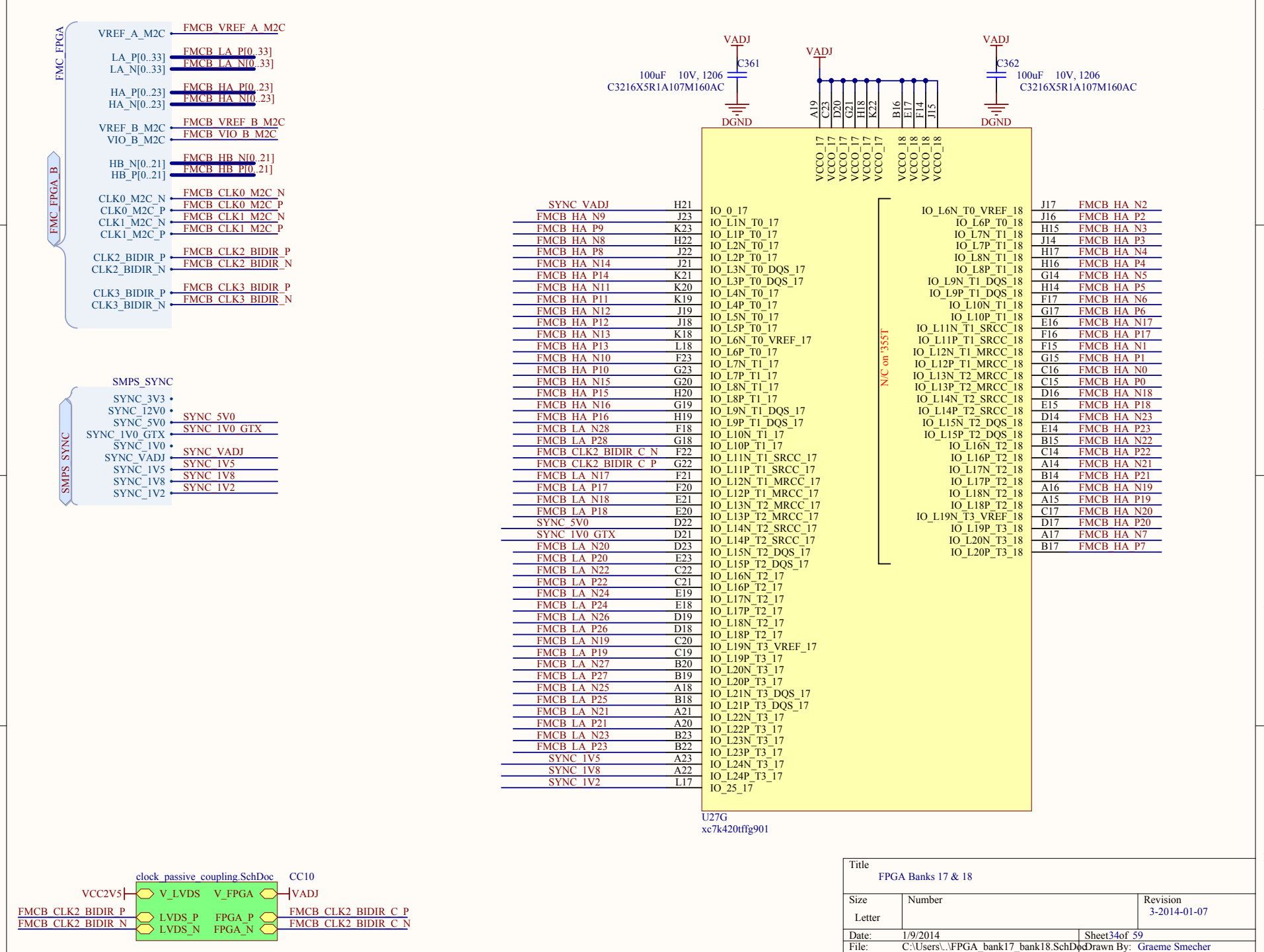
Title: FPGA Power Sheet		
Size: Letter	Number	Revision: 3-2014-01-07
Date: 1/9/2014		Sheet 30 of 59
File: C:\Users\..\FPGA_ground.SchDoc		Drawn By: Graeme Smecher

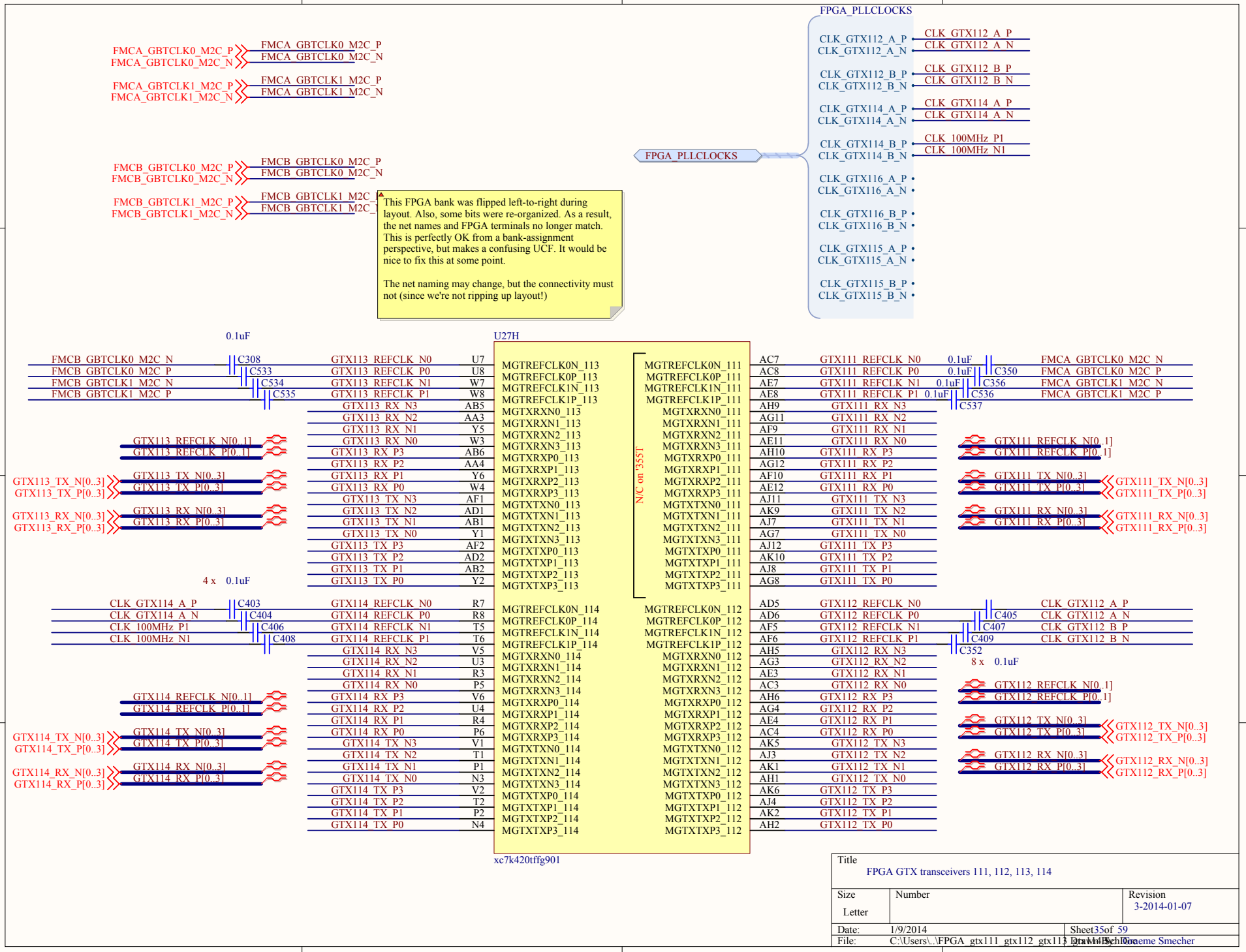

FMC_FPGA
VREF_A_M2C
FMCA_VREF_A_M2C
LA_P[0..33]
LA_N[0..33]
FMCA_LA_P[0..33]
FMCA_LA_N[0..33]
HA_P[0..23]
HA_N[0..23]
FMCA_HA_P[0..23]
FMCA_HA_N[0..23]
VREF_B_M2C
VIO_B_M2C
FMCA_VREF_B_M2C
FMCA_VIO_B_M2C
FMC_FPGA_A
HB_N[0..21]
HB_P[0..21]
FMCA_HB_N[0..21]
FMCA_HB_P[0..21]
CLK0_M2C_N
CLK0_M2C_P
CLK1_M2C_N
CLK1_M2C_P
FMCA_CLK0_M2C_N
FMCA_CLK0_M2C_P
FMCA_CLK1_M2C_N
FMCA_CLK1_M2C_P
CLK2_BIDIR_P
CLK2_BIDIR_N
FMCA_CLK2_BIDIR_P
FMCA_CLK2_BIDIR_N
CLK3_BIDIR_P
CLK3_BIDIR_N
FMCA_CLK3_BIDIR_P
FMCA_CLK3_BIDIR_N
FMCA_VIO_B_M2C
C354
100uF 10V, 1206
C3216X5R1A107M160AC
DGND
VADJ
C355
100uF 10V, 1206
C3216X5R1A107M160AC
DGND
U27D
xc7k420tffg901
TODO: coupling network for CLK_M2Cs
TODO: pair/bus routing annotations
DNU: Not powered when VCCIOB not provided by FMC0
DNU: Not powered when VCCIOB not provided by FMC0
N/C on '355T
UART1_RX_V
UART1_TX_V
RS232
UART1
TXD
RXD
VCC3V3
U28
VCCB
VCCA
GND
OE
B1
B2
A1
A2
TXB0102DCUR
DGND
VADJ
C294
C295
0.47uF
0.47uF
DGND
DGND
Connects to RS-232 port on ARM (and possibly GPIO on ARM)
clock_passive_coupling.SchDoc
CC5
VCC2V5
V_LVDS
V_FPGA
LVDS_P
LVDS_N
FPGA_P
FPGA_N
FMCA_CLK3_BIDIR_C_P
FMCA_CLK3_BIDIR_C_N
Title
FPGA Banks 11 & 12
Size
Letter
Number
Revision
3-2014-01-07
Date: 1/9/2014
Sheet 31 of 59
File: C:\Users\..\FPGA_bank11_bank12.SchDoc
Drawn By: Graeme Smecher



Title	FPGA Banks 13 & 14	
Size: Letter	Number	Revision: 3-2014-01-07
Date: 1/9/2014	Sheet 32 of 59	
File: C:\Users\..\FPGA_bank13_bank14.SchDoc	Drawn By: Graeme Smecher	

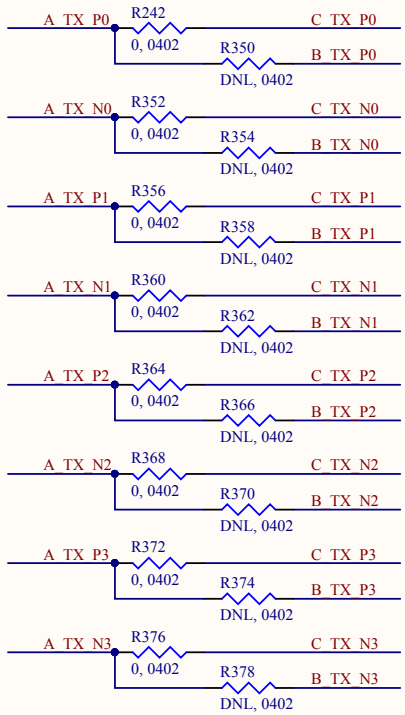

FMC_FPGA
VREF_A_M2C
FMCB_VREF_A_M2C
LA_P[0..33]
LA_N[0..33]
FMCB_LA_P[0..33]
FMCB_LA_N[0..33]
HA_P[0..23]
HA_N[0..23]
FMCB_HA_P[0..23]
FMCB_HA_N[0..23]
VREF_B_M2C
VIO_B_M2C
FMCB_VREF_B_M2C
FMCB_VIO_B_M2C
FMC_FPGA_B
HB_N[0..21]
HB_P[0..21]
FMCB_HB_N[0..21]
FMCB_HB_P[0..21]
CLK0_M2C_N
CLK0_M2C_P
CLK1_M2C_N
CLK1_M2C_P
FMCB_CLK0_M2C_N
FMCB_CLK0_M2C_P
FMCB_CLK1_M2C_N
FMCB_CLK1_M2C_P
CLK2_BIDIR_P
CLK2_BIDIR_N
FMCB_CLK2_BIDIR_P
FMCB_CLK2_BIDIR_N
CLK3_BIDIR_P
CLK3_BIDIR_N
FMCB_CLK3_BIDIR_P
FMCB_CLK3_BIDIR_N
SMPS_SYNC
SYNC_3V3
SYNC_12V0
SYNC_5V0
SYNC_1V0_GTX
SYNC_1V0
SYNC_VADJ
SYNC_1V5
SYNC_1V8
SYNC_1V2
SYNC_1V0
clock_passive_coupling.SchDoc
CC8
VCC2V5
V_LVDS
V_FPGA
FMCB_VIO_B_M2C
LVDS_P
LVDS_N
FPGA_P
FPGA_N
FMCB_CLK3_BIDIR_C_P
FMCB_CLK3_BIDIR_C_N
VADJ
D17
Schottky Diode CDBU00340
FPGA_CLK_RAW
R81
100
CLK_RAW
R39
300
DGND
FMCB_VIO_B_M2C
100uF 10V, 1206
C3216X5R1A107M160AC
C359
VADJ
C360
100uF 10V, 1206
C3216X5R1A107M160AC
P30 P20 N23 M26 L29 L19
J25 H28 F24 E27 D30 B26 A29
VCCO_15
VCCO_16
L22
K29
K28
K30
L30
M30
N30
M29
N29
P29
R29
L28
M28
K26
K25
N26
P26
M27
N27
P28
P27
L25
M25
L27
L26
M24
M23
N25
N24
K24
L23
M22
N22
P24
P23
L21
L20
M20
N20
N21
P21
M19
M18
N19
P19
M17
N17
P18
P17
P22
FMCB_HB_N1
FMCB_HB_P1
FMCB_HB_N2
FMCB_HB_P2
FMCB_HB_N3
FMCB_HB_P3
FMCB_HB_N4
FMCB_HB_P4
FMCB_HB_N5
FMCB_HB_P5
FMCB_HB_N9
FMCB_HB_P9
FMCB_HB_N7
FMCB_HB_P7
FMCB_HB_N8
FMCB_HB_P8
FMCB_HB_N10
FMCB_HB_P10
FMCB_HB_N6
FMCB_HB_P6
FMCB_HB_N0
FMCB_HB_P0
FMCB_HB_N17
FMCB_HB_P17
FMCB_CLK3_BIDIR_C_N
FMCB_CLK3_BIDIR_C_P
FMCB_HB_N13
FMCB_HB_P13
FMCB_HB_N19
FMCB_HB_P19
FMCB_HB_N11
FMCB_HB_P11
FMCB_HB_N15
FMCB_HB_P15
FMCB_HB_N16
FMCB_HB_P16
FMCB_HB_N20
FMCB_HB_P20
FMCB_HB_N21
FMCB_HB_P21
FMCB_HB_N12
FMCB_HB_P12
FMCB_HB_N18
FMCB_HB_P18
FMCB_HB_N14
FMCB_HB_P14
IO_0_15
IO_L1N_T0_AD0N_15
IO_L1P_T0_AD0P_15
IO_L2N_T0_AD8N_15
IO_L2P_T0_AD8P_15
IO_L3N_T0_DQS_AD1N_15
IO_L3P_T0_DQS_AD1P_15
IO_L4N_T0_AD9N_15
IO_L4P_T0_AD9P_15
IO_L5N_T0_AD2N_15
IO_L5P_T0_AD2P_15
IO_L6N_T0_VREF_15
IO_L6P_T0_15
IO_L7N_T1_AD10N_15
IO_L7P_T1_AD10P_15
IO_L8N_T1_AD3N_15
IO_L8P_T1_AD3P_15
IO_L9N_T1_DQS_AD11N_15
IO_L9P_T1_DQS_AD11P_15
IO_L10N_T1_AD4N_15
IO_L10P_T1_AD4P_15
IO_L11N_T1_SRCC_AD12N_15
IO_L11P_T1_SRCC_AD12P_15
IO_L12N_T1_MRCC_AD5N_15
IO_L12P_T1_MRCC_AD5P_15
IO_L13N_T2_MRCC_15
IO_L13P_T2_MRCC_15
IO_L14N_T2_SRCC_15
IO_L14P_T2_SRCC_15
IO_L15N_T2_DQS_ADV_B_15
IO_L15P_T2_DQS_15
IO_L16N_T2_A27_15
IO_L16P_T2_A28_15
IO_L17N_T2_A25_15
IO_L17P_T2_A26_15
IO_L18N_T2_A23_15
IO_L18P_T2_A24_15
IO_L19N_T3_A21_VREF_15
IO_L19P_T3_A22_15
IO_L20N_T3_A19_15
IO_L20P_T3_A20_15
IO_L21N_T3_DQS_A18_15
IO_L21P_T3_DQS_15
IO_L22N_T3_A16_15
IO_L22P_T3_A17_15
IO_L23N_T3_FWE_B_15
IO_L23P_T3_FOE_B_15
IO_L24N_T3_RS0_15
IO_L24P_T3_RS1_15
IO_25_15
IO_0_16
IO_L1N_T0_16
IO_L1P_T0_16
IO_L2N_T0_16
IO_L2P_T0_16
IO_L3N_T0_DQS_16
IO_L3P_T0_DQS_16
IO_L4N_T0_16
IO_L4P_T0_16
IO_L5N_T0_16
IO_L5P_T0_16
IO_L6N_T0_VREF_16
IO_L6P_T0_16
IO_L7N_T1_16
IO_L7P_T1_16
IO_L8N_T1_16
IO_L8P_T1_16
IO_L9N_T1_DQS_16
IO_L9P_T1_DQS_16
IO_L10N_T1_16
IO_L10P_T1_16
IO_L11N_T1_SRCC_16
IO_L11P_T1_SRCC_16
IO_L12N_T1_MRCC_16
IO_L12P_T1_MRCC_16
IO_L13N_T2_MRCC_16
IO_L13P_T2_MRCC_16
IO_L14N_T2_SRCC_16
IO_L14P_T2_SRCC_16
IO_L15N_T2_DQS_16
IO_L15P_T2_DQS_16
IO_L16N_T2_16
IO_L16P_T2_16
IO_L17N_T2_16
IO_L17P_T2_16
IO_L18N_T2_16
IO_L18P_T2_16
IO_L19N_T3_VREF_16
IO_L19P_T3_16
IO_L20N_T3_16
IO_L20P_T3_16
IO_L21N_T3_DQS_16
IO_L21P_T3_DQS_16
IO_L22N_T3_16
IO_L22P_T3_16
IO_L23N_T3_16
IO_L23P_T3_16
IO_L24N_T3_16
IO_L24P_T3_16
IO_25_16
E24 CLK_RAW
C24 FMCB_LA_N11
D24 FMCB_LA_P11
B25 FMCB_LA_N2
B24 FMCB_LA_P2
C26 FMCB_LA_N6
C25 FMCB_LA_P6
A26 FMCB_LA_N4
A25 FMCB_LA_P4
B27 FMCB_LA_N5
C27 FMCB_LA_P5
D27 FMCB_LA_N3
D26 FMCB_LA_P3
G24 FMCB_LA_N7
H24 FMCB_LA_P7
G25 FMCB_LA_N8
H25 FMCB_LA_P8
J27 FMCB_LA_N9
J26 FMCB_LA_P9
H27 FMCB_LA_N10
H26 FMCB_LA_P10
F26 FMCB_CLK0_M2C_N
F25 FMCB_CLK0_M2C_P
E26 FMCB_LA_N0
E25 FMCB_LA_P0
F27 FMCB_LA_N1
G27 FMCB_LA_P1
F28 FMCB_CLK1_M2C_N
G28 FMCB_CLK1_M2C_P
G29 FMCB_LA_N15
H29 FMCB_LA_P15
E30 FMCB_LA_N16
F30 FMCB_LA_P16
G30 FMCB_LA_N12
H30 FMCB_LA_P12
J29 FMCB_LA_N13
J28 FMCB_LA_P13
D28 FMCB_LA_N14
E28 FMCB_LA_P14
D29 FMCB_LA_N30
E29 FMCB_LA_P30
C30 FMCB_LA_N29
C29 FMCB_LA_P29
B29 FMCB_LA_N31
B28 FMCB_LA_P31
A30 FMCB_LA_N32
B30 FMCB_LA_P32
A28 FMCB_LA_N33
A27 FMCB_LA_P33
J24 SYNC_1V0
U27F
xc7k420tffg901
Title
FPGA Banks 15 & 16
Size
Letter
Number
Revision
3-2014-01-07
Date: 1/9/2014
Sheet33of 59
File: C:\Users\..\FPGA_bank15_bank16.SchDoc
Drawn By: Graeme Smecher

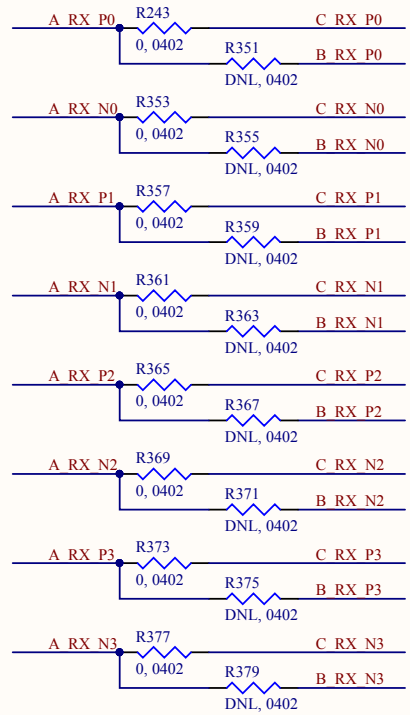

FMC_FPGA
VREF_A_M2C
FMCB_VREF_A_M2C
LA_P[0..33]
LA_N[0..33]
FMCB_LA_P[0..33]
FMCB_LA_N[0..33]
HA_P[0..23]
HA_N[0..23]
FMCB_HA_P[0..23]
FMCB_HA_N[0..23]
VREF_B_M2C
VIO_B_M2C
FMCB_VREF_B_M2C
FMCB_VIO_B_M2C
FMC_FPGA_B
HB_N[0..21]
HB_P[0..21]
FMCB_HB_N[0..21]
FMCB_HB_P[0..21]
CLK0_M2C_N
CLK0_M2C_P
CLK1_M2C_N
CLK1_M2C_P
FMCB_CLK0_M2C_N
FMCB_CLK0_M2C_P
FMCB_CLK1_M2C_N
FMCB_CLK1_M2C_P
CLK2_BIDIR_P
CLK2_BIDIR_N
FMCB_CLK2_BIDIR_P
FMCB_CLK2_BIDIR_N
CLK3_BIDIR_P
CLK3_BIDIR_N
FMCB_CLK3_BIDIR_P
FMCB_CLK3_BIDIR_N
SMPS_SYNC
SMPS_SYNC
SYNC_3V3
SYNC_12V0
SYNC_5V0
SYNC_1V0_GTX
SYNC_1V0
SYNC_VADJ
SYNC_1V5
SYNC_1V8
SYNC_1V2
SYNC_5V0
SYNC_1V0_GTX
SYNC_VADJ
SYNC_1V5
SYNC_1V8
SYNC_1V2
VADJ
C361
100uF 10V, 1206
C3216X5R1A107M160AC
DGND
VADJ
A19
C23
D20
G21
H18
K22
B16
E17
F14
J15
VADJ
C362
100uF 10V, 1206
C3216X5R1A107M160AC
DGND
VCCO_17
VCCO_17
VCCO_17
VCCO_17
VCCO_17
VCCO_17
VCCO_18
VCCO_18
VCCO_18
VCCO_18
SYNC_VADJ H21
FMCB_HA_N9 J23
FMCB_HA_P9 K23
FMCB_HA_N8 H22
FMCB_HA_P8 J22
FMCB_HA_N14 J21
FMCB_HA_P14 K21
FMCB_HA_N11 K20
FMCB_HA_P11 K19
FMCB_HA_N12 J19
FMCB_HA_P12 J18
FMCB_HA_N13 K18
FMCB_HA_P13 L18
FMCB_HA_N10 F23
FMCB_HA_P10 G23
FMCB_HA_N15 G20
FMCB_HA_P15 H20
FMCB_HA_N16 G19
FMCB_HA_P16 H19
FMCB_LA_N28 F18
FMCB_LA_P28 G18
FMCB_CLK2_BIDIR_C_N F22
FMCB_CLK2_BIDIR_C_P G22
FMCB_LA_N17 F21
FMCB_LA_P17 F20
FMCB_LA_N18 E21
FMCB_LA_P18 E20
SYNC_5V0 D22
SYNC_1V0_GTX D21
FMCB_LA_N20 D23
FMCB_LA_P20 E23
FMCB_LA_N22 C22
FMCB_LA_P22 C21
FMCB_LA_N24 E19
FMCB_LA_P24 E18
FMCB_LA_N26 D19
FMCB_LA_P26 D18
FMCB_LA_N19 C20
FMCB_LA_P19 C19
FMCB_LA_N27 B20
FMCB_LA_P27 B19
FMCB_LA_N25 A18
FMCB_LA_P25 B18
FMCB_LA_N21 A21
FMCB_LA_P21 A20
FMCB_LA_N23 B23
FMCB_LA_P23 B22
SYNC_1V5 A23
SYNC_1V8 A22
SYNC_1V2 L17
IO_0_17
IO_L1N_T0_17
IO_L1P_T0_17
IO_L2N_T0_17
IO_L2P_T0_17
IO_L3N_T0_DQS_17
IO_L3P_T0_DQS_17
IO_L4N_T0_17
IO_L4P_T0_17
IO_L5N_T0_17
IO_L5P_T0_17
IO_L6N_T0_VREF_17
IO_L6P_T0_17
IO_L7N_T1_17
IO_L7P_T1_17
IO_L8N_T1_17
IO_L8P_T1_17
IO_L9N_T1_DQS_17
IO_L9P_T1_DQS_17
IO_L10N_T1_17
IO_L10P_T1_17
IO_L11N_T1_SRCC_17
IO_L11P_T1_SRCC_17
IO_L12N_T1_MRCC_17
IO_L12P_T1_MRCC_17
IO_L13N_T2_MRCC_17
IO_L13P_T2_MRCC_17
IO_L14N_T2_SRCC_17
IO_L14P_T2_SRCC_17
IO_L15N_T2_DQS_17
IO_L15P_T2_DQS_17
IO_L16N_T2_17
IO_L16P_T2_17
IO_L17N_T2_17
IO_L17P_T2_17
IO_L18N_T2_17
IO_L18P_T2_17
IO_L19N_T3_VREF_17
IO_L19P_T3_17
IO_L20N_T3_17
IO_L20P_T3_17
IO_L21N_T3_DQS_17
IO_L21P_T3_DQS_17
IO_L22N_T3_17
IO_L22P_T3_17
IO_L23N_T3_17
IO_L23P_T3_17
IO_L24N_T3_17
IO_L24P_T3_17
IO_25_17
N/C on 355T
IO_L6N_T0_VREF_18 J17 FMCB_HA_N2
IO_L6P_T0_18 J16 FMCB_HA_P2
IO_L7N_T1_18 H15 FMCB_HA_N3
IO_L7P_T1_18 J14 FMCB_HA_P3
IO_L8N_T1_18 H17 FMCB_HA_N4
IO_L8P_T1_18 H16 FMCB_HA_P4
IO_L9N_T1_DQS_18 G14 FMCB_HA_N5
IO_L9P_T1_DQS_18 H14 FMCB_HA_P5
IO_L10N_T1_18 F17 FMCB_HA_N6
IO_L10P_T1_18 G17 FMCB_HA_P6
IO_L11N_T1_SRCC_18 E16 FMCB_HA_N17
IO_L11P_T1_SRCC_18 F16 FMCB_HA_P17
IO_L12N_T1_MRCC_18 F15 FMCB_HA_N1
IO_L12P_T1_MRCC_18 G15 FMCB_HA_P1
IO_L13N_T2_MRCC_18 C16 FMCB_HA_N0
IO_L13P_T2_MRCC_18 C15 FMCB_HA_P0
IO_L14N_T2_SRCC_18 D16 FMCB_HA_N18
IO_L14P_T2_SRCC_18 E15 FMCB_HA_P18
IO_L15N_T2_DQS_18 D14 FMCB_HA_N23
IO_L15P_T2_DQS_18 E14 FMCB_HA_P23
IO_L16N_T2_18 B15 FMCB_HA_N22
IO_L16P_T2_18 C14 FMCB_HA_P22
IO_L17N_T2_18 A14 FMCB_HA_N21
IO_L17P_T2_18 B14 FMCB_HA_P21
IO_L18N_T2_18 A16 FMCB_HA_N19
IO_L18P_T2_18 A15 FMCB_HA_P19
IO_L19N_T3_VREF_18 C17 FMCB_HA_N20
IO_L19P_T3_18 D17 FMCB_HA_P20
IO_L20N_T3_18 A17 FMCB_HA_N7
IO_L20P_T3_18 B17 FMCB_HA_P7
U27G
xc7k420tffg901
clock_passive_coupling.SchDoc
CC10
VCC2V5
V_LVDS
V_FPGA
VADJ
FMCB_CLK2_BIDIR_P
FMCB_CLK2_BIDIR_N
LVDS_P
LVDS_N
FPGA_P
FPGA_N
FMCB_CLK2_BIDIR_C_P
FMCB_CLK2_BIDIR_C_N
Title
FPGA Banks 17 & 18
Size
Letter
Number
Revision
3-2014-01-07
Date: 1/9/2014
Sheet34of 59
File: C:\Users\..\FPGA_bank17_bank18.SchDoc
Drawn By: Graeme Smecher

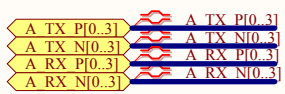

FMCA_GBTCLK0_M2C_P
FMCA_GBTCLK0_M2C_N
FMCA_GBTCLK1_M2C_P
FMCA_GBTCLK1_M2C_N
FMCB_GBTCLK0_M2C_P
FMCB_GBTCLK0_M2C_N
FMCB_GBTCLK1_M2C_P
FMCB_GBTCLK1_M2C_N
This FPGA bank was flipped left-to-right during layout. Also, some bits were re-organized. As a result, the net names and FPGA terminals no longer match. This is perfectly OK from a bank-assignment perspective, but makes a confusing UCF. It would be nice to fix this at some point.
The net naming may change, but the connectivity must not (since we're not ripping up layout!)
FPGA_PLLCLOCKS
CLK_GTX112_A_P
CLK_GTX112_A_N
CLK_GTX112_B_P
CLK_GTX112_B_N
CLK_GTX114_A_P
CLK_GTX114_A_N
CLK_100MHz_P1
CLK_100MHz_N1
CLK_GTX116_A_P
CLK_GTX116_A_N
CLK_GTX116_B_P
CLK_GTX116_B_N
CLK_GTX115_A_P
CLK_GTX115_A_N
CLK_GTX115_B_P
CLK_GTX115_B_N
U27H
0.1uF
C308
C533
C534
C535
GTX113_REFCLK_N0 U7
GTX113_REFCLK_P0 U8
GTX113_REFCLK_N1 W7
GTX113_REFCLK_P1 W8
GTX113_RX_N3 AB5
GTX113_RX_N2 AA3
GTX113_RX_N1 Y5
GTX113_RX_N0 W3
GTX113_RX_P3 AB6
GTX113_RX_P2 AA4
GTX113_RX_P1 Y6
GTX113_RX_P0 W4
GTX113_TX_N3 AF1
GTX113_TX_N2 AD1
GTX113_TX_N1 AB1
GTX113_TX_N0 Y1
GTX113_TX_P3 AF2
GTX113_TX_P2 AD2
GTX113_TX_P1 AB2
GTX113_TX_P0 Y2
4 x 0.1uF
C403
C404
C406
C408
GTX114_REFCLK_N0 R7
GTX114_REFCLK_P0 R8
GTX114_REFCLK_N1 T5
GTX114_REFCLK_P1 T6
GTX114_RX_N3 V5
GTX114_RX_N2 U3
GTX114_RX_N1 R3
GTX114_RX_N0 P5
GTX114_RX_P3 V6
GTX114_RX_P2 U4
GTX114_RX_P1 R4
GTX114_RX_P0 P6
GTX114_TX_N3 V1
GTX114_TX_N2 T1
GTX114_TX_N1 P1
GTX114_TX_N0 N3
GTX114_TX_P3 V2
GTX114_TX_P2 T2
GTX114_TX_P1 P2
GTX114_TX_P0 N4
MGTREFCLK0N_113
MGTREFCLK0P_113
MGTREFCLK1N_113
MGTREFCLK1P_113
MGTXRXN0_113
MGTXRXN1_113
MGTXRXN2_113
MGTXRXN3_113
MGTXRXP0_113
MGTXRXP1_113
MGTXRXP2_113
MGTXRXP3_113
MGTXTXN0_113
MGTXTXN1_113
MGTXTXN2_113
MGTXTXN3_113
MGTXTXP0_113
MGTXTXP1_113
MGTXTXP2_113
MGTXTXP3_113
MGTREFCLK0N_114
MGTREFCLK0P_114
MGTREFCLK1N_114
MGTREFCLK1P_114
MGTXRXN0_114
MGTXRXN1_114
MGTXRXN2_114
MGTXRXN3_114
MGTXRXP0_114
MGTXRXP1_114
MGTXRXP2_114
MGTXRXP3_114
MGTXTXN0_114
MGTXTXN1_114
MGTXTXN2_114
MGTXTXN3_114
MGTXTXP0_114
MGTXTXP1_114
MGTXTXP2_114
MGTXTXP3_114
N/C on '355T
MGTREFCLK0N_111
MGTREFCLK0P_111
MGTREFCLK1N_111
MGTREFCLK1P_111
MGTXRXN0_111
MGTXRXN1_111
MGTXRXN2_111
MGTXRXN3_111
MGTXRXP0_111
MGTXRXP1_111
MGTXRXP2_111
MGTXRXP3_111
MGTXTXN0_111
MGTXTXN1_111
MGTXTXN2_111
MGTXTXN3_111
MGTXTXP0_111
MGTXTXP1_111
MGTXTXP2_111
MGTXTXP3_111
MGTREFCLK0N_112
MGTREFCLK0P_112
MGTREFCLK1N_112
MGTREFCLK1P_112
MGTXRXN0_112
MGTXRXN1_112
MGTXRXN2_112
MGTXRXN3_112
MGTXRXP0_112
MGTXRXP1_112
MGTXRXP2_112
MGTXRXP3_112
MGTXTXN0_112
MGTXTXN1_112
MGTXTXN2_112
MGTXTXN3_112
MGTXTXP0_112
MGTXTXP1_112
MGTXTXP2_112
MGTXTXP3_112
xc7k420tffg901
AC7 GTX111_REFCLK_N0
AC8 GTX111_REFCLK_P0
AE7 GTX111_REFCLK_N1
AE8 GTX111_REFCLK_P1
AH9 GTX111_RX_N3
AG11 GTX111_RX_N2
AF9 GTX111_RX_N1
AE11 GTX111_RX_N0
AH10 GTX111_RX_P3
AG12 GTX111_RX_P2
AF10 GTX111_RX_P1
AE12 GTX111_RX_P0
AJ11 GTX111_TX_N3
AK9 GTX111_TX_N2
AJ7 GTX111_TX_N1
AG7 GTX111_TX_N0
AJ12 GTX111_TX_P3
AK10 GTX111_TX_P2
AJ8 GTX111_TX_P1
AG8 GTX111_TX_P0
0.1uF
C350
C356
C536
C537
AD5 GTX112_REFCLK_N0
AD6 GTX112_REFCLK_P0
AF5 GTX112_REFCLK_N1
AF6 GTX112_REFCLK_P1
AH5 GTX112_RX_N3
AG3 GTX112_RX_N2
AE3 GTX112_RX_N1
AC3 GTX112_RX_N0
AH6 GTX112_RX_P3
AG4 GTX112_RX_P2
AE4 GTX112_RX_P1
AC4 GTX112_RX_P0
AK5 GTX112_TX_N3
AJ3 GTX112_TX_N2
AK1 GTX112_TX_N1
AH1 GTX112_TX_N0
AK6 GTX112_TX_P3
AJ4 GTX112_TX_P2
AK2 GTX112_TX_P1
AH2 GTX112_TX_P0
C405
C407
C409
C352
8 x 0.1uF
Title
FPGA GTX transceivers 111, 112, 113, 114
Size Letter
Number
Revision 3-2014-01-07
Date: 1/9/2014
Sheet 35 of 59
File: C:\Users\..\FPGA_gtx111_gtx112_gtx113_gtx114.SchDoc
Drawn By: Duane Smecher

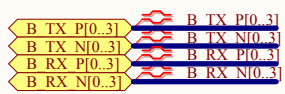

GTX_MUX_113
FPGA_gtx_mux4.SchDoc
GTX113_TX_P[0..3]
GTX113_TX_N[0..3]
GTX113_RX_P[0..3]
GTX113_RX_N[0..3]
A_TX_P[0..3]
A_TX_N[0..3]
A_RX_P[0..3]
A_RX_N[0..3]
B_TX_P[0..3]
B_TX_N[0..3]
B_RX_P[0..3]
B_RX_N[0..3]
C_TX_P[0..3]
C_TX_N[0..3]
C_RX_P[0..3]
C_RX_N[0..3]
FMCB_GTX_TX_P[0..3]
FMCB_GTX_TX_N[0..3]
FMCB_GTX_RX_P[0..3]
FMCB_GTX_RX_N[0..3]
QUAD_TX_P[1..4]
QUAD_TX_N[1..4]
QUAD_RX_P[1..4]
QUAD_RX_N[1..4]
GTX_MUX_111
FPGA_gtx_mux4.SchDoc
GTX111_TX_P[0..3]
GTX111_TX_N[0..3]
GTX111_RX_P[0..3]
GTX111_RX_N[0..3]
FMCA_GTX_TX_P[0..3]
FMCA_GTX_TX_N[0..3]
FMCA_GTX_RX_P[0..3]
FMCA_GTX_RX_N[0..3]
GTX_MUX_111B_TX_P[0..3]
GTX_MUX_111B_TX_N[0..3]
SHUFFLE_RX_P[5..8]
SHUFFLE_RX_N[5..8]
GTX_MUX_114
FPGA_gtx_mux4.SchDoc
GTX114_TX_P[0..3]
GTX114_TX_N[0..3]
GTX114_RX_P[0..3]
GTX114_RX_N[0..3]
FMCB_GTX_TX_P[4..7]
FMCB_GTX_TX_N[4..7]
FMCB_GTX_RX_P[4..7]
FMCB_GTX_RX_N[4..7]
GTX_MUX_114B_TX_P[0..3]
GTX_MUX_114B_TX_N[0..3]
SHUFFLE_RX_P[1..4]
SHUFFLE_RX_N[1..4]
GTX_MUX_112
FPGA_gtx_mux4.SchDoc
GTX112_TX_P[0..3]
GTX112_TX_N[0..3]
GTX112_RX_P[0..3]
GTX112_RX_N[0..3]
FMCA_GTX_TX_P[4..7]
FMCA_GTX_TX_N[4..7]
FMCA_GTX_RX_P[4..7]
FMCA_GTX_RX_N[4..7]
GTX_MUX_112B_TX_P[0..3]
GTX_MUX_112B_TX_N[0..3]
SHUFFLE_RX_P[9..12]
SHUFFLE_RX_N[9..12]
BP_GTX
SHUFFLE_TX_P[1..15]
SHUFFLE_TX_N[1..15]
SHUFFLE_RX_P[1..15]
SHUFFLE_RX_N[1..15]
QUAD_TX_P[1..4]
QUAD_TX_N[1..4]
QUAD_RX_P[1..4]
QUAD_RX_N[1..4]
BP_GTX
FMC_GTX
FMCA_GTX_TX_P[0..7]
FMCA_GTX_TX_N[0..7]
FMCA_GTX_RX_P[0..7]
FMCA_GTX_RX_N[0..7]
FMCA_GBTCLK0_M2C_P
FMCA_GBTCLK0_M2C_N
FMCA_GBTCLK1_M2C_P
FMCA_GBTCLK1_M2C_N
TX_P[0..7]
TX_N[0..7]
RX_P[0..7]
RX_N[0..7]
GBTCLK0_M2C_P
GBTCLK0_M2C_N
GBTCLK1_M2C_P
GBTCLK1_M2C_N
FMCA_GTX
FMC_GTX
FMCB_GTX_TX_P[0..7]
FMCB_GTX_TX_N[0..7]
FMCB_GTX_RX_P[0..7]
FMCB_GTX_RX_N[0..7]
FMCB_GBTCLK0_M2C_P
FMCB_GBTCLK0_M2C_N
FMCB_GBTCLK1_M2C_P
FMCB_GBTCLK1_M2C_N
TX_P[0..7]
TX_N[0..7]
RX_P[0..7]
RX_N[0..7]
GBTCLK0_M2C_P
GBTCLK0_M2C_N
GBTCLK1_M2C_P
GBTCLK1_M2C_N
FMCB_GTX
GTX_MUX_114B_TX_P0
GTX_MUX_114B_TX_P1
GTX_MUX_114B_TX_P2
GTX_MUX_114B_TX_P3
GTX_MUX_114B_TX_N0
GTX_MUX_114B_TX_N1
GTX_MUX_114B_TX_N2
GTX_MUX_114B_TX_N3
C424
C425
C427
C429
C431
C432
C434
C436
SHUFFLE_TX_P1
SHUFFLE_TX_P2
SHUFFLE_TX_P3
SHUFFLE_TX_P4
SHUFFLE_TX_N1
SHUFFLE_TX_N2
SHUFFLE_TX_N3
SHUFFLE_TX_N4
GTX_MUX_111B_TX_P0
GTX_MUX_111B_TX_P1
GTX_MUX_111B_TX_P2
GTX_MUX_111B_TX_P3
GTX_MUX_111B_TX_N0
GTX_MUX_111B_TX_N1
GTX_MUX_111B_TX_N2
GTX_MUX_111B_TX_N3
C391
C393
C395
C336
C398
C400
C402
C439
SHUFFLE_TX_P5
SHUFFLE_TX_P6
SHUFFLE_TX_P7
SHUFFLE_TX_P8
SHUFFLE_TX_N5
SHUFFLE_TX_N6
SHUFFLE_TX_N7
SHUFFLE_TX_N8
GTX_MUX_112B_TX_P0
GTX_MUX_112B_TX_P1
GTX_MUX_112B_TX_P2
GTX_MUX_112B_TX_P3
GTX_MUX_112B_TX_N0
GTX_MUX_112B_TX_N1
GTX_MUX_112B_TX_N2
GTX_MUX_112B_TX_N3
C426
C428
C430
C376
C433
C435
C437
C438
SHUFFLE_TX_P9
SHUFFLE_TX_P10
SHUFFLE_TX_P11
SHUFFLE_TX_P12
SHUFFLE_TX_N9
SHUFFLE_TX_N10
SHUFFLE_TX_N11
SHUFFLE_TX_N12
24 x 0.1uF
There are several conventions for coupling caps:
* For PCIe, coupling caps are placed wherever there's a transceiver.
* For QSFP/SFP, coupling caps are placed in the module.
* For FMC, caps are placed (if needed) on the mezzanine.
* For backplane shuffles, we follow PCIe convention.
Title
FPGA GTX transceivers 111, 112, 113, 114 (Multiplexing)
Size
Letter
Number
Revision
3-2014-01-07
Date:
1/9/2014
Sheet36of 59
File:
C:\Users\..\FPGA_gtx111_gtx112_gtx113_gtx114_Multiplexing.SchDoc
Drawn By:

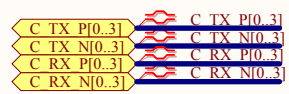
We connect A <-> C by default. This connects GTXes to the backplane.











Title: GTX Multiplexer		
Size: Letter	Number:	Revision: 3-2014-01-07
Date: 1/9/2014		Sheet37of 59
File: C:\Users\..\FPGA_gtx_mux4.SchDoc		Drawn By: Graeme Smecher

CLK_BANK_14 now goes to GTX clock 115A
CLK_BANK_17 now goes to GTX clock 115B

U27I

CLK_GTX115_A_N 0.1uF GTX115_REFCLK_N0 J7
CLK_GTX115_A_P C646 0.1uF C645 GTX115_REFCLK_P0 J8
CLK_GTX115_B_N C648 C647 GTX115_REFCLK_N1 L7
CLK_GTX115_B_P 0.1uF GTX115_REFCLK_P1 L8

SHUFFLE_RX_N13 M5
SHUFFLE_RX_N14 0.1uF L3
SHUFFLE_RX_N15 K5
PCI_SFP_RX_N J3
SHUFFLE_RX_P13 M6
SHUFFLE_RX_P14 L4
SHUFFLE_RX_P15 K6
PCI_SFP_RX_P J4

SHUFFLE_TX_N13 C327 GTX115_TX_N0 M1
SHUFFLE_TX_N14 C329 GTX115_TX_N1 K1
SHUFFLE_TX_N15 C331 GTX115_TX_N2 H1
PCI_SFP_TX_N F1
SHUFFLE_TX_P13 C335 GTX115_TX_P0 M2
SHUFFLE_TX_P14 C337 GTX115_TX_P1 K2
SHUFFLE_TX_P15 C339 GTX115_TX_P2 H2
PCI_SFP_TX_P F2

0.1uF x10
CLK_GTX116_A_P C344 GTX116_REFCLK_N0 F5
CLK_GTX116_A_N C345 GTX116_REFCLK_P0 F6
CLK_GTX116_B_P C347 GTX116_REFCLK_N1 H5
CLK_GTX116_B_N C349 GTX116_REFCLK_P1 H6

QSFPA_RX_N1 G3
QSFPA_RX_N2 E3
QSFPA_RX_N3 D5
QSFPA_RX_N4 C3
QSFPA_RX_P1 G4
QSFPA_RX_P2 E4
QSFPA_RX_P3 D6
QSFPA_RX_P4 C4
QSFPA_TX_N1 D1
QSFPA_TX_N2 B1
QSFPA_TX_N3 A3
QSFPA_TX_N4 B5
QSFPA_TX_P1 D2
QSFPA_TX_P2 B2
QSFPA_TX_P3 A4
QSFPA_TX_P4 B6

MGTREFCLK0N_115
MGTREFCLK0P_115
MGTREFCLK1N_115
MGTREFCLK1P_115
MGTXRXN0_115
MGTXRXN1_115
MGTXRXN2_115
MGTXRXN3_115
MGTXRXP0_115
MGTXRXP1_115
MGTXRXP2_115
MGTXRXP3_115
MGTXTXN0_115
MGTXTXN1_115
MGTXTXN2_115
MGTXTXN3_115
MGTXTXP0_115
MGTXTXP1_115
MGTXTXP2_115
MGTXTXP3_115

MGTREFCLK0N_116
MGTREFCLK0P_116
MGTREFCLK1N_116
MGTREFCLK1P_116
MGTXRXN0_116
MGTXRXN1_116
MGTXRXN2_116
MGTXRXN3_116
MGTXRXP0_116
MGTXRXP1_116
MGTXRXP2_116
MGTXRXP3_116
MGTXTXN0_116
MGTXTXN1_116
MGTXTXN2_116
MGTXTXN3_116
MGTXTXP0_116
MGTXTXP1_116
MGTXTXP2_116 B6
MGTXTXP3_116 A4

MGTREFCLK0N_117 E7
MGTREFCLK0P_117 E8
MGTREFCLK1N_117 G7
MGTREFCLK1P_117 G8
MGTXRXN0_117 F9
MGTXRXN1_117 D9
MGTXRXN2_117 E11
MGTXRXN3_117 C11
MGTXRXP0_117 F10
MGTXRXP1_117 D10
MGTXRXP2_117 E12
MGTXRXP3_117 C12
MGTXTXN0_117 C7
MGTXTXN1_117 A7
MGTXTXN2_117 B9
MGTXTXN3_117 A11
MGTXTXP0_117 C8
MGTXTXP1_117 A8
MGTXTXP2_117 B10
MGTXTXP3_117 A12

MGTAVTTRCAL_115 N8
MGTRREF_115 N7

xc7k420tffg901

QSFPB_RX_N1
QSFPB_RX_N2
QSFPB_RX_N3
QSFPB_RX_N4
QSFPB_RX_P1
QSFPB_RX_P2
QSFPB_RX_P3
QSFPB_RX_P4
QSFPB_TX_N1
QSFPB_TX_N2
QSFPB_TX_N3
QSFPB_TX_N4
QSFPB_TX_P1
QSFPB_TX_P2
QSFPB_TX_P3
QSFPB_TX_P4

GTX115_TX_N[0..2]
GTX115_TX_P[0..2]
GTX115_REFCLK_N[0..1]
GTX115_REFCLK_P[0..1]
GTX116_REFCLK_N[0..1]
GTX116_REFCLK_P[0..1]

VCC1V2
R214
100

Layout guidelines in UG476, fig. 5-3

SEL_SFP
SEL_SFP

BP_GTX
SHUFFLE_TX_P[1..15]
SHUFFLE_TX_N[1..15]
SHUFFLE_RX_P[1..15]
SHUFFLE_RX_N[1..15]
• QUAD_TX_P[1..4]
• QUAD_TX_N[1..4]
• QUAD_RX_P[1..4]
• QUAD_RX_N[1..4]
BP_GTX

FPGA_PLLCLOCKS
CLK_GTX112_A_P
CLK_GTX112_A_N
CLK_GTX112_B_P
CLK_GTX112_B_N
CLK_GTX114_A_P
CLK_GTX114_A_N
CLK_GTX114_B_P
CLK_GTX114_B_N
CLK_GTX116_A_P
CLK_GTX116_A_N
CLK_GTX116_B_P
CLK_GTX116_B_N
CLK_GTX115_A_P
CLK_GTX115_A_N
CLK_GTX115_B_P
CLK_GTX115_B_N
FPGA_PLLCLOCKS

CLK_GTX116_A_P
CLK_GTX116_A_N
CLK_GTX116_B_P
CLK_GTX116_B_N
CLK_GTX115_A_P
CLK_GTX115_A_N
CLK_GTX115_B_P
CLK_GTX115_B_N

VCC3V3
4.7k R15
4.7k R166
DNP R194
DGND
SEL_SFP

U40
CBTL02043A
XSD
SEL
A0_P
A0_N
A1_P
A1_N
B0_P
B0_N
B1_P
B1_N
C0_P
C0_N
C1_P
C1_N
VDD
GND

PCI_SFP_TX_P
PCI_SFP_TX_N
PCI_SFP_RX_P
PCI_SFP_RX_N

PCI TX/RX perspective is from the ARM core

PCIe_RX_P
PCIe_RX_N
PCIe_TX_P
PCIe_TX_N
0.1uF x4
C317
C325
C333
C341
PCIe_RX_C_P
PCIe_RX_C_N
PCIe_TX_C_P
PCIe_TX_C_N
RX_P
RX_N
TX_P
TX_N
PCIe

SFP_TX_P
SFP_TX_N
SFP_RX_P
SFP_RX_N
TX_P
TX_N
RX_P
RX_N
SFP_FPGA

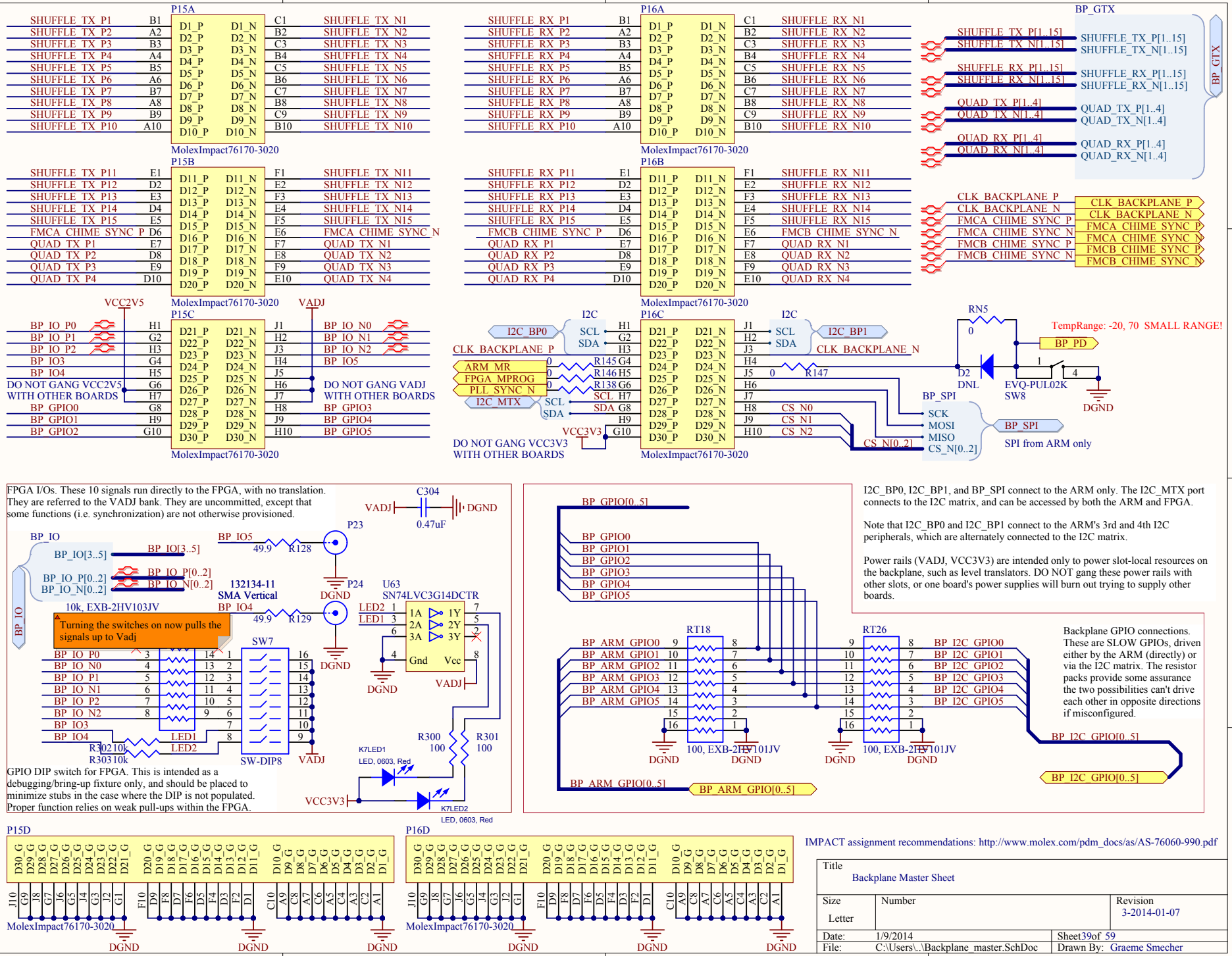
QSFP_FPGA
QSFPA_TX_N[1..4]
QSFPA_TX_P[1..4]
QSFPA_RX_N[1..4]
QSFPA_RX_P[1..4]
TX_N[1..4]
TX_P[1..4]
RX_N[1..4]
RX_P[1..4]
QSFP_FPGA_A

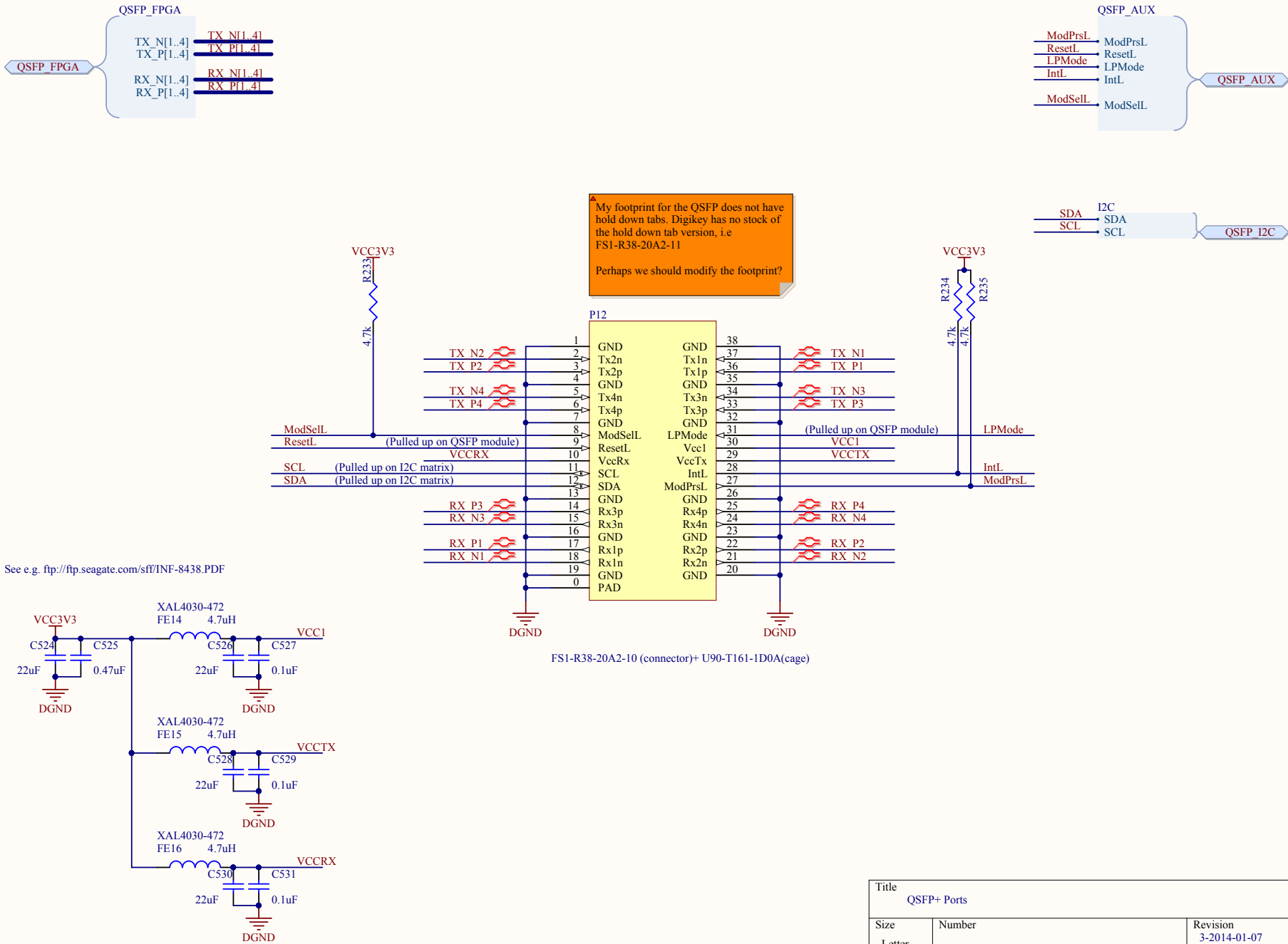
QSFP_FPGA
QSFPB_TX_N[1..4]
QSFPB_TX_P[1..4]
QSFPB_RX_N[1..4]
QSFPB_RX_P[1..4]
TX_N[1..4]
TX_P[1..4]
RX_N[1..4]
RX_P[1..4]
QSFP_FPGA_B

Selects between PCIe and SFP connections. Defaults to SFP.

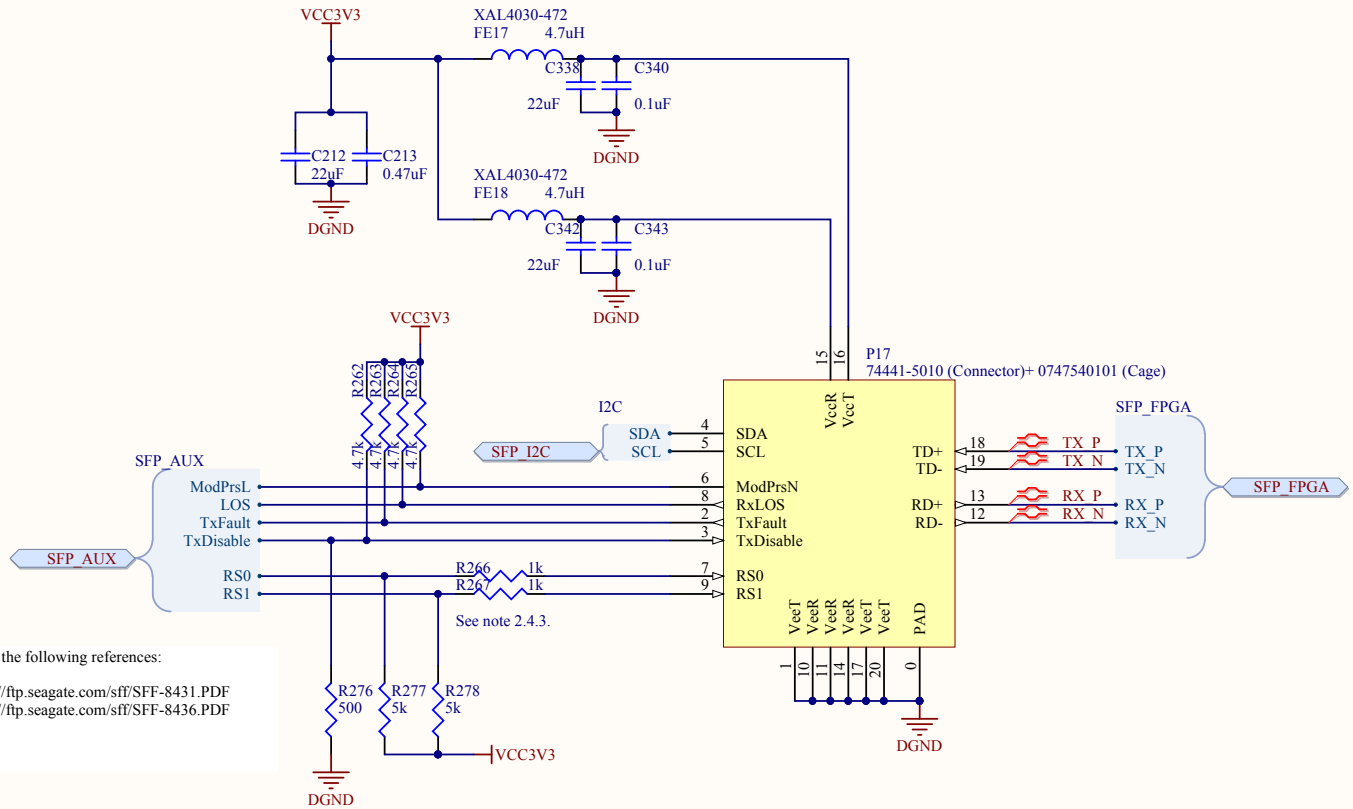
VCC3V3
C314 47uF, 6.3V
T520B476M006ATE070
C315
C316
C318
0.47uF
DGND

Title: FPGA GTX transceivers 115, 116, 117		
Size: Letter	Number	Revision: 3-2014-01-07
Date: 1/9/2014		Sheet 38 of 59
File: C:\Users\...\FPGA_gtx115_gtx116_gtx117.SchDoc		Drawn By: Graeme Smecher

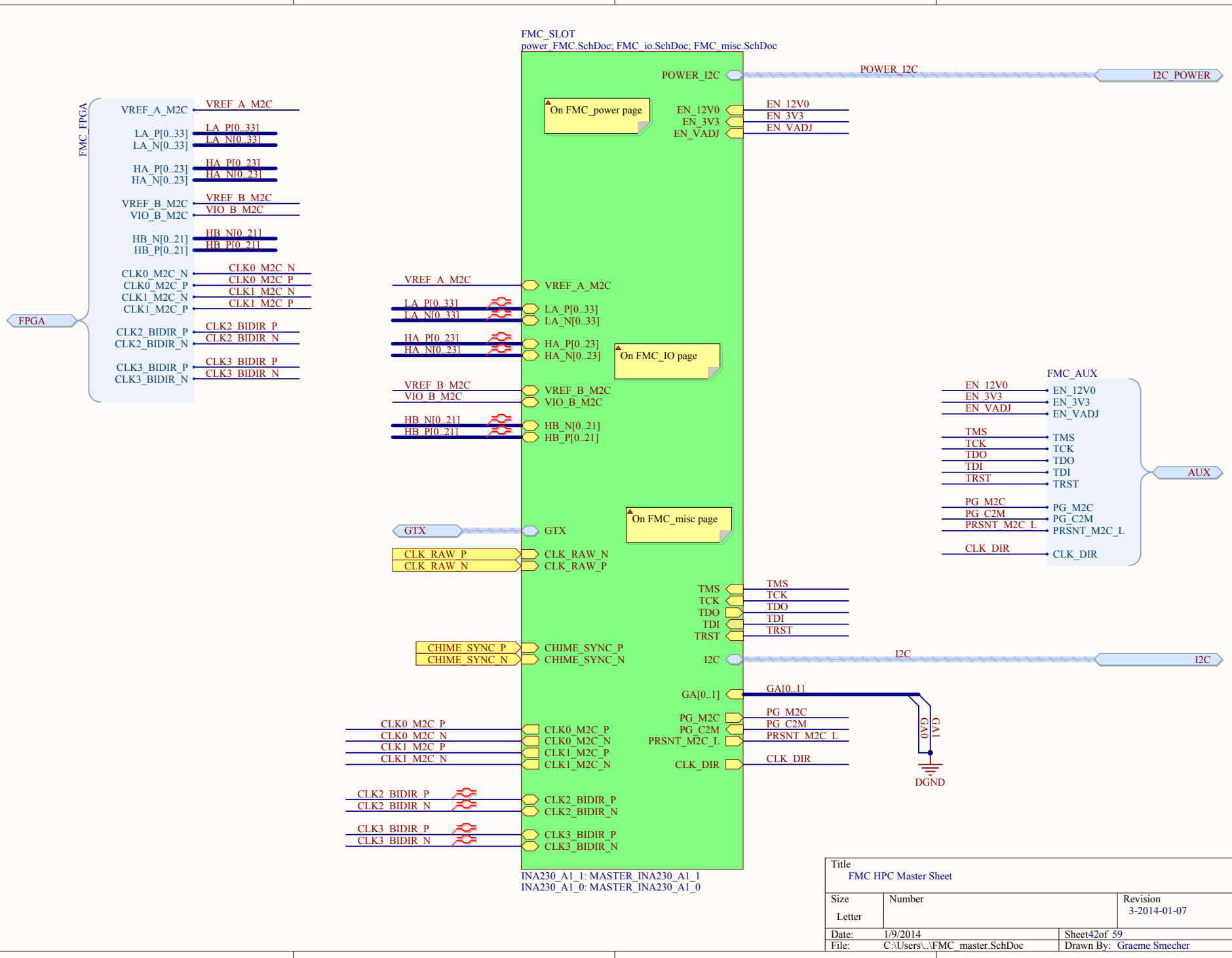

P15A
SHUFFLE_TX_P1 B1
SHUFFLE_TX_P2 A2
SHUFFLE_TX_P3 B3
SHUFFLE_TX_P4 A4
SHUFFLE_TX_P5 B5
SHUFFLE_TX_P6 A6
SHUFFLE_TX_P7 B7
SHUFFLE_TX_P8 A8
SHUFFLE_TX_P9 B9
SHUFFLE_TX_P10 A10
D1_P D1_N
D2_P D2_N
D3_P D3_N
D4_P D4_N
D5_P D5_N
D6_P D6_N
D7_P D7_N
D8_P D8_N
D9_P D9_N
D10_P D10_N
C1 SHUFFLE_TX_N1
B2 SHUFFLE_TX_N2
C3 SHUFFLE_TX_N3
B4 SHUFFLE_TX_N4
C5 SHUFFLE_TX_N5
B6 SHUFFLE_TX_N6
C7 SHUFFLE_TX_N7
B8 SHUFFLE_TX_N8
C9 SHUFFLE_TX_N9
B10 SHUFFLE_TX_N10
MolexImpact76170-3020
P15B
SHUFFLE_TX_P11 E1
SHUFFLE_TX_P12 D2
SHUFFLE_TX_P13 E3
SHUFFLE_TX_P14 D4
SHUFFLE_TX_P15 E5
FMCA_CHIME_SYNC_P D6
QUAD_TX_P1 E7
QUAD_TX_P2 D8
QUAD_TX_P3 E9
QUAD_TX_P4 D10
D11_P D11_N
D12_P D12_N
D13_P D13_N
D14_P D14_N
D15_P D15_N
D16_P D16_N
D17_P D17_N
D18_P D18_N
D19_P D19_N
D20_P D20_N
F1 SHUFFLE_TX_N11
E2 SHUFFLE_TX_N12
F3 SHUFFLE_TX_N13
E4 SHUFFLE_TX_N14
F5 SHUFFLE_TX_N15
E6 FMCA_CHIME_SYNC_N
F7 QUAD_TX_N1
E8 QUAD_TX_N2
F9 QUAD_TX_N3
E10 QUAD_TX_N4
MolexImpact76170-3020
P15C
VCC2V5
VADJ
BP_IO_P0 H1
BP_IO_P1 G2
BP_IO_P2 H3
BP_IO3 G4
BP_IO4 H5
DO NOT GANG VCC2V5
WITH OTHER BOARDS G6
H7
BP_GPIO0 G8
BP_GPIO1 H9
BP_GPIO2 G10
D21_P D21_N
D22_P D22_N
D23_P D23_N
D24_P D24_N
D25_P D25_N
D26_P D26_N
D27_P D27_N
D28_P D28_N
D29_P D29_N
D30_P D30_N
J1 BP_IO_N0
H2 BP_IO_N1
J3 BP_IO_N2
H4 BP_IO5
J5
H6 DO NOT GANG VADJ
J7 WITH OTHER BOARDS
H8 BP_GPIO3
J9 BP_GPIO4
H10 BP_GPIO5
MolexImpact76170-3020
P16A
SHUFFLE_RX_P1 B1
SHUFFLE_RX_P2 A2
SHUFFLE_RX_P3 B3
SHUFFLE_RX_P4 A4
SHUFFLE_RX_P5 B5
SHUFFLE_RX_P6 A6
SHUFFLE_RX_P7 B7
SHUFFLE_RX_P8 A8
SHUFFLE_RX_P9 B9
SHUFFLE_RX_P10 A10
D1_P D1_N
D2_P D2_N
D3_P D3_N
D4_P D4_N
D5_P D5_N
D6_P D6_N
D7_P D7_N
D8_P D8_N
D9_P D9_N
D10_P D10_N
C1 SHUFFLE_RX_N1
B2 SHUFFLE_RX_N2
C3 SHUFFLE_RX_N3
B4 SHUFFLE_RX_N4
C5 SHUFFLE_RX_N5
B6 SHUFFLE_RX_N6
C7 SHUFFLE_RX_N7
B8 SHUFFLE_RX_N8
C9 SHUFFLE_RX_N9
B10 SHUFFLE_RX_N10
MolexImpact76170-3020
P16B
SHUFFLE_RX_P11 E1
SHUFFLE_RX_P12 D2
SHUFFLE_RX_P13 E3
SHUFFLE_RX_P14 D4
SHUFFLE_RX_P15 E5
FMCB_CHIME_SYNC_P D6
QUAD_RX_P1 E7
QUAD_RX_P2 D8
QUAD_RX_P3 E9
QUAD_RX_P4 D10
D11_P D11_N
D12_P D12_N
D13_P D13_N
D14_P D14_N
D15_P D15_N
D16_P D16_N
D17_P D17_N
D18_P D18_N
D19_P D19_N
D20_P D20_N
F1 SHUFFLE_RX_N11
E2 SHUFFLE_RX_N12
F3 SHUFFLE_RX_N13
E4 SHUFFLE_RX_N14
F5 SHUFFLE_RX_N15
E6 FMCB_CHIME_SYNC_N
F7 QUAD_RX_N1
E8 QUAD_RX_N2
F9 QUAD_RX_N3
E10 QUAD_RX_N4
MolexImpact76170-3020
P16C
I2C
I2C_BP0
SCL G2
SDA H3
CLK_BACKPLANE_P
ARM_MR R145 G4
FPGA_MPROG R146 H5
PLL_SYNC_N R138 G6
I2C_MTX SCL H7
SDA G8
H9
VCC3V3 G10
DO NOT GANG VCC3V3
WITH OTHER BOARDS
D21_P D21_N
D22_P D22_N
D23_P D23_N
D24_P D24_N
D25_P D25_N
D26_P D26_N
D27_P D27_N
D28_P D28_N
D29_P D29_N
D30_P D30_N
J1 I2C
H2 SCL
J3 SDA
H4
J5 0 R147
H6
J7
H8 CS_N0
J9 CS_N1
H10 CS_N2
I2C_BP1
CLK_BACKPLANE_N
CS_N[0..2]
MolexImpact76170-3020
RN5
0
BP_PD
1
4
D2
DNL
EVQ-PUL02K
SW8
DGND
TempRange: -20, 70 SMALL RANGE!
BP_SPI
SCK
MOSI
MISO
CS_N[0..2]
BP_SPI
SPI from ARM only
BP_GTX
SHUFFLE_TX_P[1..15]
SHUFFLE_TX_N[1..15]
SHUFFLE_RX_P[1..15]
SHUFFLE_RX_N[1..15]
QUAD_TX_P[1..4]
QUAD_TX_N[1..4]
QUAD_RX_P[1..4]
QUAD_RX_N[1..4]
CLK_BACKPLANE_P
CLK_BACKPLANE_N
FMCA_CHIME_SYNC_P
FMCA_CHIME_SYNC_N
FMCB_CHIME_SYNC_P
FMCB_CHIME_SYNC_N
FPGA I/Os. These 10 signals run directly to the FPGA, with no translation. They are referred to the VADJ bank. They are uncommitted, except that some functions (i.e. synchronization) are not otherwise provisioned.
C304
VADJ
0.47uF
DGND
P23
BP_IO
BP_IO[3..5]
BP_IO_P[0..2]
BP_IO_N[0..2]
BP_IO5
49.9 R128
P24
DGND
132134-11
SMA Vertical
BP_IO4
49.9 R129
DGND
U63
SN74LVC3G14DCTR
LED2 1
LED1 3
1A 1Y 7
2A 2Y 5
3A 3Y 2
Gnd 4
Vcc 8
VADJ
DGND
10k, EXB-2HV103JV
Turning the switches on now pulls the signals up to Vadj
SW7
BP_IO_P0 3 14 1 16
BP_IO_N0 4 13 2 15
BP_IO_P1 5 12 3 14
BP_IO_N1 6 11 4 13
BP_IO_P2 7 10 5 12
BP_IO_N2 8 9 6 11
BP_IO3 7 10
BP_IO4 8 9
R302 10k
R303 10k
LED1
LED2
SW-DIP8
VADJ
R300 100
R301 100
K7LED1
LED, 0603, Red
VCC3V3
K7LED2
LED, 0603, Red
GPIO DIP switch for FPGA. This is intended as a debugging/bring-up fixture only, and should be placed to minimize stubs in the case where the DIP is not populated. Proper function relies on weak pull-ups within the FPGA.
BP_GPIO[0..5]
BP_GPIO0
BP_GPIO1
BP_GPIO2
BP_GPIO3
BP_GPIO4
BP_GPIO5
RT18
RT26
BP_ARM_GPIO0 9 8
BP_ARM_GPIO1 10 7
BP_ARM_GPIO2 11 6
BP_ARM_GPIO3 12 5
BP_ARM_GPIO4 13 4
BP_ARM_GPIO5 14 3
15 2
16 1
100, EXB-2HV101JV
DGND
BP_I2C_GPIO0
BP_I2C_GPIO1
BP_I2C_GPIO2
BP_I2C_GPIO3
BP_I2C_GPIO4
BP_I2C_GPIO5
BP_I2C_GPIO[0..5]
BP_ARM_GPIO[0..5]
I2C_BP0, I2C_BP1, and BP_SPI connect to the ARM only. The I2C_MTX port connects to the I2C matrix, and can be accessed by both the ARM and FPGA.
Note that I2C_BP0 and I2C_BP1 connect to the ARM's 3rd and 4th I2C peripherals, which are alternately connected to the I2C matrix.
Power rails (VADJ, VCC3V3) are intended only to power slot-local resources on the backplane, such as level translators. DO NOT gang these power rails with other slots, or one board's power supplies will burn out trying to supply other boards.
Backplane GPIO connections. These are SLOW GPIOs, driven either by the ARM (directly) or via the I2C matrix. The resistor packs provide some assurance the two possibilities can't drive each other in opposite directions if misconfigured.
P15D
P16D
D30_G D29_G D28_G D27_G D26_G D25_G D24_G D23_G D22_G D21_G
D20_G D19_G D18_G D17_G D16_G D15_G D14_G D13_G D12_G D11_G
D10_G D9_G D8_G D7_G D6_G D5_G D4_G D3_G D2_G D1_G
J10 G9 J8 G7 J6 G5 J4 G3 J2 G1
F10 D9 F8 D7 F6 D5 F4 D3 F2 D1
C10 A9 C8 A7 C6 A5 C4 A3 C2 A1
MolexImpact76170-3020
DGND
IMPACT assignment recommendations: http://www.molex.com/pdm_docs/as/AS-76060-990.pdf
Title
Backplane Master Sheet
Size
Letter
Number
Revision
3-2014-01-07
Date: 1/9/2014
Sheet 39 of 59
File: C:\Users\..\Backplane_master.SchDoc
Drawn By: Graeme Smecher

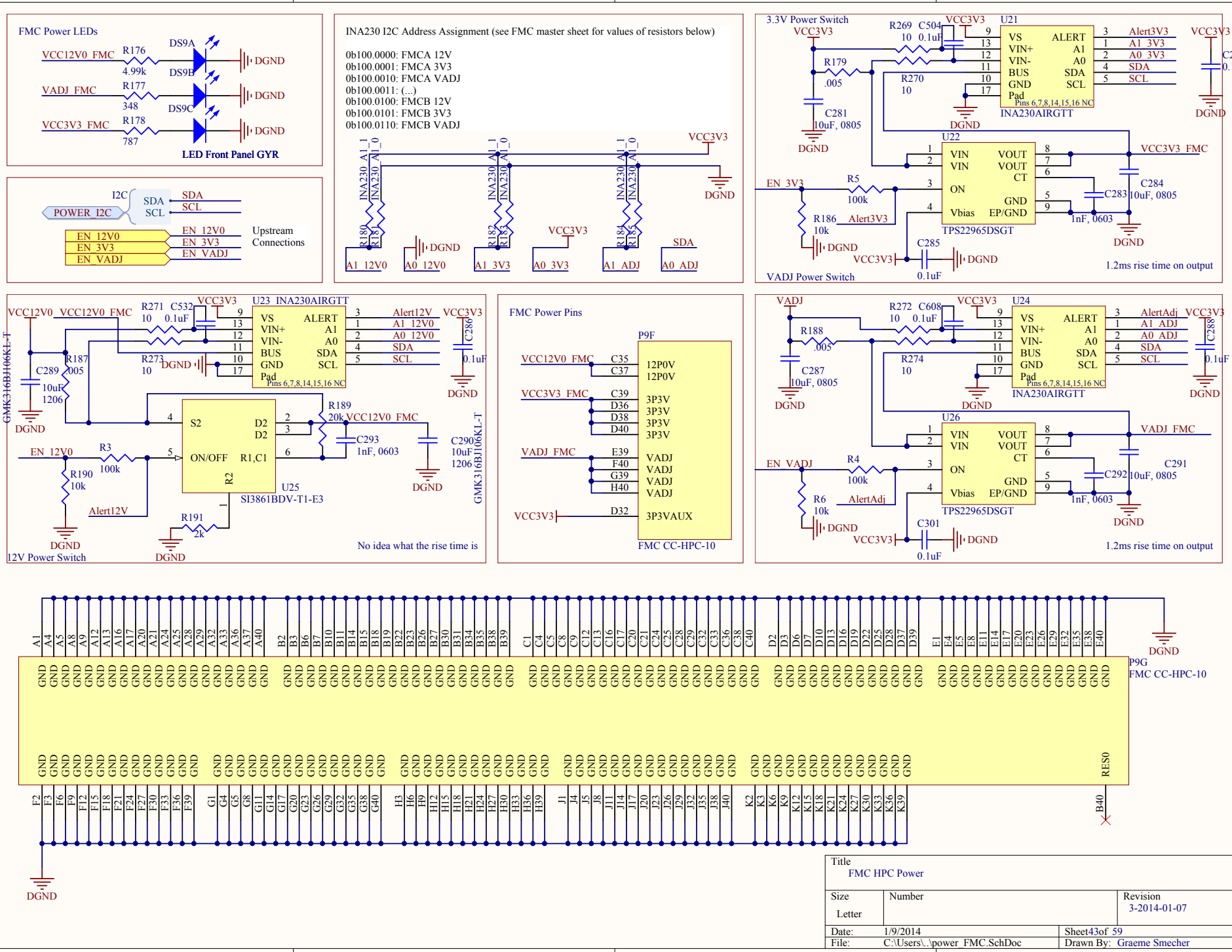


Title: QSFP+ Ports		
Size: Letter	Number:	Revision: 3-2014-01-07
Date: 1/9/2014		Sheet 40 of 59
File: C:\Users\..\QSFP.SchDoc		Drawn By: Graeme Smecher

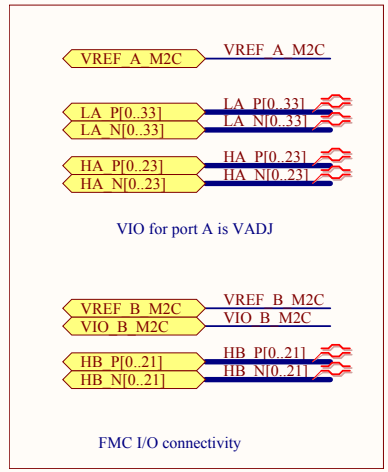


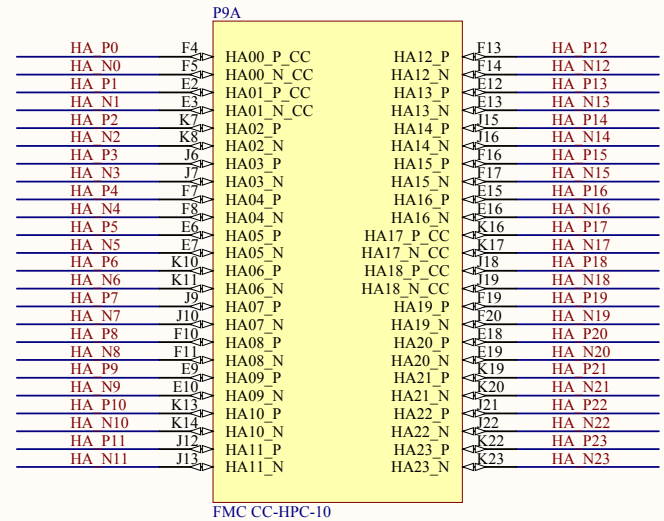
Title: SFP+ Port		
Size: Letter	Number:	Revision: 3-2014-01-07
Date: 1/9/2014		Sheet 41 of 59
File: C:\Users\..\SFP.SchDoc		Drawn By: Graeme Smecher

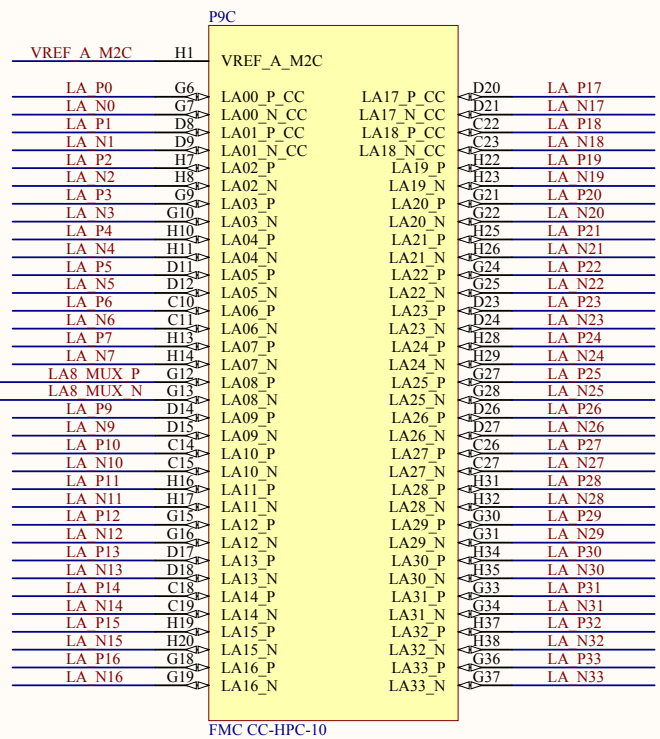

FMC_SLOT
power_FMC.SchDoc; FMC_io.SchDoc; FMC_misc.SchDoc
POWER_I2C
I2C_POWER
On FMC_power page
EN_12V0
EN_3V3
EN_VADJ
FMC_FPGA
VREF_A_M2C
LA_P[0..33]
LA_N[0..33]
HA_P[0..23]
HA_N[0..23]
VREF_B_M2C
VIO_B_M2C
HB_N[0..21]
HB_P[0..21]
CLK0_M2C_N
CLK0_M2C_P
CLK1_M2C_N
CLK1_M2C_P
CLK2_BIDIR_P
CLK2_BIDIR_N
CLK3_BIDIR_P
CLK3_BIDIR_N
FPGA
On FMC_IO page
On FMC_misc page
GTX
CLK_RAW_P
CLK_RAW_N
TMS
TCK
TDO
TDI
TRST
CHIME_SYNC_P
CHIME_SYNC_N
I2C
GA[0..1]
GA0
GA1
DGND
PG_M2C
PG_C2M
PRSNT_M2C_L
CLK_DIR
FMC_AUX
AUX
INA230_A1_1: MASTER_INA230_A1_1
INA230_A1_0: MASTER_INA230_A1_0
Title
FMC HPC Master Sheet
Size
Letter
Number
Revision
3-2014-01-07
Date: 1/9/2014
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File: C:\Users\..\FMC_master.SchDoc
Drawn By: Graeme Smecher

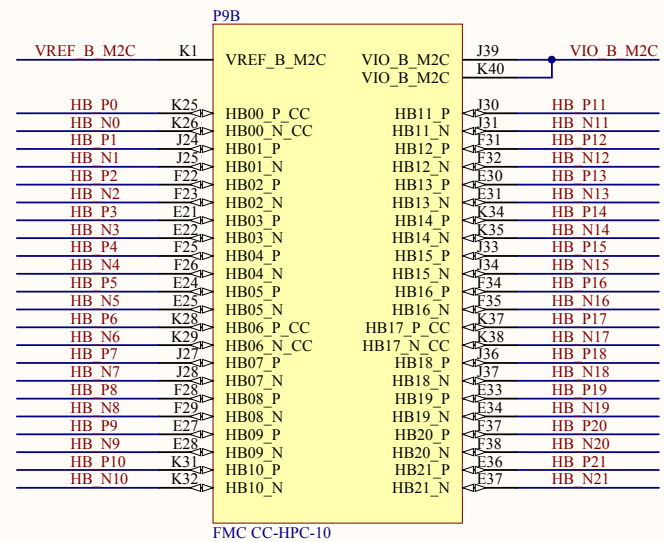

FMC Power LEDs
VCC12V0_FMC
R176
4.99k
DS9A
DS9B
DGND
VADJ_FMC
R177
348
DS9C
DGND
VCC3V3_FMC
R178
787
DGND
LED Front Panel GYR
I2C
POWER_I2C
SDA
SCL
EN_12V0
EN_3V3
EN_VADJ
Upstream Connections
INA230 I2C Address Assignment (see FMC master sheet for values of resistors below)
0b100.0000: FMCA 12V
0b100.0001: FMCA 3V3
0b100.0010: FMCA VADJ
0b100.0011: (...)
0b100.0100: FMCB 12V
0b100.0101: FMCB 3V3
0b100.0110: FMCB VADJ
INA230_A1_1
INA230_A1_0
R180
R181
R182
R183
R184
R185
VCC3V3
DGND
A1_12V0
A0_12V0
A1_3V3
A0_3V3
A1_ADJ
A0_ADJ
SDA
3.3V Power Switch
VCC3V3
R269
10
C504
0.1uF
R179
.005
C281
10uF, 0805
R270
10
U21
INA230AIRGTT
Alert3V3
A1_3V3
A0_3V3
SDA
SCL
C282
0.1uF
U22
TPS22965DSGT
EN_3V3
R5
100k
R186
10k
Alert3V3
C285
0.1uF
C283
1nF, 0603
C284
10uF, 0805
VCC3V3_FMC
1.2ms rise time on output
VADJ Power Switch
VCC12V0_VCC12V0_FMC
R271
10
C532
0.1uF
U23_INA230AIRGTT
Alert12V
A1_12V0
A0_12V0
SDA
SCL
C286
0.1uF
GMK316BJ106KL-T
C289
10uF
1206
R187
.005
R273
10
R189
20k
C293
1nF, 0603
C290
10uF
1206
S2
D2
ON/OFF
R1,C1
R2
U25
SI3861BDV-T1-E3
EN_12V0
R3
100k
R190
10k
R191
2k
Alert12V
No idea what the rise time is
12V Power Switch
FMC Power Pins
P9F
VCC12V0_FMC
C35
C37
12P0V
VCC3V3_FMC
C39
D36
D38
D40
3P3V
VADJ_FMC
E39
F40
G39
H40
VADJ
D32
3P3VAUX
FMC CC-HPC-10
VADJ
R188
.005
C287
10uF, 0805
R272
10
C608
0.1uF
U24
INA230AIRGTT
AlertAdj
A1_ADJ
A0_ADJ
SDA
SCL
C288
0.1uF
U26
TPS22965DSGT
EN_VADJ
R4
100k
R6
10k
AlertAdj
C301
0.1uF
C292
1nF, 0603
C291
10uF, 0805
VADJ_FMC
1.2ms rise time on output
P9G
FMC CC-HPC-10
RES0
B40
DGND
Title
FMC HPC Power
Size
Letter
Number
Revision
3-2014-01-07
Date:
1/9/2014
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File:
C:\Users\..\power_FMC.SchDoc
Drawn By:
Graeme Smecher

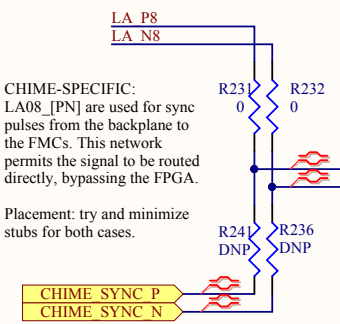
FMC ERRATA: HA18 is also clock-capable, and should be labeled as such









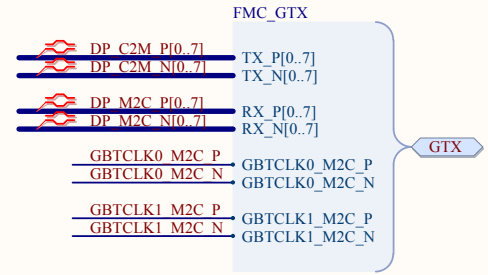


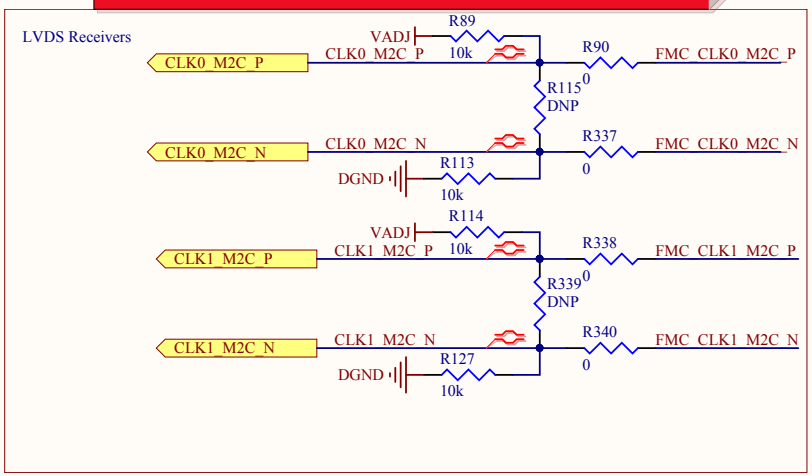
CHIME-SPECIFIC:
LA08_[PN] are used for sync pulses from the backplane to the FMCs. This network permits the signal to be routed directly, bypassing the FPGA.

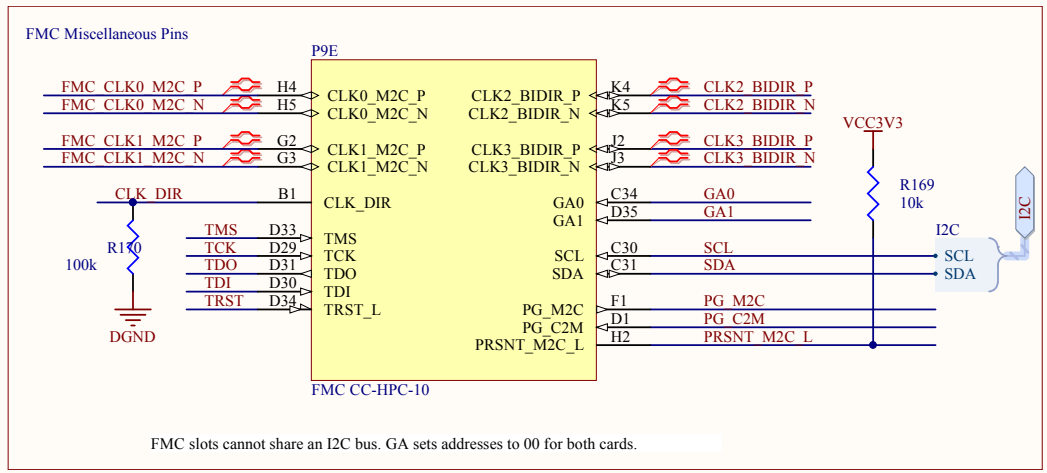
Placement: try and minimize stubs for both cases.

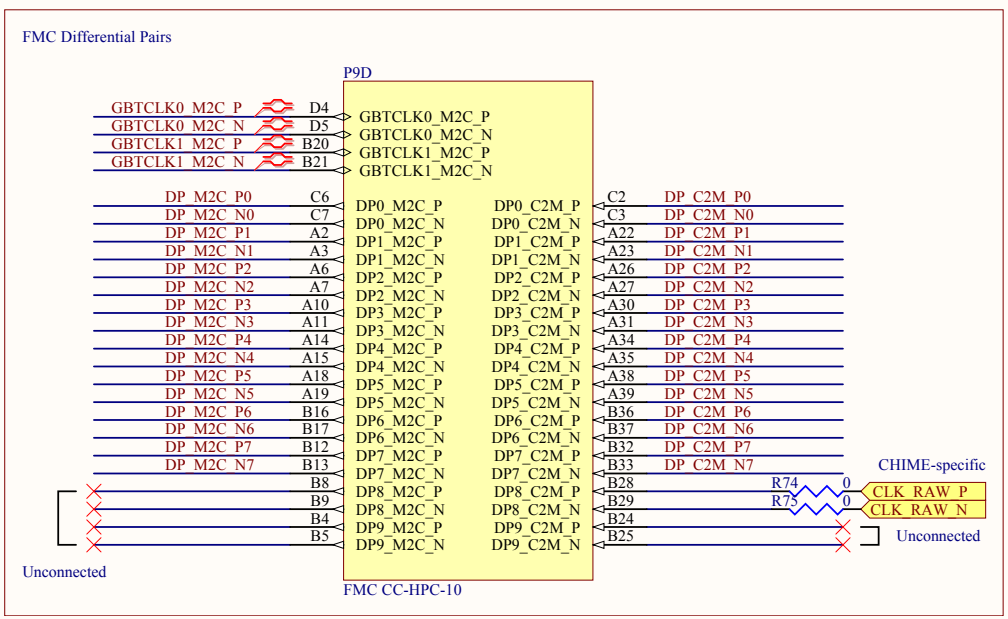
Title: FMC HPC I/O Pins		
Size: Letter	Number:	Revision: 3-2014-01-07
Date: 1/9/2014		Sheet 44 of 59
File: C:\Users\..\FMC_io.SchDoc		Drawn By: Graeme Smecher

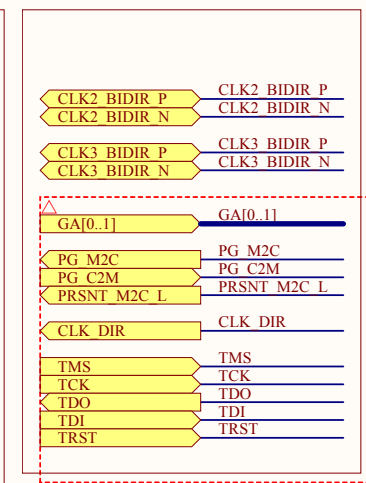
Are we okay with this buffering?
Not quite clock passive coupling network but close. Didn't want to use clock passive coupling because want DC coupling by default

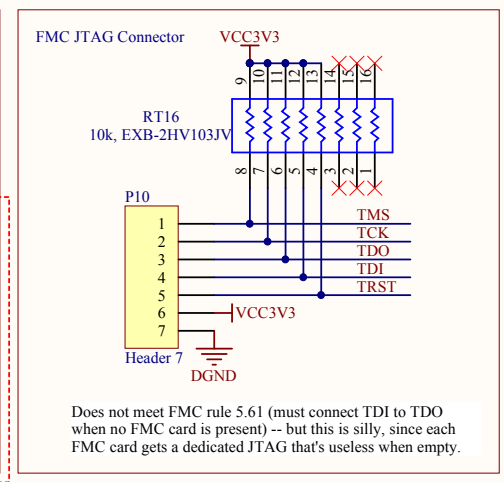




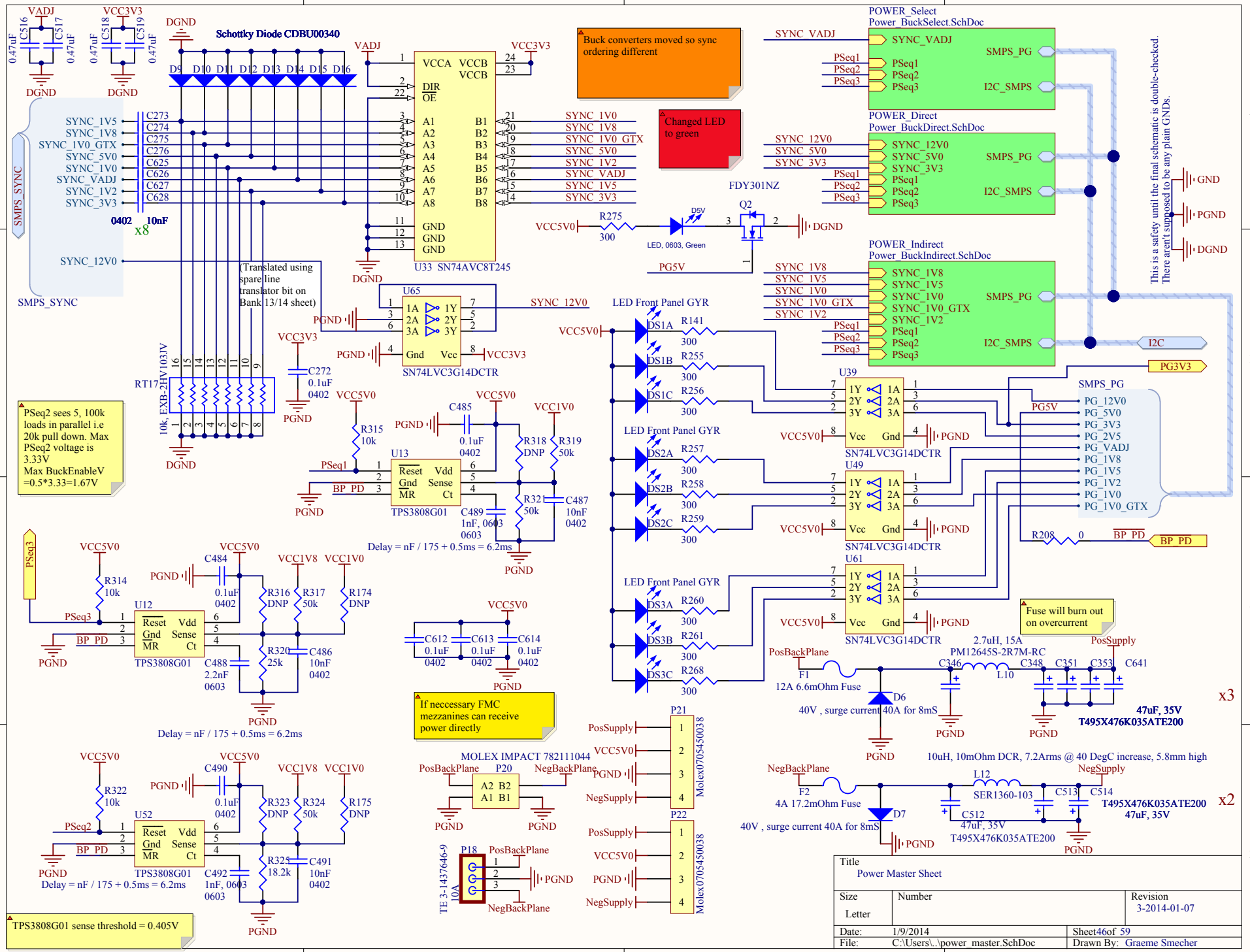


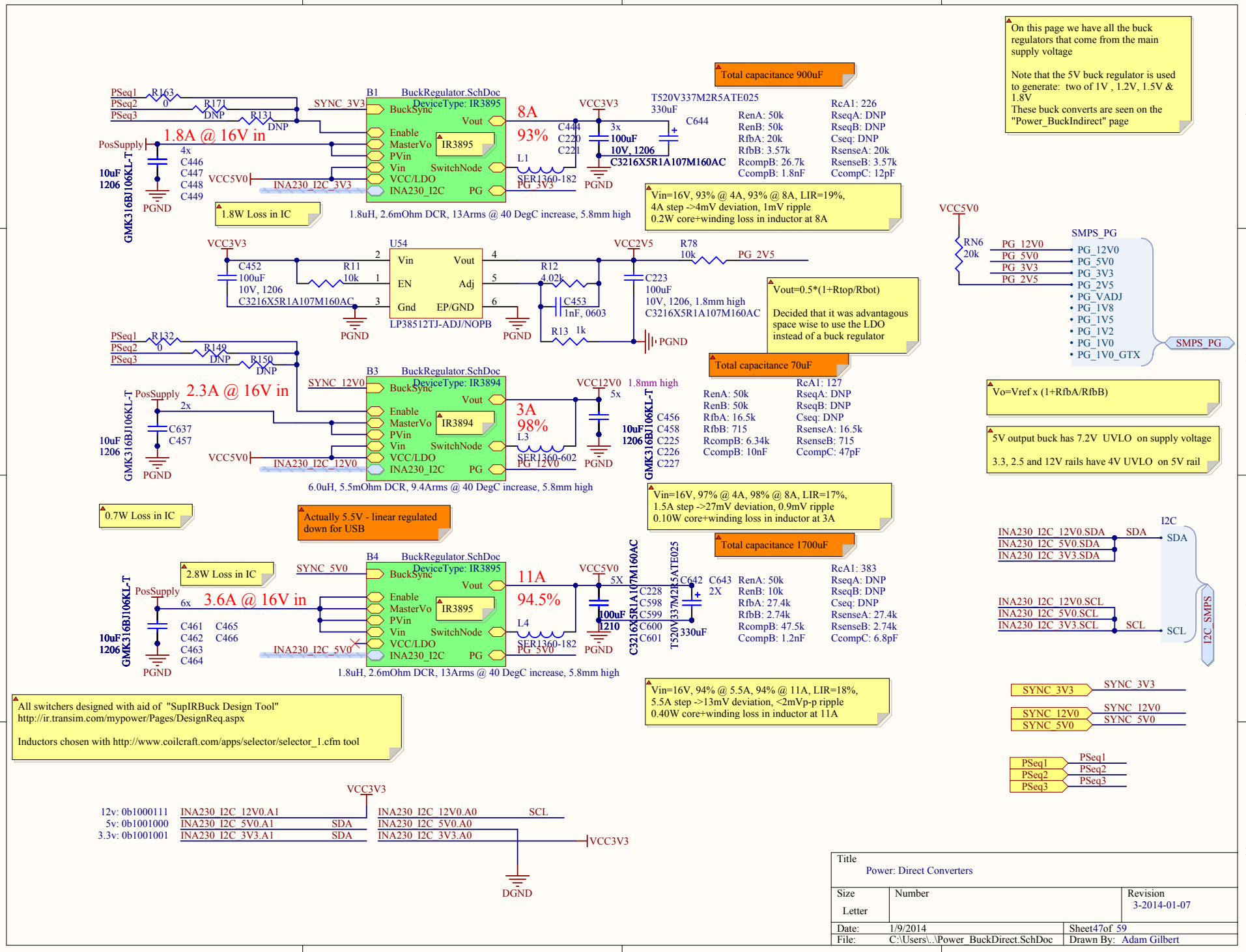


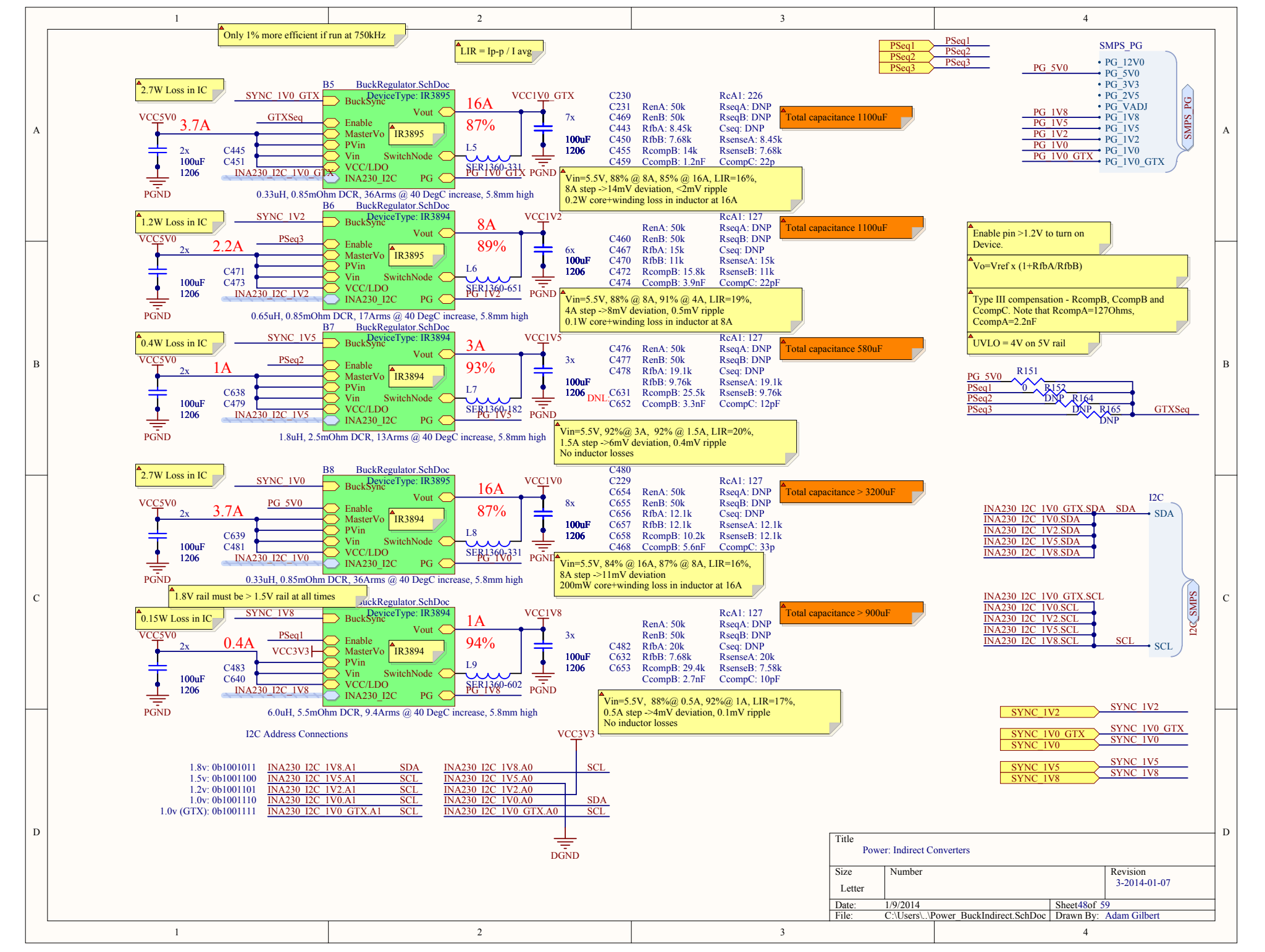


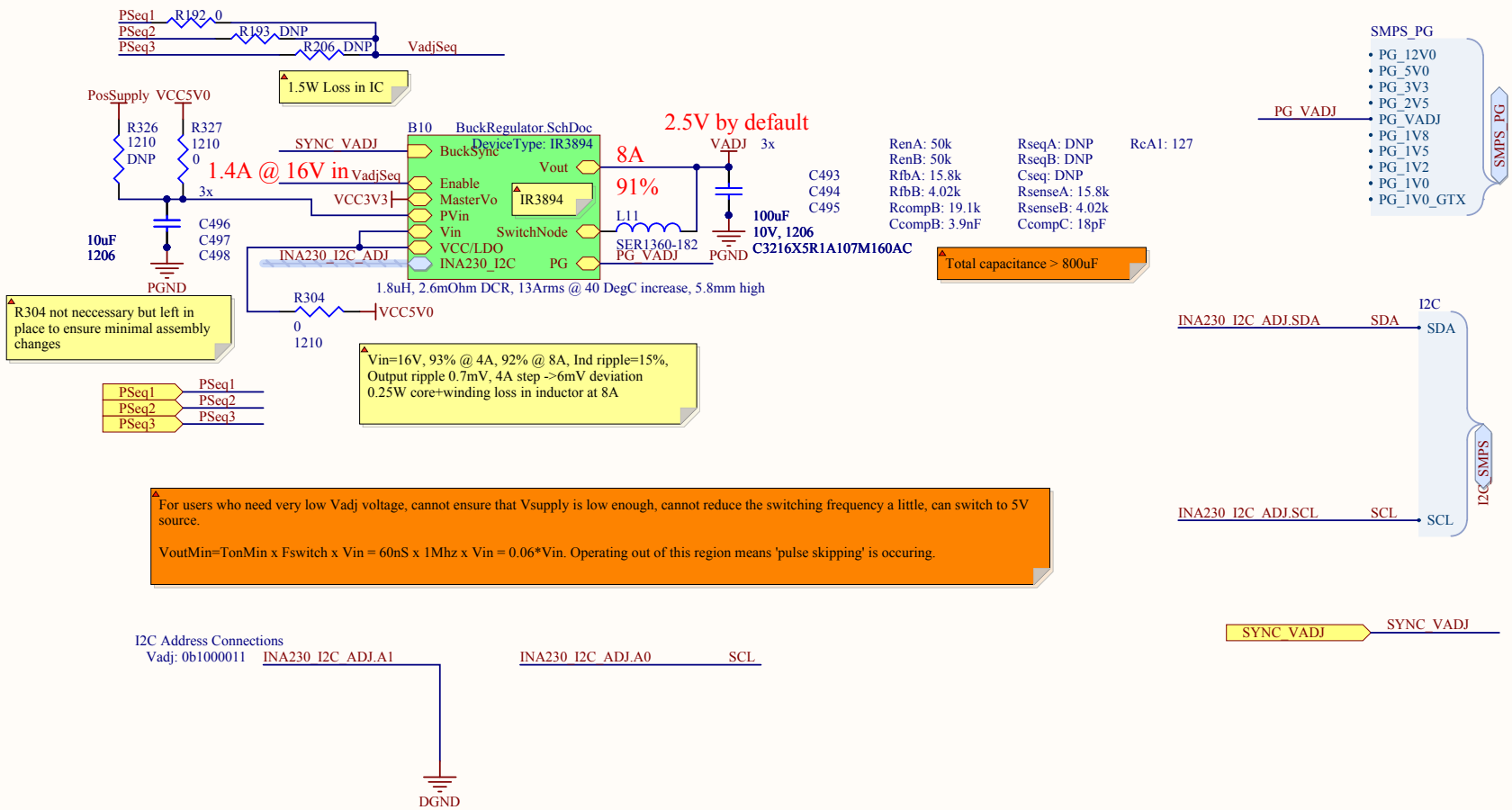


Title: FMC Miscellaneous Pins		
Size: Letter	Number:	Revision: 3-2014-01-07
Date: 1/9/2014		Sheet45of 59
File: C:\Users\..\FMC_misc.SchDoc		Drawn By: Graeme Smecher

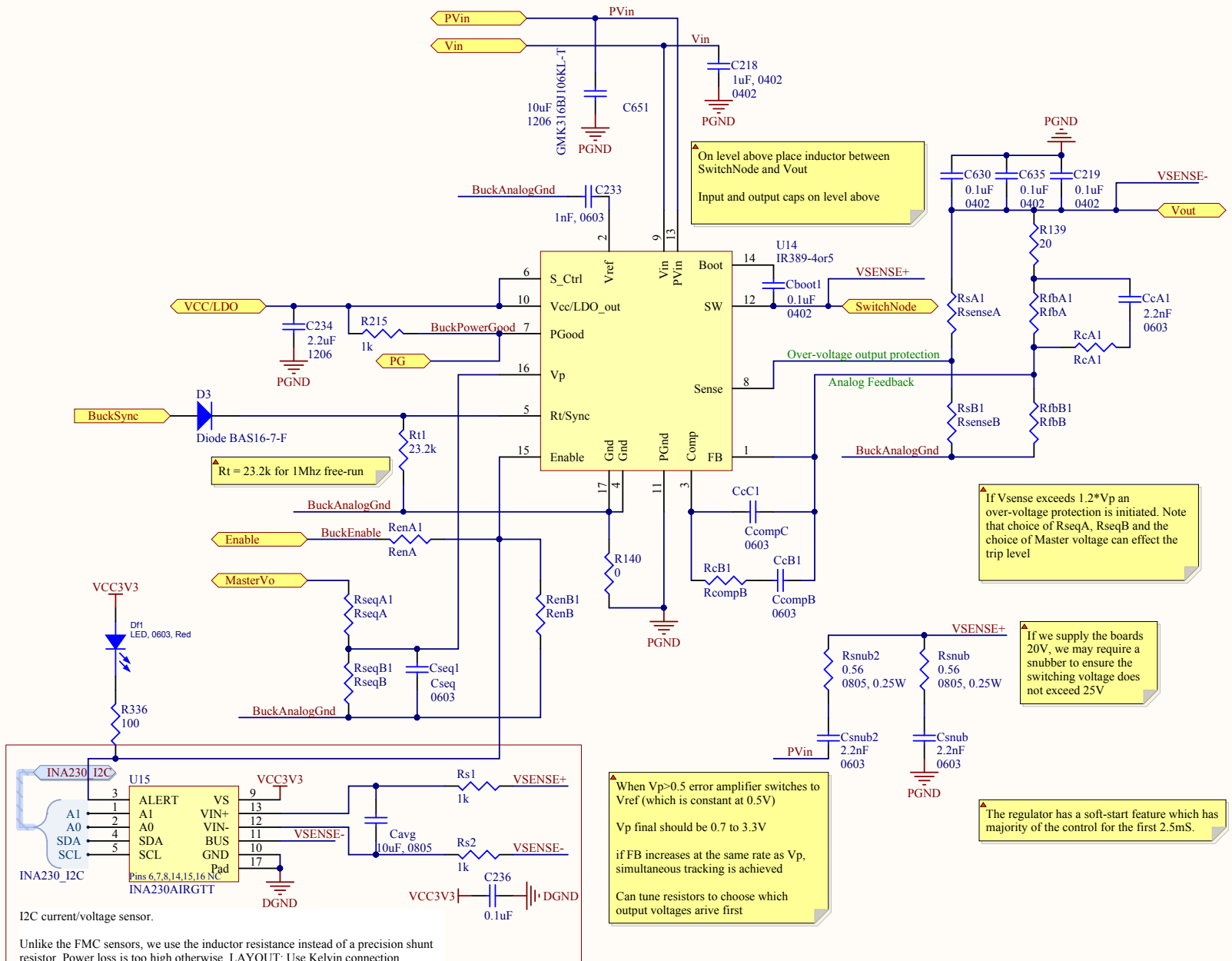

Schottky Diode CDBU00340
Buck converters moved so sync ordering different
Changed LED to green
POWER_Select
Power_BuckSelect.SchDoc
POWER_Direct
Power_BuckDirect.SchDoc
POWER_Indirect
Power_BuckIndirect.SchDoc
FDY301NZ
U33 SN74AVC8T245
U65
SN74LVC3G14DCTR
U13
TPS3808G01
U12
U52
U39
U49
U61
LED Front Panel GYR
(Translated using spare line translator bit on Bank 13/14 sheet)
PSeq2 sees 5, 100k loads in parallel i.e 20k pull down. Max PSeq2 voltage is 3.33V
Max BuckEnableV =0.5*3.33=1.67V
Delay = nF / 175 + 0.5ms = 6.2ms
If neccessary FMC mezzanines can receive power directly
MOLEX IMPACT 782111044
Fuse will burn out on overcurrent
2.7uH, 15A PM12645S-2R7M-RC
10uH, 10mOhm DCR, 7.2Arms @ 40 DegC increase, 5.8mm high
SER1360-103
47uF, 35V T495X476K035ATE200
TPS3808G01 sense threshold = 0.405V
This is a safety until the final schematic is double-checked. There aren't supposed to be any plain GNDs.
Title
Power Master Sheet
Size Letter
Number
Revision 3-2014-01-07
Date: 1/9/2014
Sheet46of 59
File: C:\Users\..\power_master.SchDoc
Drawn By: Graeme Smecher


On this page we have all the buck regulators that come from the main supply voltage
Note that the 5V buck regulator is used to generate: two of 1V , 1.2V, 1.5V & 1.8V
These buck converts are seen on the "Power_BuckIndirect" page
B1 BuckRegulator.SchDoc
DeviceType: IR3895
1.8A @ 16V in
8A
93%
1.8W Loss in IC
1.8uH, 2.6mOhm DCR, 13Arms @ 40 DegC increase, 5.8mm high
Total capacitance 900uF
Vin=16V, 93% @ 4A, 93% @ 8A, LIR=19%,
4A step ->4mV deviation, 1mV ripple
0.2W core+winding loss in inductor at 8A
U54
LP38512TJ-ADJ/NOPB
Vout=0.5*(1+Rtop/Rbot)
Decided that it was advantagous space wise to use the LDO instead of a buck regulator
Total capacitance 70uF
B3 BuckRegulator.SchDoc
DeviceType: IR3894
2.3A @ 16V in
3A
98%
6.0uH, 5.5mOhm DCR, 9.4Arms @ 40 DegC increase, 5.8mm high
0.7W Loss in IC
Actually 5.5V - linear regulated down for USB
Vin=16V, 97% @ 4A, 98% @ 8A, LIR=17%,
1.5A step ->27mV deviation, 0.9mV ripple
0.10W core+winding loss in inductor at 3A
Total capacitance 1700uF
B4 BuckRegulator.SchDoc
DeviceType: IR3895
2.8W Loss in IC
3.6A @ 16V in
11A
94.5%
1.8uH, 2.6mOhm DCR, 13Arms @ 40 DegC increase, 5.8mm high
Vin=16V, 94% @ 5.5A, 94% @ 11A, LIR=18%,
5.5A step ->13mV deviation, <2mVp-p ripple
0.40W core+winding loss in inductor at 11A
All switchers designed with aid of "SupIRBuck Design Tool"
http://ir.transim.com/mypower/Pages/DesignReq.aspx
Inductors chosen with http://www.coilcraft.com/apps/selector/selector_1.cfm tool
Vo=Vref x (1+RfbA/RfbB)
5V output buck has 7.2V UVLO on supply voltage
3.3, 2.5 and 12V rails have 4V UVLO on 5V rail
SMPS_PG
I2C
I2C_SMPS
12v: 0b1000111
5v: 0b1001000
3.3v: 0b1001001
DGND
Title
Power: Direct Converters
Size
Letter
Number
Revision
3-2014-01-07
Date: 1/9/2014
Sheet 47 of 59
File: C:\Users\..\Power_BuckDirect.SchDoc
Drawn By: Adam Gilbert

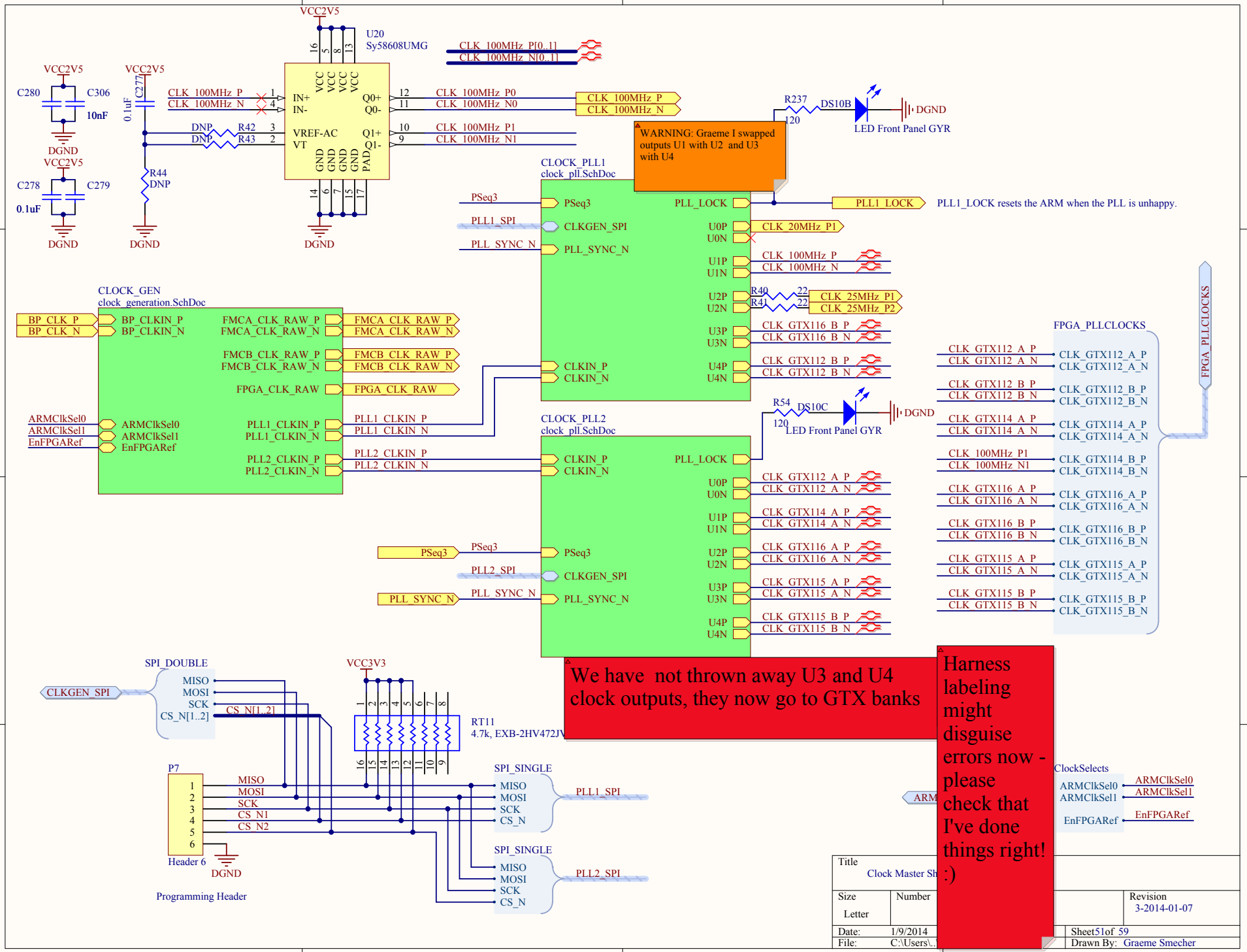

Only 1% more efficient if run at 750kHz
LIR = Ip-p / I avg
PSeq1
PSeq2
PSeq3
SMPS_PG
PG_12V0
PG_5V0
PG_3V3
PG_2V5
PG_VADJ
PG_1V8
PG_1V5
PG_1V2
PG_1V0
PG_1V0_GTX
2.7W Loss in IC
B5 BuckRegulator.SchDoc
DeviceType: IR3895
SYNC_1V0_GTX
VCC5V0
3.7A
GTXSeq
2x
100uF
1206
C445
C451
INA230_I2C_1V0_GTX
PGND
BuckSync
Vout
Enable
MasterVo
PVin
Vin
SwitchNode
VCC/LDO
INA230_I2C
PG
IR3895
16A
87%
L5
SER1360-331
VCC1V0_GTX
PG_1V0_GTX
PGND
0.33uH, 0.85mOhm DCR, 36Arms @ 40 DegC increase, 5.8mm high
C230
7x
100uF
1206
C231
C469
C443
C450
C455
C459
RenA: 50k
RenB: 50k
RfbA: 8.45k
RfbB: 7.68k
RcompB: 14k
CcompB: 1.2nF
RcA1: 226
RseqA: DNP
RseqB: DNP
Cseq: DNP
RsenseA: 8.45k
RsenseB: 7.68k
CcompC: 22p
Total capacitance 1100uF
Vin=5.5V, 88% @ 8A, 85% @ 16A, LIR=16%,
8A step ->14mV deviation, <2mV ripple
0.2W core+winding loss in inductor at 16A
1.2W Loss in IC
B6 BuckRegulator.SchDoc
DeviceType: IR3894
SYNC_1V2
VCC5V0
2.2A
PSeq3
2x
100uF
1206
C471
C473
INA230_I2C_1V2
PGND
IR3895
8A
89%
L6
SER1360-651
VCC1V2
PG_1V2
PGND
0.65uH, 0.85mOhm DCR, 17Arms @ 40 DegC increase, 5.8mm high
C460
6x
100uF
1206
C467
C470
C472
C474
RenA: 50k
RenB: 50k
RfbA: 15k
RfbB: 11k
RcompB: 15.8k
CcompB: 3.9nF
RcA1: 127
RseqA: DNP
RseqB: DNP
Cseq: DNP
RsenseA: 15k
RsenseB: 11k
CcompC: 22pF
Total capacitance 1100uF
Vin=5.5V, 88% @ 8A, 91% @ 4A, LIR=19%,
4A step ->8mV deviation, 0.5mV ripple
0.1W core+winding loss in inductor at 8A
Enable pin >1.2V to turn on Device.
Vo=Vref x (1+RfbA/RfbB)
Type III compensation - RcompB, CcompB and CcompC. Note that RcompA=127Ohms, CcompA=2.2nF
UVLO = 4V on 5V rail
0.4W Loss in IC
B7 BuckRegulator.SchDoc
DeviceType: IR3894
SYNC_1V5
VCC5V0
1A
PSeq2
2x
100uF
1206
C638
C479
INA230_I2C_1V5
PGND
IR3894
3A
93%
L7
SER1360-182
VCC1V5
PG_1V5
PGND
1.8uH, 2.5mOhm DCR, 13Arms @ 40 DegC increase, 5.8mm high
C476
3x
100uF
1206
C477
C478
C631
C652
DNL
RenA: 50k
RenB: 50k
RfbA: 19.1k
RfbB: 9.76k
RcompB: 25.5k
CcompB: 3.3nF
RcA1: 127
RseqA: DNP
RseqB: DNP
Cseq: DNP
RsenseA: 19.1k
RsenseB: 9.76k
CcompC: 12pF
Total capacitance 580uF
Vin=5.5V, 92%@ 3A, 92% @ 1.5A, LIR=20%,
1.5A step ->6mV deviation, 0.4mV ripple
No inductor losses
PG_5V0
PSeq1
PSeq2
PSeq3
R151
0
R152
DNP
R164
DNP
R165
DNP
GTXSeq
2.7W Loss in IC
B8 BuckRegulator.SchDoc
DeviceType: IR3895
SYNC_1V0
VCC5V0
3.7A
PG_5V0
2x
100uF
1206
C639
C481
INA230_I2C_1V0
PGND
IR3894
16A
87%
L8
SER1360-331
VCC1V0
PG_1V0
PGND
0.33uH, 0.85mOhm DCR, 36Arms @ 40 DegC increase, 5.8mm high
C480
C229
8x
100uF
1206
C654
C655
C656
C657
C658
C468
RenA: 50k
RenB: 50k
RfbA: 12.1k
RfbB: 12.1k
RcompB: 10.2k
CcompB: 5.6nF
RcA1: 127
RseqA: DNP
RseqB: DNP
Cseq: DNP
RsenseA: 12.1k
RsenseB: 12.1k
CcompC: 33p
Total capacitance > 3200uF
Vin=5.5V, 84% @ 16A, 87% @ 8A, LIR=16%,
8A step ->11mV deviation
200mW core+winding loss in inductor at 16A
1.8V rail must be > 1.5V rail at all times
0.15W Loss in IC
B9 BuckRegulator.SchDoc
DeviceType: IR3894
SYNC_1V8
VCC5V0
0.4A
PSeq1
VCC3V3
2x
100uF
1206
C483
C640
INA230_I2C_1V8
PGND
IR3894
1A
94%
L9
SER1360-602
VCC1V8
PG_1V8
PGND
6.0uH, 5.5mOhm DCR, 9.4Arms @ 40 DegC increase, 5.8mm high
3x
100uF
1206
C482
C632
C653
RenA: 50k
RenB: 50k
RfbA: 20k
RfbB: 7.68k
RcompB: 29.4k
CcompB: 2.7nF
RcA1: 127
RseqA: DNP
RseqB: DNP
Cseq: DNP
RsenseA: 20k
RsenseB: 7.58k
CcompC: 10pF
Total capacitance > 900uF
Vin=5.5V, 88%@ 0.5A, 92%@ 1A, LIR=17%,
0.5A step ->4mV deviation, 0.1mV ripple
No inductor losses
I2C
INA230_I2C_1V0_GTX.SDA
INA230_I2C_1V0.SDA
INA230_I2C_1V2.SDA
INA230_I2C_1V5.SDA
INA230_I2C_1V8.SDA
SDA
INA230_I2C_1V0_GTX.SCL
INA230_I2C_1V0.SCL
INA230_I2C_1V2.SCL
INA230_I2C_1V5.SCL
INA230_I2C_1V8.SCL
SCL
I2C_SMPS
SYNC_1V2
SYNC_1V0_GTX
SYNC_1V0
SYNC_1V5
SYNC_1V8
I2C Address Connections
1.8v: 0b1001011
1.5v: 0b1001100
1.2v: 0b1001101
1.0v: 0b1001110
1.0v (GTX): 0b1001111
INA230_I2C_1V8.A1
INA230_I2C_1V5.A1
INA230_I2C_1V2.A1
INA230_I2C_1V0.A1
INA230_I2C_1V0_GTX.A1
INA230_I2C_1V8.A0
INA230_I2C_1V5.A0
INA230_I2C_1V2.A0
INA230_I2C_1V0.A0
INA230_I2C_1V0_GTX.A0
VCC3V3
DGND
Title
Power: Indirect Converters
Size
Letter
Number
Revision
3-2014-01-07
Date:
1/9/2014
Sheet48of 59
File:
C:\Users\..\Power_BuckIndirect.SchDoc
Drawn By:
Adam Gilbert



Title: Selectable Buck Regulators		
Size: Letter	Number:	Revision: 3-2014-01-07
Date: 1/9/2014		Sheet 49 of 59
File: C:\Users\..\Power_BuckSelect.SchDoc		Drawn By: Adam Gilbert



Title: Power: Buck Converter		
Size: Letter	Number	Revision: 3-2014-01-07
Date: 1/9/2014		Sheet 50 of 59
File: C:\Users\..\BuckRegulator.SchDoc		Drawn By: Adam Gilbert


VCC2V5
U20
Sy58608UMG
CLK_100MHz_P[0..1]
CLK_100MHz_N[0..1]
C280
C306
10nF
C277
0.1uF
C278
C279
0.1uF
DGND
CLK_100MHz_P
CLK_100MHz_N
IN+
IN-
VREF-AC
VT
Q0+
Q0-
Q1+
Q1-
CLK_100MHz_P0
CLK_100MHz_N0
CLK_100MHz_P1
CLK_100MHz_N1
DNP
R42
R43
R44
R237
120
DS10B
LED Front Panel GYR
WARNING: Graeme I swapped outputs U1 with U2 and U3 with U4
CLOCK_PLL1
clock_pll.SchDoc
PSeq3
PLL1_SPI
PLL_SYNC_N
PLL_LOCK
CLKGEN_SPI
U0P
U0N
U1P
U1N
U2P
U2N
U3P
U3N
U4P
U4N
CLK_20MHz_P1
CLK_100MHz_P
CLK_100MHz_N
R40
R41
22
CLK_25MHz_P1
CLK_25MHz_P2
CLK_GTX116_B_P
CLK_GTX116_B_N
CLK_GTX112_B_P
CLK_GTX112_B_N
CLKIN_P
CLKIN_N
PLL1_LOCK resets the ARM when the PLL is unhappy.
CLOCK_GEN
clock_generation.SchDoc
BP_CLK_P
BP_CLK_N
BP_CLKIN_P
BP_CLKIN_N
FMCA_CLK_RAW_P
FMCA_CLK_RAW_N
FMCB_CLK_RAW_P
FMCB_CLK_RAW_N
FPGA_CLK_RAW
ARMClkSel0
ARMClkSel1
EnFPGARef
PLL1_CLKIN_P
PLL1_CLKIN_N
PLL2_CLKIN_P
PLL2_CLKIN_N
R54
DS10C
CLOCK_PLL2
clock_pll.SchDoc
CLK_GTX112_A_P
CLK_GTX112_A_N
CLK_GTX114_A_P
CLK_GTX114_A_N
CLK_GTX116_A_P
CLK_GTX116_A_N
CLK_GTX115_A_P
CLK_GTX115_A_N
CLK_GTX115_B_P
CLK_GTX115_B_N
FPGA_PLLCLOCKS
CLK_100MHz_P1
CLK_100MHz_N1
SPI_DOUBLE
MISO
MOSI
SCK
CS_N[1..2]
VCC3V3
RT11
4.7k, EXB-2HV472JV
We have not thrown away U3 and U4 clock outputs, they now go to GTX banks
Harness labeling might disguise errors now - please check that I've done things right! :)
P7
Header 6
Programming Header
SPI_SINGLE
CS_N1
CS_N2
PLL1_SPI
PLL2_SPI
ClockSelects
Title
Clock Master Sh
Size
Letter
Number
Revision
3-2014-01-07
Date:
1/9/2014
File:
C:\Users\..
Sheet51of 59
Drawn By:
Graeme Smecher

Backplane clock replication

BP clockin can be LVDS or LVPECL, right now configured LVDS, just add 50Ohm to ground then LVPECL see Fig3 P4

All differential outputs LVDS

All differential outputs LVDS

WARNING: Graeme, I switched PLL2 and PLL1 clock

100 differential termination with common mode in right place
See Fig2 P3
http://www.ti.com/lit/an/scaa062/scaa062.pdf

Crystal select, SMA select, Backplane select

Local 20MHz oscillator and SMA multiplexer

Providing ClckSe1 is low, if ClkSel0 low backplane selected, if high sma selected

If ClkSel1 high, Oscilator will run and fanout output will be oscillator output

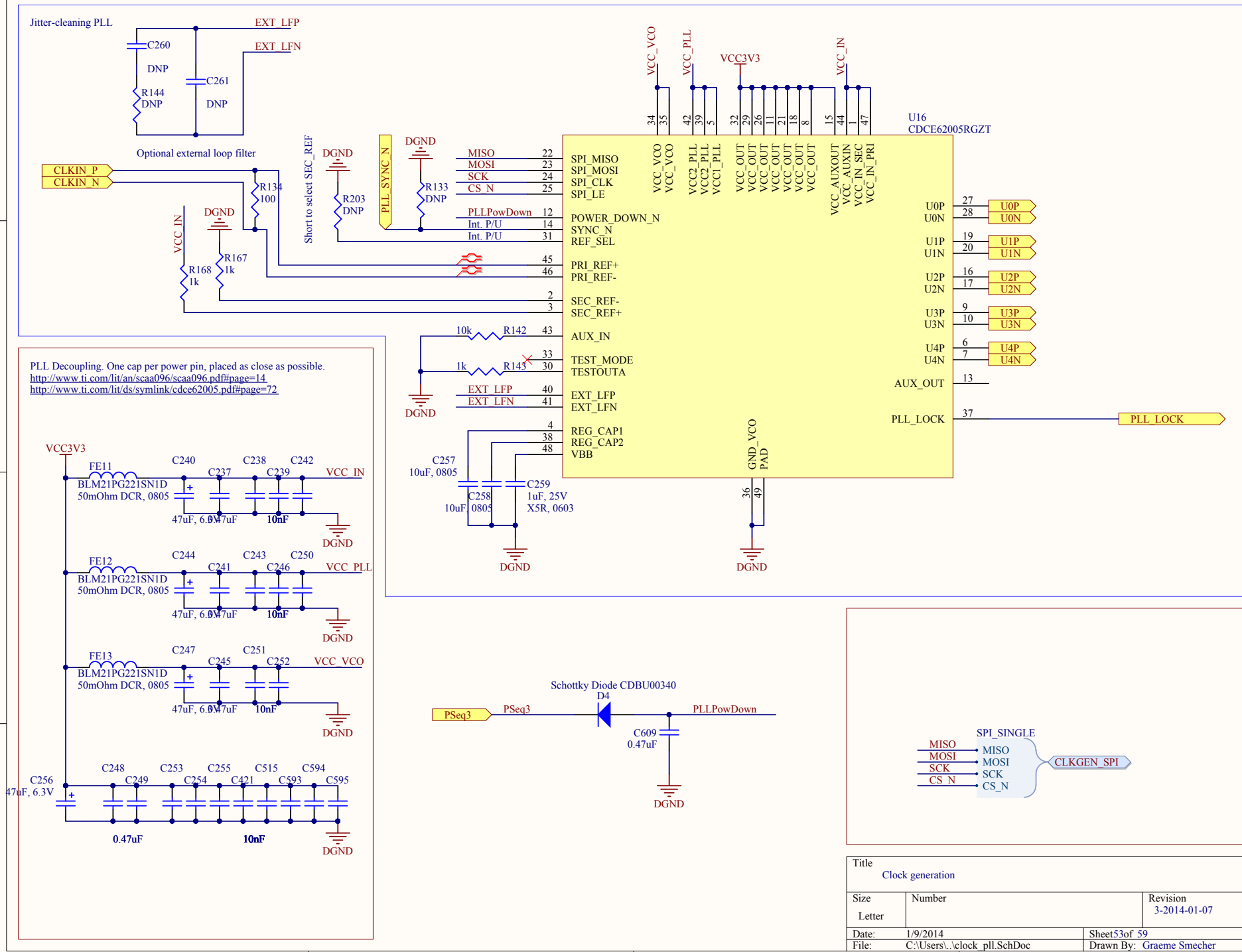
U17 LMK00304SQ

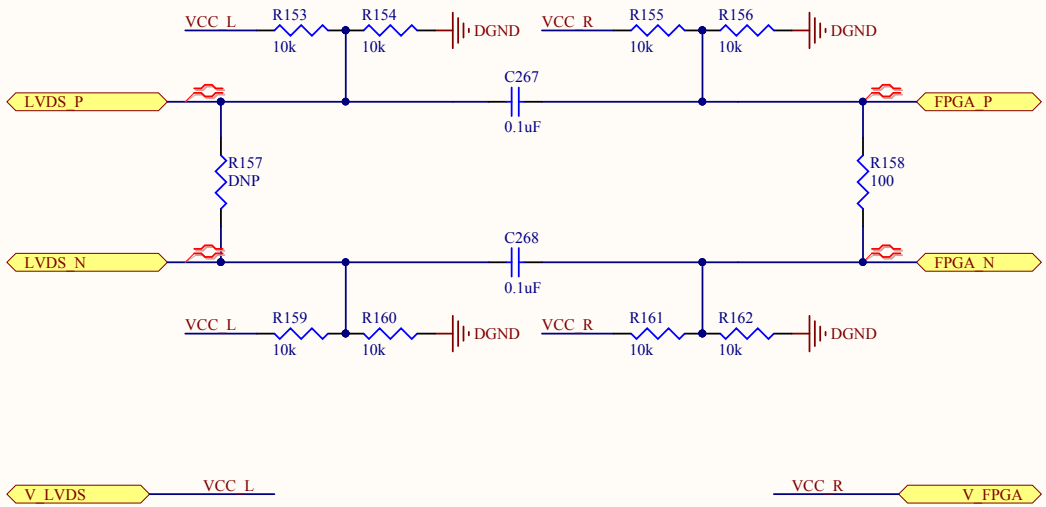
U31A MC100EPT22DTG LVPECL

U31B MC100EPT22DTG

X1 ASFL1-10.000MHZ-EK-T

Title: Clock Generation		
Size: Letter	Number:	Revision: 3-2014-01-07
Date: 1/9/2014		Sheet52of 59
File: C:\Users\..\clock_generation.SchDoc		Drawn By: Graeme Smecher

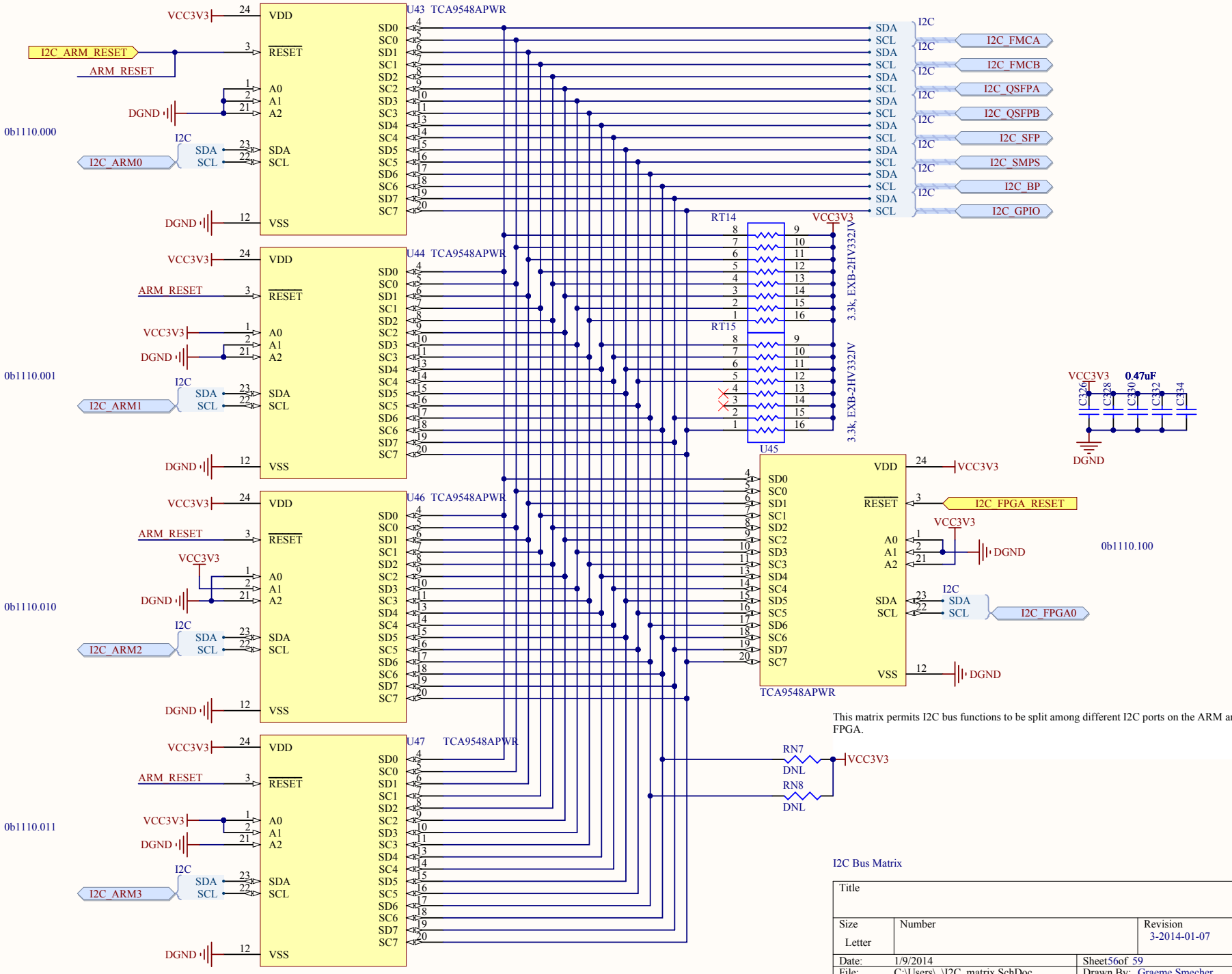

Jitter-cleaning PLL
EXT_LFP
EXT_LFN
C260
DNP
C261
DNP
R144
DNP
Optional external loop filter
CLKIN_P
CLKIN_N
R134
100
Short to select SEC_REF
DGND
R203
DNP
PLL_SYNC_N
R133
DNP
VCC_IN
R167
1k
R168
1k
VCC_VCO
VCC_PLL
VCC3V3
VCC_IN
U16
CDCE62005RGZT
MISO 22
MOSI 23
SCK 24
CS_N 25
SPI_MISO
SPI_MOSI
SPI_CLK
SPI_LE
PLLPowDown 12
Int. P/U 14
Int. P/U 31
POWER_DOWN_N
SYNC_N
REF_SEL
45
46
PRI_REF+
PRI_REF-
2
3
SEC_REF-
SEC_REF+
10k R142 43
AUX_IN
1k R143 33 30
TEST_MODE
TESTOUTA
EXT_LFP 40
EXT_LFN 41
EXT_LFP
EXT_LFN
4
38
48
REG_CAP1
REG_CAP2
VBB
C257
10uF, 0805
C258
10uF, 0805
C259
1uF, 25V
X5R, 0603
GND_VCO PAD
36
49
U0P 27
U0N 28
U1P 19
U1N 20
U2P 16
U2N 17
U3P 9
U3N 10
U4P 6
U4N 7
AUX_OUT 13
PLL_LOCK 37
PLL_LOCK
PLL Decoupling. One cap per power pin, placed as close as possible.
http://www.ti.com/lit/an/scaa096/scaa096.pdf#page=14
http://www.ti.com/lit/ds/symlink/cdce62005.pdf#page=72
FE11
BLM21PG221SN1D
50mOhm DCR, 0805
C240
C237
C238
C239
C242
47uF, 6.3V
0.47uF
10nF
FE12
C244
C241
C243
C246
C250
FE13
C247
C245
C251
C252
C248
C249
C253
C254
C255
C421
C515
C593
C594
C256
47uF, 6.3V
C595
Schottky Diode CDBU00340
D4
PSeq3
C609
0.47uF
SPI_SINGLE
CLKGEN_SPI
Title
Clock generation
Size
Letter
Number
Revision
3-2014-01-07
Date: 1/9/2014
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File: C:\Users\..\clock_pll.SchDoc
Drawn By: Graeme Smecher

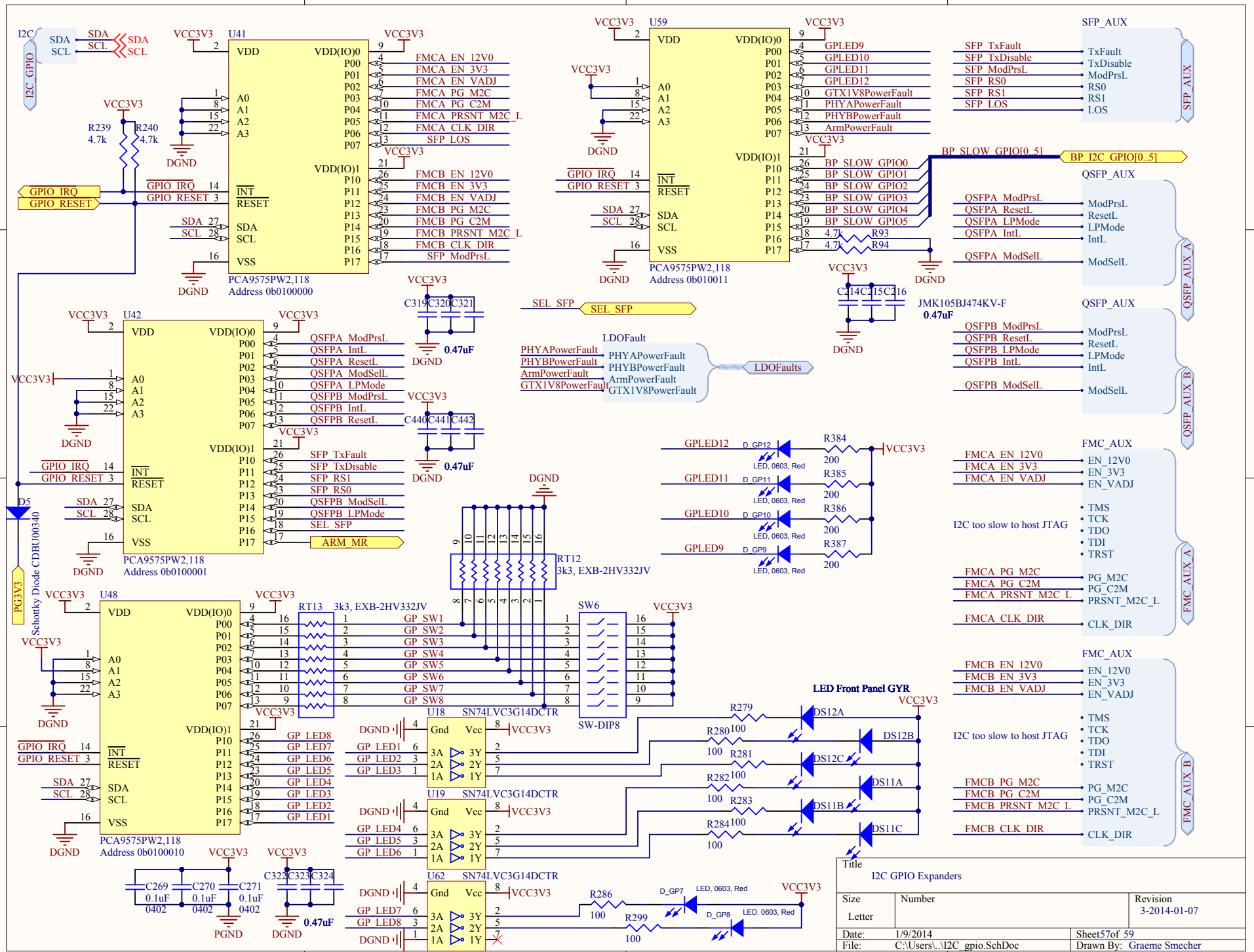


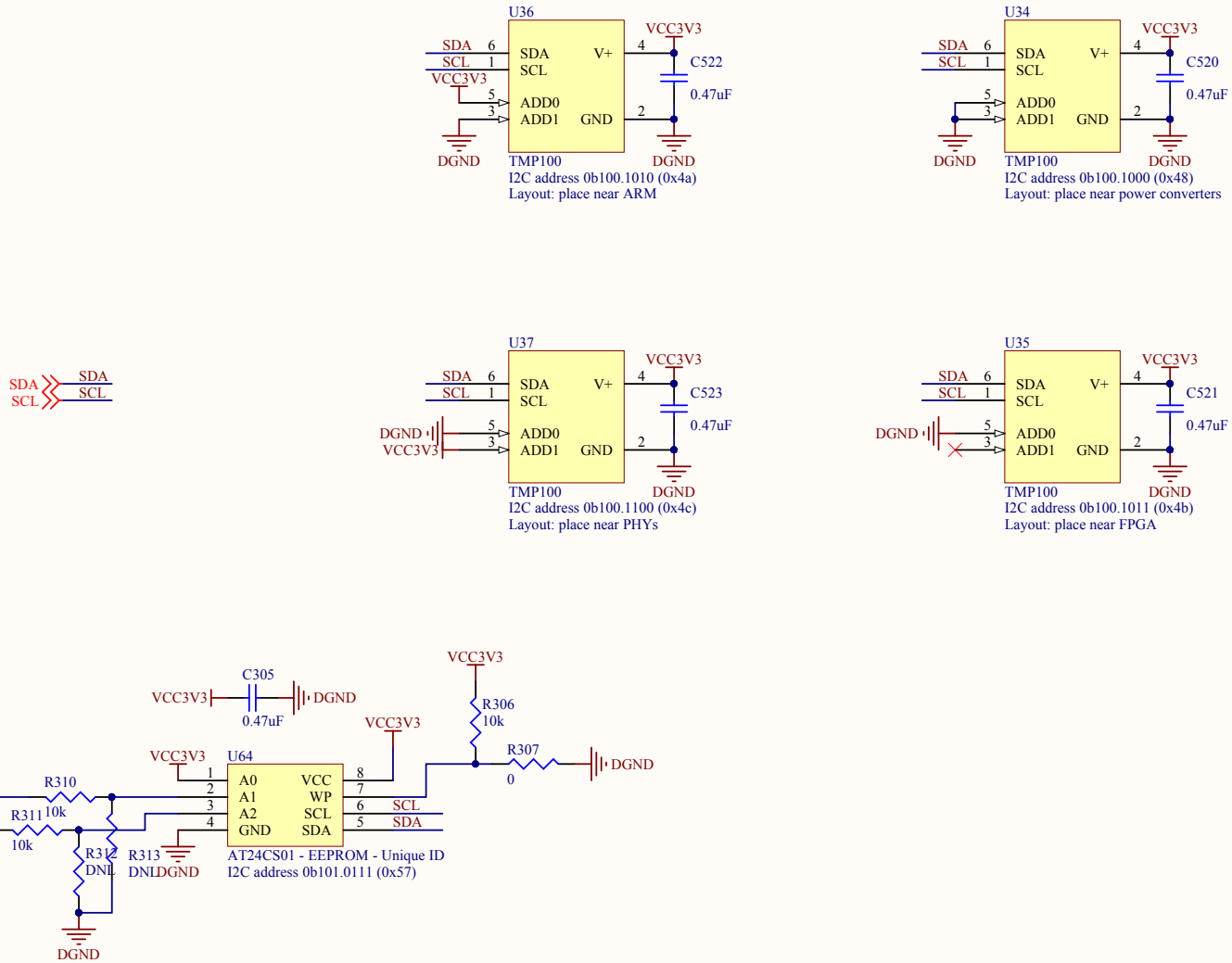
FMC clocks are 2.5v LVDS, and are hooked up to FPGA banks powered via either VADJ or VCCIOB. In both cases, the actual bank voltage isn't known a-priori, so it's necessary to provide a level-translation or coupling network. We follow the advice of UG471, p.92.

Note that some clocks are actually bidirectional, so I've modified the recommended biasing network to be symmetric.

Title: LVDS Clock Coupling		
Size: Letter	Number:	Revision: 3-2014-01-07
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U43 TCA9548APWR
U44 TCA9548APWR
U46 TCA9548APWR
U47 TCA9548APWR
U45
TCA9548APWR
VCC3V3
DGND
ARM_RESET
I2C_ARM_RESET
I2C_ARM0
I2C_ARM1
I2C_ARM2
I2C_ARM3
0b1110.000
0b1110.001
0b1110.010
0b1110.011
0b1110.100
I2C_FMCA
I2C_FMCB
I2C_QSFPA
I2C_QSFPB
I2C_SFP
I2C_SMPS
I2C_BP
I2C_GPIO
I2C_FPGA_RESET
I2C_FPGA0
RT14
RT15
3.3k, EXB-2HV332JV
C326
C328
C330
C332
C334
0.47uF
RN7
RN8
DNL
This matrix permits I2C bus functions to be split among different I2C ports on the ARM and FPGA.
I2C Bus Matrix
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I2C
SDA
SCL
I2C_GPIO
U41
PCA9575PW2,118
Address 0b0100000
U59
PCA9575PW2,118
Address 0b010011
U42
PCA9575PW2,118
Address 0b0100001
U48
PCA9575PW2,118
Address 0b0100010
VCC3V3
DGND
PGND
GPIO_IRQ
GPIO_RESET
R239 4.7k
R240 4.7k
FMCA_EN_12V0
FMCA_EN_3V3
FMCA_EN_VADJ
FMCA_PG_M2C
FMCA_PG_C2M
FMCA_PRSNT_M2C_L
FMCA_CLK_DIR
SFP_LOS
SFP_ModPrsL
FMCB_EN_12V0
FMCB_EN_3V3
FMCB_EN_VADJ
FMCB_PG_M2C
FMCB_PG_C2M
FMCB_PRSNT_M2C_L
FMCB_CLK_DIR
GPLED12
GPLED11
GPLED10
GPLED9
GTX1V8PowerFault
PHYAPowerFault
PHYBPowerFault
ArmPowerFault
SFP_TxFault
SFP_TxDisable
SFP_ModPrsL
SFP_RS0
SFP_RS1
SFP_LOS
SFP_AUX
TxFault
TxDisable
ModPrsL
RS0
RS1
LOS
BP_SLOW_GPIO0
BP_SLOW_GPIO1
BP_SLOW_GPIO2
BP_SLOW_GPIO3
BP_SLOW_GPIO4
BP_SLOW_GPIO5
BP_SLOW_GPIO[0..5]
BP_I2C_GPIO[0..5]
R93 4.7k
R94 4.7k
QSFP_AUX
QSFPA_ModPrsL
QSFPA_ResetL
QSFPA_LPMode
QSFPA_IntL
QSFPA_ModSelL
QSFP_AUX_A
QSFPB_ModPrsL
QSFPB_ResetL
QSFPB_LPMode
QSFPB_IntL
QSFPB_ModSelL
QSFP_AUX_B
C214C215C216
JMK105BJ474KV-F
0.47uF
C319C320C321
C440C441C442
SEL_SFP
LDOFault
LDOFaults
R384
R385
R386
R387
200
LED, 0603, Red
D_GP12
D_GP11
D_GP10
D_GP9
FMC_AUX
EN_12V0
EN_3V3
EN_VADJ
TMS
TCK
TDO
TDI
TRST
I2C too slow to host JTAG
PG_M2C
PG_C2M
PRSNT_M2C_L
CLK_DIR
FMC_AUX_A
FMC_AUX_B
D5
Schottky Diode CDBU00340
PG3V3
RT12
RT13
3k3, EXB-2HV332JV
SW6
SW-DIP8
GP_SW1
GP_SW2
GP_SW3
GP_SW4
GP_SW5
GP_SW6
GP_SW7
GP_SW8
GP_LED1
GP_LED2
GP_LED3
GP_LED4
GP_LED5
GP_LED6
GP_LED7
GP_LED8
U18
U19
U62
SN74LVC3G14DCTR
Gnd
Vcc
3A
2A
1A
3Y
2Y
1Y
LED Front Panel GYR
R279
R280
R281
R282
R283
R284
100
DS12A
DS12B
DS12C
DS11A
DS11B
DS11C
R286
R299
D_GP7
D_GP8
C269
C270
C271
C322C323C324
0.1uF
0402
0.47uF
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I2C GPIO Expanders
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Graeme Smecher



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Fiducial marks

FM1 Fiducial
FM2 Fiducial
FM3 Fiducial
FM4 Fiducial
FM5 Fiducial
FM6 Fiducial
FM7 Fiducial
FM8 Fiducial
FM9 Fiducial
FM10 Fiducial

Logos

Logo1
LOGO
ChimeLogo1
Logo2
LOGO
ChimeLogo1
Logo3
LOGO
McGillLogo1
Logo4
LOGO
McGillLogo1
Logo5
LOGO
McGillLogo1

FMC Front Panel Bezels

FMC Bezel
Bezel1
FMC Bezel
Bezel2
FMC Bezel
Bezel3
FMC Bezel
Bezel4

These FMC Front panel Bezels have been added to aid the mechanical design of the board

FP1
9U Front panel
FrontPanel 9U

The heatsinks are not part of the BOM at the moment - Which one should we choose to start?

HS1
Heatsink

HS2
MPW37.5-12

HS3
Heatsink

Have ensured that the PCB will be compatible with these heatsinks.

The MPA31-12 is 11.6mm in height and has 3.84, 2.43 and 1.82 degC/W at 200, 400 and 600 LFM respectively

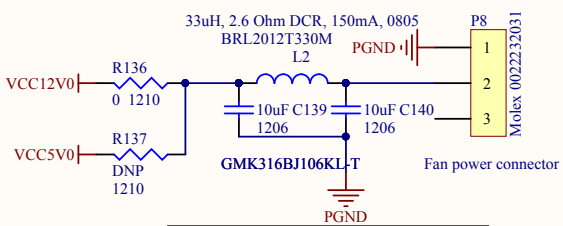
The MPW37.5-12 is 11.6mm in height and has 2.74, 1.77 and 1.4 decC/W at 200, 400 and 600 LFM respectively

The SLF-40 is 13.8mm in height and has 2 degC/W. Note that in a crate this height will cause a collision.

Front panel Support

PS1
Front panel bracket
61-960
PS2
Front panel bracket
61-960
PS3
Front panel bracket
61-960

ASSEMBLY ISSUE: Board stiffener warped board



Pi filter here will only attenuate high frequency noise

Board Stiffener

Board Stiffener
BusBarNet
BoardStiffener1
PGND

14 inches long, 1 inch spacing, half hardness, non epoxy coated, lowest profile 250mil

ITX Case mounting holes

ITXMount
ITX Mounting Holes
0
Gnd
ITXmountingholes
PGND

ARM / PHY / Memory Shield

Arm Shield
SH1
20S-CBS-4x2x0.2
PGND

Shield is 200mil high

Title: Mechanical		
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